

Mapping the Consumption Gap:
Planned Data Center Capacity versus Electricity Supply in
California

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Abstract

California’s utilities reported 23,277 MW of data center energization requests to the California Energy Commission (CEC) in December 2025, of which 20,677 MW are located in the state: 5,086 MW with signed agreements, 6,987 MW in active applications and 8,604 MW of inquiries. That is 40 percent of the record CAISO peak and about twenty times the roughly 1,000 MW of data center load that exists. This report asks how much of it is likely to be served by 2030, whether California’s supply can keep up, what a flexible load could do about the difference, and how well the state’s way of counting compares with the other regimes in use. It is an independent replication and audit of the CEC’s data center forecast method, built on public data with every input recorded in a manifest and every figure traceable to its files.

The CEC’s published confidence levels and utilization factor, with SVP’s requests exempt as the Commission states, reproduce the CEC’s own 2040 endpoints exactly and its 2030 components within 2 to 6 percent. Applied symmetrically to the supply side, a completion model estimated on 925 unit-vintages of California’s own planned generators since 2016 removes a third of the 20 GW planned list for 2030. A 10,000-draw Monte Carlo over both sides puts the 2030 gap between demand growth and supply growth at 50 TWh (19 to 78) and 7.2 GW at the net peak (4.2 to 10.1), with data centers 18 TWh and 2.3 GW of it; the proposal’s upper bound, every requested megawatt operating flat, gives 202 to 242 TWh and 25 to 30 GW. Imports, the state’s own load forecast, hydro and Diablo Canyon drive the energy gap, not the data center assumptions. Replicating the Duke Nicholas Institute’s method on CAISO hourly load gives 3.8 GW of curtailment-enabled headroom at 0.5 percent curtailment, twice the probability-weighted data center addition and half the weighted peak gap. Under six counting regimes the same requests are 20,677 MW (California and Texas rules), 5,086 MW (PJM and the Texas forecast rule), about 2,400 MW at ERCOT’s realized phase-transition rates and about 1,000 MW as measured load. The data are frozen on September 22, 2026, the repository is tagged, and the processed outputs ship with the report.

Chapter 1

Introduction

1.1 The question

At the IRTF SUSTAIN workshop in July 2026, Jukka Manner showed that Finland operates about 285 MW of data centers while its public investment registry lists 7,777 MW more, phased from feasibility study to investment decision [Manner, 2026]. The registry makes the gap between what is planned and what exists visible in one place. The United States has no such registry. What is known about planned American data centers is scattered across state energy agencies, grid operators, utility filings and commercial trackers, each counting in its own units and at its own stage of maturity, and the same project can appear in several of them at once.

California is the largest state without a registry and the one whose numbers travel furthest. In December 2025 its utilities reported 23,277 MW of data center energization requests to the California Energy Commission (CEC), sorted into three tiers: 5,086 MW with signed service agreements, 9,587 MW in active applications and 8,604 MW of inquiries [California Energy Commission, 2026b,d]. The CEC does not count those requests at face value. It applies a confidence level to each tier, a utilization factor and a ramp, and carries the result into the adopted demand forecast as 1,743 MW at the 2030 CAISO peak in its Planning forecast and 4,377 MW in its Local Reliability scenario [California Energy Commission, 2026e]. Whether those discounts are right is an empirical question that the state’s own record, and the records of Texas and the Mid-Atlantic, can inform.

1.2 Research questions and scope

The proposal set four research questions, restated here for a California-only scope after Finland was dropped as a replication target and kept only as the motivating example:

- **RQ1.** How large is California’s planned data center pipeline by stage, in absolute terms and relative to the grid it would connect to?
- **RQ2.** What is the gap between planned demand and supply through 2030, raw and probability-weighted, in annual energy and at the net peak?
- **RQ3.** How do the data and measurement regimes that count planned loads, the CEC tiers, ERCOT’s interconnection phases, PJM’s firm and non-firm vetting, the EIA pilot survey, Texas SB 6 and FERC Docket RM26-4, differ, and what would California’s headline number be under each?

- **RQ4.** How much of the gap could load flexibility absorb, measured as the curtailment-enabled headroom of the existing system?

Alongside those, the report answers the questions asked during the work: how much of the state’s electricity data centers use today, how high data center construction has climbed since the release of ChatGPT, where the grid stands now, and how California’s generation has grown. Texas and PJM enter as data rather than as narrative. ERCOT publishes the only monthly, phase-level series of large-load requests large enough to estimate how a pipeline actually moves, and PJM publishes the only explicit firm and non-firm rule; both are used to test the CEC’s discounts rather than to compare states.

The geographic boundary is stated wherever it matters. Hourly load, prices and carbon intensity are CAISO’s; generation, capacity, retirements and the demand forecast are statewide; the tiers are California-only, with the 2,600 MW that Valley Electric Association reported for Nevada carried as a memo line because it sits inside the CAISO balancing area but outside the state. The request data are the December 2025 vintage, the newest complete set at the data freeze.

1.3 Contribution

The contribution is an independent replication and audit of the CEC’s method with transparent, tested assumptions rather than a new forecast. Five things are new. First, the CEC’s published parameters are shown to reproduce its own forecast, including the exemption of Silicon Valley Power’s queue that the methodology memo states but does not quantify, so the forecast can be rebuilt and varied by anyone (Chapter 5). Second, the supply side is discounted with the same discipline as the demand side, using a completion model estimated on every California generator planned since 2016 and the outcome of each by December 2025 (Chapter 4). Third, the CEC’s confidence levels are set against two external yardsticks: monthly phase-transition rates estimated from ERCOT’s queue reports and the firm-only rule PJM applies (Chapter 5). Fourth, the gap is reported as a distribution with a stated upper bound, with a sensitivity analysis that names the assumptions that move it. Fifth, the Duke Nicholas Institute’s headroom method is replicated on CAISO load and expressed as a share of the gap, and six counting regimes are scored on a common rubric with the same requests restated under each rule.

Every number in the report can be followed back to a file. Raw downloads are immutable and recorded with their URL, access date and SHA-256 in a manifest; the processed outputs, figures and LaTeX tables are regenerated by one command per chapter; every figure carries its source line, naming the processed files, manifest ids and code that produced it; the appendix repeats those lines as tables; and the data are frozen on a stated date, the repository tagged, and the processed outputs exported with the report.

1.4 Related work in brief

The national picture is set by the Lawrence Berkeley National Laboratory’s usage reports and EPRI’s state table, which put data centers at a few percent of United States consumption in 2023 with California among the larger states [Electric Power Research Institute, 2024]; Chapter 2 uses EPRI’s California share as one of four estimates of existing use. The CEC’s own methodology memo, its preliminary and final forecast decks and the CED 2025 forms are the primary documents this report audits [California Energy Commission, 2026d, 2025, Smith, 2026, California Energy Commission, 2026c]. On attrition, LBNL’s interconnection-queue studies and NERC’s large-load

assessments document how few requests reach operation and how often the same project is submitted in several places; this report estimates California’s own generator attrition from EIA-860M vintages instead of borrowing a national rate [U.S. Energy Information Administration, 2026b]. The Duke Nicholas Institute’s headroom study supplies the flexibility method and the CAISO values that Chapter 5 replicates [Norris et al., 2025]. The break-test and forecasting methods of Chapter 3 follow Chow and Bai and Perron [Chow, 1960, Bai and Perron, 1998, 2003], and the sensitivity analysis uses Sobol indices as implemented in SALib [Sobol’, 2001, Herman and Usher, 2017].

1.5 Structure of the report

Chapter 2 measures the grid as it stands and the range of existing data center use. Chapter 3 measures data center growth, reproduces the Census construction chart with break tests, builds California proxies and answers RQ1 from the tier vintages. Chapter 4 builds the supply side: generation and imports since 2010, capacity, storage and curtailment, the retirement schedule, the realization model and probability-weighted supply for 2030. Chapter 5 puts the two sides together for RQ2, replicates the headroom method for RQ4 and compares the counting regimes for RQ3. Chapter 6 restates the contribution, sets out what the audit found about each of the CEC’s assumptions, draws the policy implications and states the limits. Appendix A lists the provenance of every figure.

Chapter 2

Current Status of Data Centers and the Grid in California

This chapter establishes the baseline that the gap analysis in later chapters is measured against. It answers two questions: what the CAISO grid looks like hour by hour today, and how much electricity California’s existing data centers already use. Every number is computed from the raw files listed in `data/raw/manifest.csv`; the notebook `notebooks/01_baseline_grid.ipynb` reproduces each table and figure.

2.1 Grid baseline: CAISO demand, net load and imports, 2019–2025

Hourly demand, generation by fuel and net interchange for the CAISO balancing authority come from the EIA-930 Hourly Electric Grid Monitor, using EIA’s “Adjusted” series [U.S. Energy Information Administration, 2026d]. Net load is demand minus utility-scale solar and wind. Two definitional points matter. First, EIA-930 demand for CISO includes battery charging and pumping load, whereas CAISO’s own reported demand excludes charging; in 2025 the hourly gap between the two series tracks battery charging almost one for one (slope 0.91, correlation 0.87). Both definitions are kept in the processed table. Second, EIA’s series contains reporting artifacts, chiefly in early 2019, where demand falls implausibly below CAISO’s demand plus charging; those hours are nulled rather than corrected, leaving 8,435 valid hours in 2019 (325 missing, 3.7 percent) and at most 27 missing hours in any later year; annual energy is the mean of the valid hours times the calendar hours, which imputes the missing hours at the annual mean, and the duration curves and profiles below are drawn over valid hours only. Third, EIA-930 carries no CISO interchange at all, in any of its reported, imputed or adjusted variants, from July 2 to November 3, 2024, and no hydro for parts of 2019 and 2020; those hours are filled from CAISO’s own five-minute imports and hydro series through a per-year linear calibration fitted on the overlapping hours (correlation 0.986 or better; the fill log is `ch1_eia930_fill_log.csv`). Fourth, EIA warns that CAISO changed its data collection for natural gas in December 2023 and that the EIA-930 gas series is not comparable across that date [U.S. Energy Information Administration, 2025b]. The warning is confirmed here: through November 2023 EIA’s CISO gas generation matches CAISO’s own five-minute fuel mix to within 0.1 TWh a month, and from December 2023 it exceeds it by 0.7 to 1.8 TWh a month, 20 TWh in 2024 and 25 TWh in 2025 (`ch1_gas_series_comparability_monthly.csv`). Gas figures in Tables 2.2 and 2.3 are EIA’s series and must not be compared across that month; Table 2.1 shows the month-by-month comparison.

Table 2.1: EIA-930 CISO natural gas generation against CAISO’s own fuel-mix gas series around the December 2023 reporting change, with the gas CO₂ factor implied by CAISO’s accounting.

Month	EIA-930 CISO gas (TWh)	CAISO fuel-mix gas (TWh)	EIA minus CAISO (TWh)	CAISO gas share of demand (%)	Implied gas CO ₂ factor, CAISO accounting (kg/MWh)
Sep 2023	7.09	7.09	+0.01	38.0	440
Oct 2023	9.05	9.06	-0.01	50.8	441
Nov 2023	8.05	7.97	+0.08	48.8	438
Dec 2023	8.94	8.25	+0.69	47.8	432
Jan 2024	9.21	7.94	+1.26	47.9	439
Feb 2024	6.34	5.13	+1.21	33.6	440
Mar 2024	4.85	3.36	+1.49	21.9	440
Apr 2024	4.49	2.71	+1.78	18.6	452
2023 total	88.0	87.2	+0.8	—	—
2024 total	85.5	65.2	+20.3	—	—
2025 total	79.0	54.2	+24.8	—	—

Table 2.2: CAISO annual baseline from EIA-930 (Adjusted series), 2019–2025. Demand energy is the mean of valid hours times calendar hours. Storage charging is the estimated net charging of the CAISO battery fleet from the five-minute fuel mix. Import share is net imports divided by demand. Gas is EIA’s series, not comparable across December 2023.

Year	Demand (TWh)	Peak (GW)	Load factor	Peak net load (GW)	Min net load (GW)	Solar (TWh)	Wind (TWh)	Storage charging (TWh)	Net imports (TWh)	Imports (% of demand)
2019	215.8	43.8	0.56	40.5	5.2	27.7	15.9	0.1	53.8	25
2020	218.0	46.6	0.53	44.0	5.6	28.6	14.9	0.1	59.4	27
2021	219.2	43.6	0.57	40.5	3.0	32.8	18.3	0.9	54.5	25
2022	223.7	51.1	0.50	47.5	1.9	35.8	17.4	1.7	48.7	22
2023	218.2	44.0	0.57	41.8	1.7	37.2	16.4	4.4	28.8	13
2024	224.1	47.6	0.54	42.4	0.3	44.8	20.1	9.5	31.7	14
2025	224.1	43.9	0.58	39.8	-3.3	49.7	19.8	14.2	36.1	16

Table 2.2 shows a grid whose annual demand has been flat since 2019 while its composition changed. Demand ranged between 216 and 224 TWh in every year. Utility-scale solar output rose from 27.7 TWh to 49.7 TWh and the combined solar and wind share of demand went from a fifth to nearly a third. Net imports fell from a quarter of demand in 2019 to 13 percent in 2023 and 14 to 16 percent in 2024 and 2025. Battery charging, negligible in 2019, reached 14 TWh in 2025, and battery discharge peaked above 10 GW (Table 2.6). The hourly demand record remains 51.1 GW, set on September 6, 2022 at 18:00.

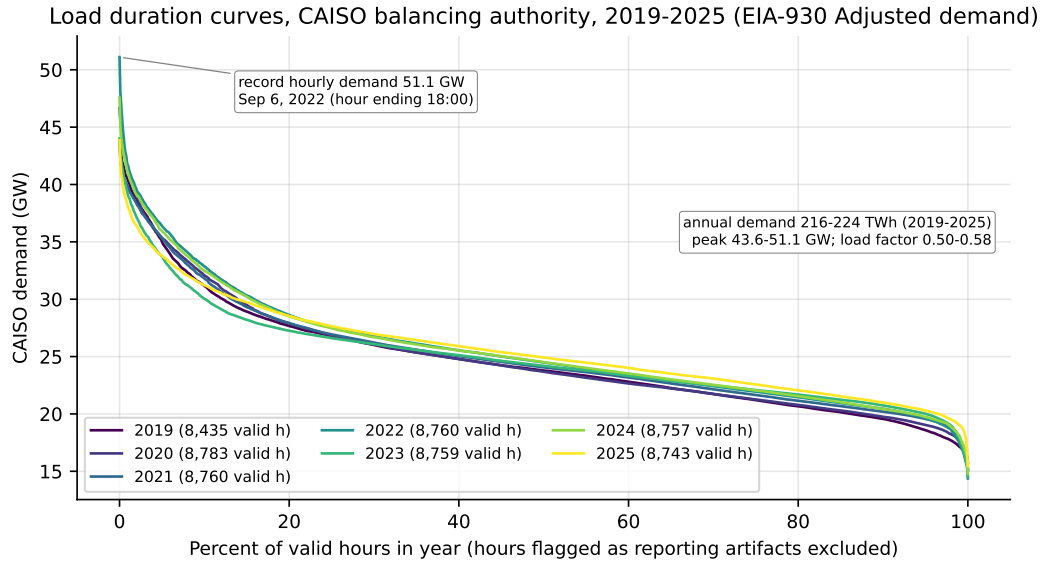


Figure 2.1: Load duration curves over valid hours (counts in the legend), CAISO, 2019–2025. Hours flagged as reporting artifacts are excluded and interchange and hydro gaps are filled from CAISO series. Source: EIA-930 Adjusted demand.

Source line: fig1_01. Data: ch1_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet. Sources (manifest ids): eia930_balance_2019_jan_jun .. 2025_jul_dec; caiso_outlook demand and fuelsource (fills, artifact filter). Code: ch1_baseline.load_eia930_ciso, clean_demand_against_caiso, fill_from_caiso, duration_curves; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

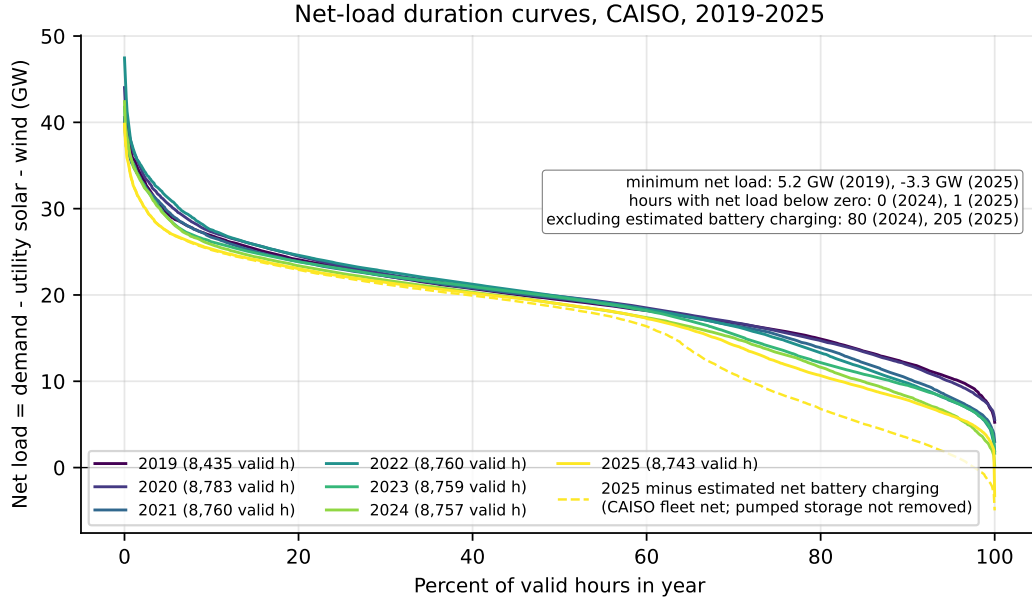


fig1_02. Data: ch1_net_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet. Sources (manifest ids): eia930 balance files; caiso_outlook fuelsource (Batteries). Code: ch1_baseline.add_storage_adjusted, duration_curves; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 2.2: Net-load duration curves over valid hours, CAISO, 2019–2025. The dashed line subtracts the estimated net battery charging of the CAISO fleet from 2025 demand; it does not remove pumped storage and cannot recover gross charging in hours when part of the fleet discharges. Source: EIA-930, CAISO fuel mix.

Source line: fig1_02. Data: ch1_net_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet. Sources (manifest ids): eia930 balance files; caiso_outlook fuelsource (Batteries). Code: ch1_baseline.add_storage_adjusted, duration_curves; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figures 2.1 and 2.2 make the contrast visible. The load duration curves are nearly indistinguishable across years. The net-load duration curves diverge at the low end: by 2024 and 2025 the lowest-net-load hours approach zero, and once battery charging, which is itself a response to the surplus, is removed from demand, utility solar and wind alone exceeded end-use demand in 80 hours of 2024 and 205 hours of 2025. This midday surplus is the physical basis for the flexibility question in Chapter 5.

Average daily profiles by season, CAISO: demand (top) and net load (bottom), 2019 / 2022 / 2025

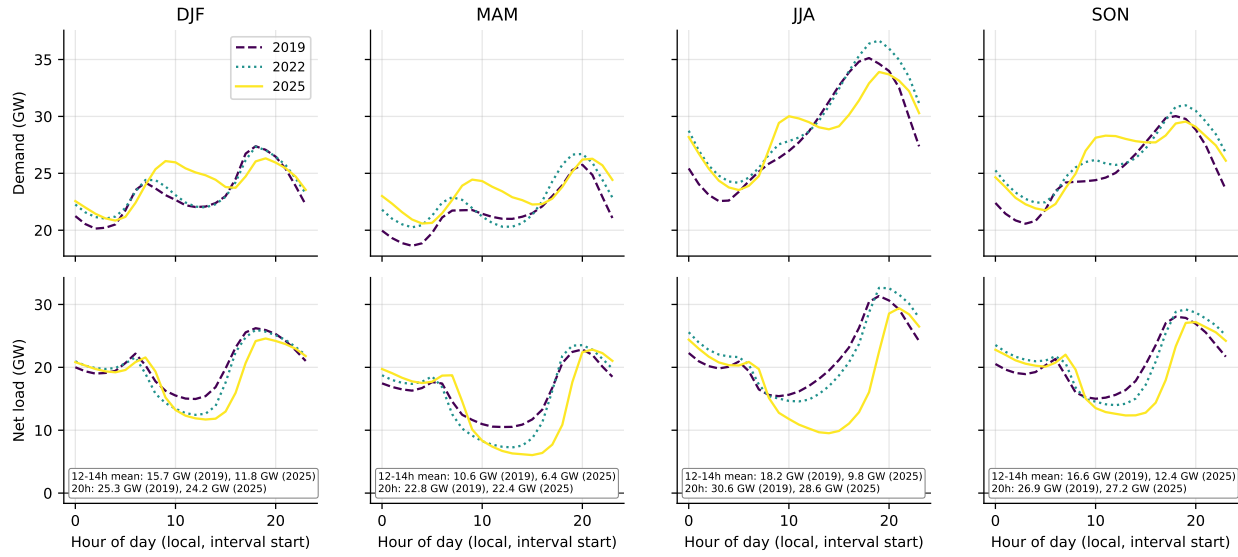


fig1_03. Data: ch1_seasonal_profiles.csv. Sources (manifest ids): eia930 balance files. Code: ch1_baseline.seasonal_profiles; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 2.3: Average daily profiles by season: demand (top) and net load (bottom), 2019, 2022 and 2025. Means over valid hours; descriptive averages, not weather-normalised, and demand includes storage charging and pumping. Source: EIA-930.

Source line: fig1_03. Data: ch1_seasonal_profiles.csv. Sources (manifest ids): eia930 balance files. Code: ch1_baseline.seasonal_profiles; run_chapter1 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

The seasonal profiles in Figure 2.3 show the mechanism. Demand profiles are similar across years, with a modest midday rise in 2025 that reflects storage charging. Net-load profiles deepen at midday and steepen in the evening: in spring the average net load between 12:00 and 14:00 fell from 10.6 GW in 2019 to 6.4 GW in 2025 while the 20:00 value stayed near 22 GW, and in summer the average net load between 12:00 and 14:00 fell from about 18 GW to under 10 GW (the values printed on the figure are the 12:00 to 14:00 and 20:00 means for 2019 and 2025).

Table 2.3: The 100 highest net-load hours of each year: when they occur and what serves them. Mean gas is EIA's series (comparability note in the text). Source: EIA-930.

Year	Net load range (GW)	Distinct days	Jul-Sep (%)	17:00–21:00 (%)	Median start	Mean solar (GW)	Mean imports (GW)	Mean gas (GW)
2019	35.6–40.5	34	90	99	18:00	0.9	8.5	18.5
2020	36.4–44.0	30	92	86	18:00	1.2	7.4	22.6
2021	34.7–40.5	42	92	100	19:00	0.6	6.3	21.0
2022	36.6–47.5	27	95	82	19:00	1.2	7.4	22.5
2023	35.0–41.8	35	96	93	20:00	0.6	3.6	21.8
2024	34.7–42.4	37	91	79	20:00	0.9	4.7	20.9
2025	32.3–39.8	33	98	78	20:00	0.7	4.6	17.3

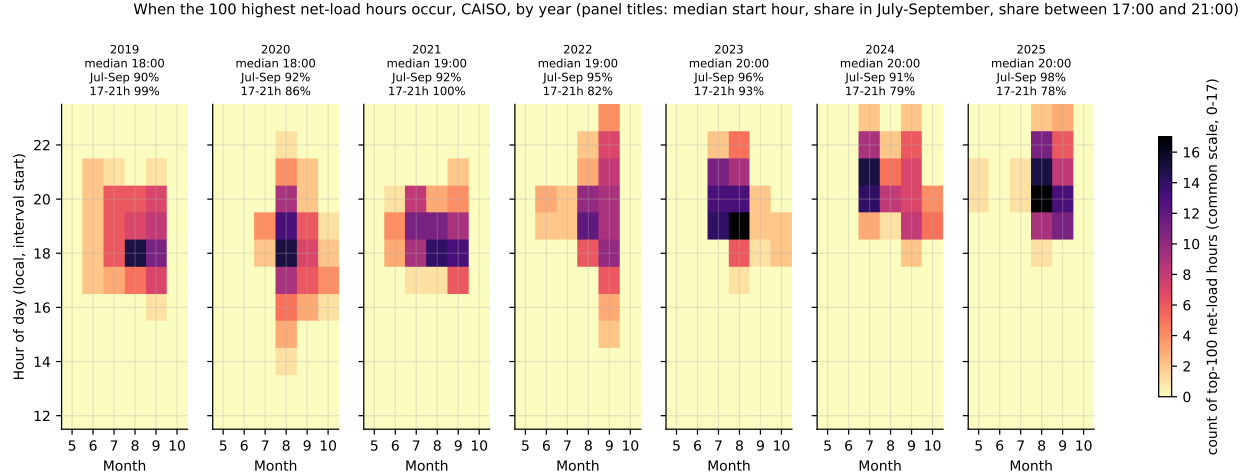


Figure 2.4: Month and hour of the 100 highest net-load hours, by year, on one common colour scale. Source: EIA-930.

Source line: fig1_04. Data: ch1_top100_net_load_hours.csv; ch1_top100_net_load_timing.csv. Sources (manifest ids):
 eia930 balance files. Code: ch1_baseline.top_hours, top_hours_timing; run_chapter1 step 1. Data frozen 2026-09-22;
 data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 2.3 and Figure 2.4 identify the hours that set capacity needs. In every year at least 90 percent of the top 100 net-load hours fall in July through September and at least 78 percent between 17:00 and 21:00. Their median start time moved from 18:00 in 2019 and 2020 to 19:00 in 2021 and 2022 and to 20:00 from 2023 onward, the signature of solar pushing the net peak past sunset and of storage shaving the early evening. Solar contributes between 0.6 and 1.2 GW on average in these hours; gas contributes 17 to 23 GW and imports 4 to 9 GW. The implication for a flat load such as a data center is direct: each megawatt adds one for one to hours that solar does not serve, and the relevant capacity resources are storage, gas and imports.

2.2 Day-ahead prices and negative-price hours

Hourly day-ahead prices at the three investor-owned utility default load aggregation points come from CAISO OASIS [California Independent System Operator, 2026c]. OASIS retains roughly 39 months of data, so hourly prices run from July 2023; 2024 and 2025 are complete years and 2026 covers January 1 to September 22. EIA's daily ICE hub prices at NP15 and SP15 extend to 2015 but are daily block averages [U.S. Energy Information Administration, 2026e].

Table 2.4: Day-ahead LMP statistics at CAISO default load aggregation points. Source: CAISO OASIS PRC_LMP, DAM.

Node	Year	Hours	Mean	Median	P5	P95	Min	Hours < \$0	Hours ≤ \$5	Negative hours 10:00–16:00 (%)
(\$/MWh)										
PG&E DLAP	2024	8784	43.0	40.1	9.0	76.2	-33.1	190	327	88
PG&E DLAP	2025	8760	40.6	42.4	8.7	63.8	-36.8	218	337	94
PG&E DLAP	2026*	6359	31.5	31.9	0.4	60.1	-14.2	294	480	87
SCE DLAP	2024	8784	34.0	37.0	-20.0	67.7	-66.6	1131	1369	77
SCE DLAP	2025	8760	34.4	40.2	-8.3	60.5	-40.5	877	1109	78
SCE DLAP	2026*	6359	28.7	30.3	-9.0	64.8	-17.9	890	1123	72
SDG&E DLAP	2024	8784	36.2	38.2	-17.3	72.5	-63.1	989	1227	77
SDG&E DLAP	2025	8760	37.5	42.4	-6.8	65.8	-40.1	748	955	80
SDG&E DLAP	2026*	6359	32.3	33.1	-7.2	70.3	-17.2	706	950	74

* partial year: 2026 covers January 1 to September 22.

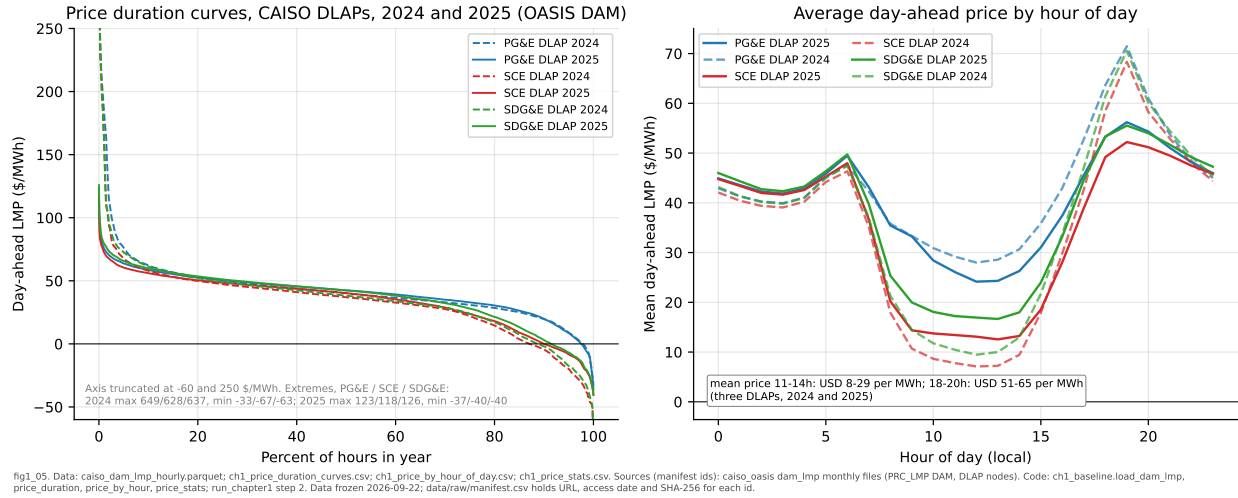


Figure 2.5: Price duration curves (left) and average price by hour of day (right), 2024 and 2025; one colour per DLAP, dashed 2024 and solid 2025. The duration axis is truncated at -60 and 250 $\$/\text{MWh}$ and the extremes are printed in the panel (2024 maxima of 628 to 649 $\$/\text{MWh}$). Source: CAISO OASIS.

Source line: fig1_05. Data: caiso_dam_lmp_hourly.parquet; ch1_price_duration_curves.csv; ch1_price_by_hour_of_day.csv; ch1_price_stats.csv. Sources (manifest ids): caiso_oasis dam_lmp monthly files (PRC_LMP DAM, DLAP nodes).

Code: ch1_baseline.load_dam_lmp, price_duration, price_by_hour, price_stats; run_chapter1 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Negative-price hours by DLAP: complete years and matched Jan 1-Sep 22 windows
(labels: share of hours in the window; OASIS retention starts July 2023)

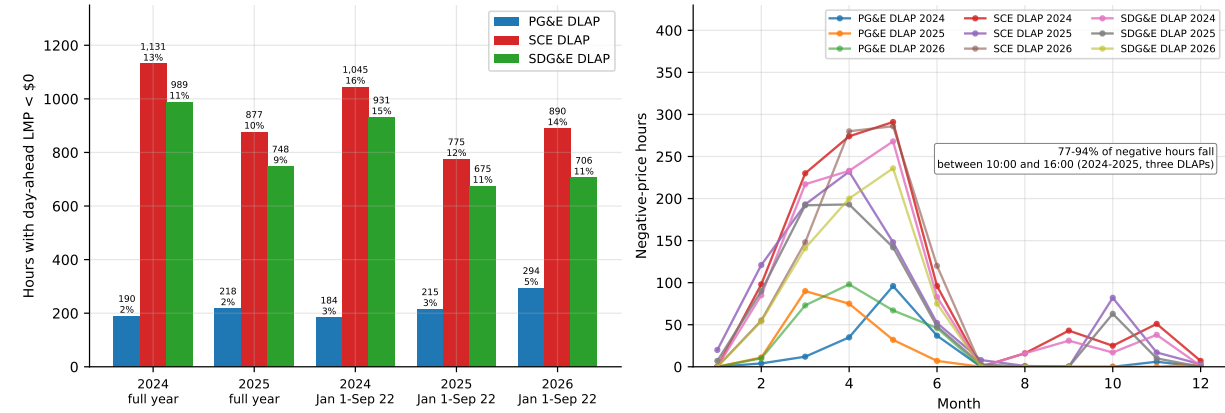


fig1_06. Data: ch1_negative_price_hours_windows.csv; ch1_negative_price_hours_by_month.csv; ch1_price_stats.csv. Sources (manifest ids): caiso_oasis dam_lmp monthly files. Code: ch1_baseline.negative_hours_by_month, price_stats; run_chapter1 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 2.6: Negative-price hours at the three DLAPs. Left: the complete years 2024 and 2025 and matched January 1 to September 22 windows for 2024, 2025 and 2026, with the share of hours in each window; 2023 (July to December only) is omitted from the comparison. Right: by month. Source: CAISO OASIS.

Source line: fig1_06. Data: ch1_negative_price_hours_windows.csv; ch1_negative_price_hours_by_month.csv; ch1_price_stats.csv. Sources (manifest ids): caiso_oasis dam_lmp monthly files. Code: ch1_baseline.negative_hours_by_month, price_stats; run_chapter1 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Negative day-ahead prices are now routine in Southern California (Table 2.4, Figures 2.5 and 2.6). The SCE aggregation point cleared below zero in 1,131 hours of 2024 and 877 hours of 2025, and in 890 hours of 2026 through September 22; SDG&E's point saw 989 and 748 hours; PG&E's 190 and 218. On the matched January 1 to September 22 window the SCE point had 1,045 negative hours in 2024, 775 in 2025 and 890 in 2026, 16, 12 and 14 percent of the hours in the window, so 2026 sits between the two complete years rather than above them (ch1_negative_price_hours_windows.csv). Between 72 and 94 percent of those hours fall between 10:00 and 16:00, concentrated in February through June, and the average price by hour of day drops to \$7–\$30 per MWh at midday against \$50–\$70 in the evening peak. The daily ICE block prices show almost no negative days, which is why the hourly series is required. This is the California-specific fact that Chapter 5 uses: a data center able to place consumption in those hours faces a zero or negative marginal energy price, and one that cannot adds to the evening hours where prices and net load are highest.

2.3 Hourly carbon intensity

CAISO's Today's Outlook reports five-minute CO₂ by source, with imports carried at CAISO's default factor and export emissions netted against them [California Independent System Operator, 2026d]. The indicator used here is that accounting total divided by CAISO demand, energy-weighted per year: an accounting rate for the supply serving CAISO demand, not a measured consumption footprint. Three properties of the data are made explicit. First, because exports are netted, net CO₂ is negative in export-heavy hours (374 in 2024, 425 in 2025); the headline series keeps those hours at their negative values, and a variant that floors them at zero is shown as a sensitivity (it raises 2025 from 186.3 to 187.7 g/kWh). Second, CAISO publishes a fixed grid of

288 five-minute rows for every calendar day, including the two daylight-saving transition days, so the repeated autumn hour and the skipped spring hour cannot be separated; an hour is counted as valid when at least six of its twelve intervals exist for both CO₂ and demand, hours with no source data are left missing rather than counted as zero, and coverage is 99.8 percent of calendar hours or better in every year. Third, EIA’s warning that CAISO’s natural gas data collection changed in December 2023 [U.S. Energy Information Administration, 2025b] applies to the EIA-930 gas series, not to CAISO’s own accounting: the emission factor implied by CAISO’s gas CO₂ over CAISO’s gas generation stays between 432 and 452 kg/MWh in every month from June 2023 to June 2024 and between 437 and 458 in every year, and CAISO’s gas share of demand moves from 48.8 to 47.8 to 47.9 percent across November, December and January. The EPA eGRID output rate for the CAMX subregion is drawn as a reference with a different boundary: a generation-based rate for the whole subregion with no import factor [U.S. Environmental Protection Agency, 2025].

Table 2.5: CO₂ accounting intensity of CAISO demand: CAISO’s CO₂ accounting (imports net of exports) divided by CAISO demand, energy-weighted, with the floored variant, the number of negative hours and valid hours as a share of the calendar. eGRID CAMX is a generation output rate with a different boundary. Source: CAISO Today’s Outlook, eGRID 2023.

Year	Accounting (g/kWh)	Floored (g/kWh)	Negative hours	P5 (g/kWh)	P95 (g/kWh)	CO ₂ (Mt)	Imports share (%)	Gas share (%)	Valid hours (%)
2019	256.7	256.8	8	109	361	55.9	42	53	100.0
2020	277.4	277.4	0	132	368	60.4	42	54	100.0
2021	274.5	274.5	0	125	361	59.7	37	59	100.0
2022	264.4	264.4	13	108	355	58.1	34	63	100.0
2023	241.9	242.0	19	85	344	51.1	21	75	100.0
2024	199.4	200.7	374	7	322	42.4	27	68	99.8
2025	186.3	187.7	425	1	319	38.8	35	61	100.0
eGRID 2023 CAMX generation output rate	198.1	–	–	–	–	47.3	–	–	–

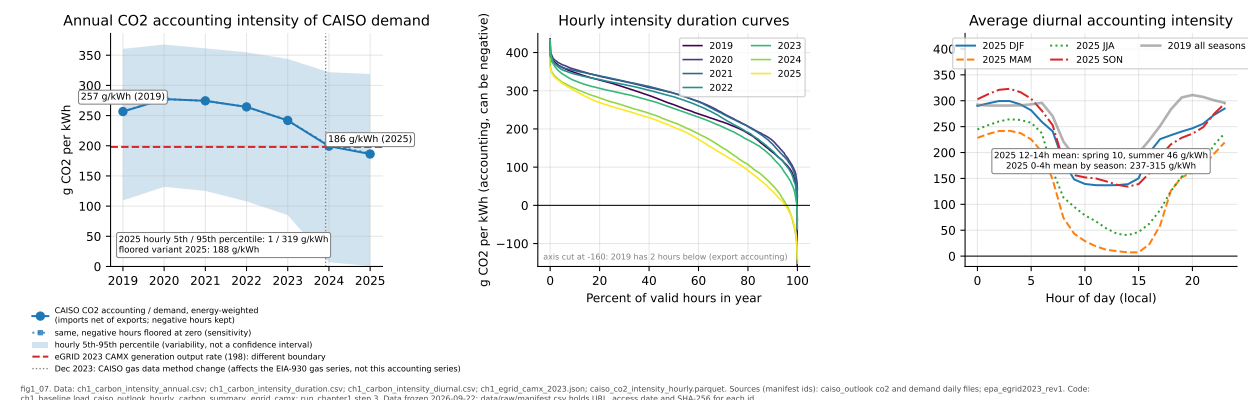


Figure 2.7: Annual, duration-curve and diurnal views of the CAISO CO₂ accounting intensity. Left: headline series with negative hours kept, the floored sensitivity, the hourly 5th to 95th percentile band (variability, not a confidence interval), the eGRID reference and the December 2023 EIA-930 gas reporting change. Middle: hourly duration curves, axis cut at –160 g/kWh. Right: 2025 seasonal diurnal means against 2019. Source: CAISO Today’s Outlook, eGRID 2023.

Source line: fig1_07. Data: ch1_carbon_intensity_annual.csv; ch1_carbon_intensity_duration.csv; ch1_carbon_intensity_diurnal.csv; ch1_egrid_camx_2023.json; caiso_co2_intensity_hourly.parquet. Sources (manifest ids): caiso_outlook co2 and demand daily files; epa_egrid2023_rev1. Code: ch1_baseline.load_caiso_outlook_hourly, carbon_summary, egrid_camx; run_chapter1 step 3. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 2.6: CAISO battery fleet energy and peak power from the five-minute fuel mix.

Year	Discharge (TWh)	Charge (TWh)	Peak discharge (GW)	Peak charge (GW)
2019	0.05	0.08	0.1	0.1
2020	0.06	0.08	0.2	0.2
2021	0.83	0.92	1.4	1.2
2022	2.73	1.70	3.0	2.1
2023	4.34	4.45	4.8	4.3
2024	8.18	9.45	7.8	7.2
2025	12.29	14.21	10.7	9.3

The accounting intensity fell from 257 g/kWh in 2019 to 186 g/kWh in 2025 (Table 2.5, Figure 2.7); the floored variant gives 257 and 188. The hourly spread widened: in 2025 the fifth percentile was 1 g/kWh and the ninety-fifth 319 g/kWh, and the average midday hour was about 10 g/kWh in spring and 46 g/kWh in summer against roughly 240 to 315 g/kWh at night. eGRID’s 2023 CAMX generation rate of 198 g/kWh is lower than the 2023 accounting figure of 242 g/kWh because it excludes the import factor and covers the whole subregion; it is a single-year reference, not a trend. The decline is consistent with the solar and battery additions of Table 2.2, but the series remains an accounting construct: a validated consumption-emissions series would need CAISO’s greenhouse gas tracking reports as an independent check, which is deferred to the supply chapter. For a flat load the annual factor is the relevant one; for a flexible load the diurnal ratio, about five to one in summer and far larger in spring, is the relevant number, and an average intensity is not a marginal emissions factor for load shifting.

2.4 How much of the state’s power do data centers use

No agency meters California’s data center consumption, and the four items in Table 2.8 and Figure 2.8 are not four comparable estimates of one quantity. They are grouped by what they are: one statewide historical estimate, one assumed conversion of a published peak, one illustrative count-based sensitivity and one utility subset. Shares are computed against EIA-861 California retail sales for the year that matches each item, 239.5 TWh in 2023 and 245.7 TWh in 2024 (Parts A, C and D, which avoids counting delivery-only sales twice) [U.S. Energy Information Administration, 2025a].

Table 2.7: Silicon Valley Power fact-sheet series used for the utility-subset estimate.

Fact sheet year	Peak demand (MW)	System load factor (%)	Retail sales (TWh)	Purchased and generated supply (TWh)	Peak $\times$ load factor $\times$ 8760 (TWh)
2017	586.6	72.5	–	3.73	3.73
2018	526.2	80.1	–	3.69	3.69
2019	587.8	72.4	–	3.73	3.73
2020	586.3	75.0	3.74	3.85	3.85
2021	594.8	79.3	4.03	4.13	4.13
2022	693.9	74.3	4.41	4.52	4.52
2023	669.2	78.3	4.48	4.59	4.59

Silicon Valley Power utility fact sheets; retail sales are reported from 2020. Data center shares applied to 2023 retail sales: 53 percent (SVP data center page), 55 percent (SVP Assembly deck, January 2026), about 60 percent (Santa Clara officials, 2025); SVP reports 64 to 67 percent utilization of requested capacity for its data center customers.

Table 2.8: Evidence and assumptions for existing California data center electricity use, by type. Shares use EIA-861 retail sales of the year in parentheses.

Group	Method	Data year	TWh per year	Share of retail sales, % (year)
A. Statewide estimate	EPRI 2024 state table, 2023 consumption	2023	9.3	3.9 (2023)
B. Assumed conversion	CEC 1,000 MW existing peak (Dec 2025) x assumed annual load factor 0.80-0.95	2025	7.0–8.3	2.9–3.4 (2024)
C. Illustrative sensitivity	Count-based: 321 Kollar-Grady facility points x assumed energy per facility	2023	6.8–13.6	2.8–5.7 (2023)
D. Utility subset	Silicon Valley Power only: 53-60% of SVP 2023 retail sales	2023	2.4–2.7	1.0–1.1 (2023)

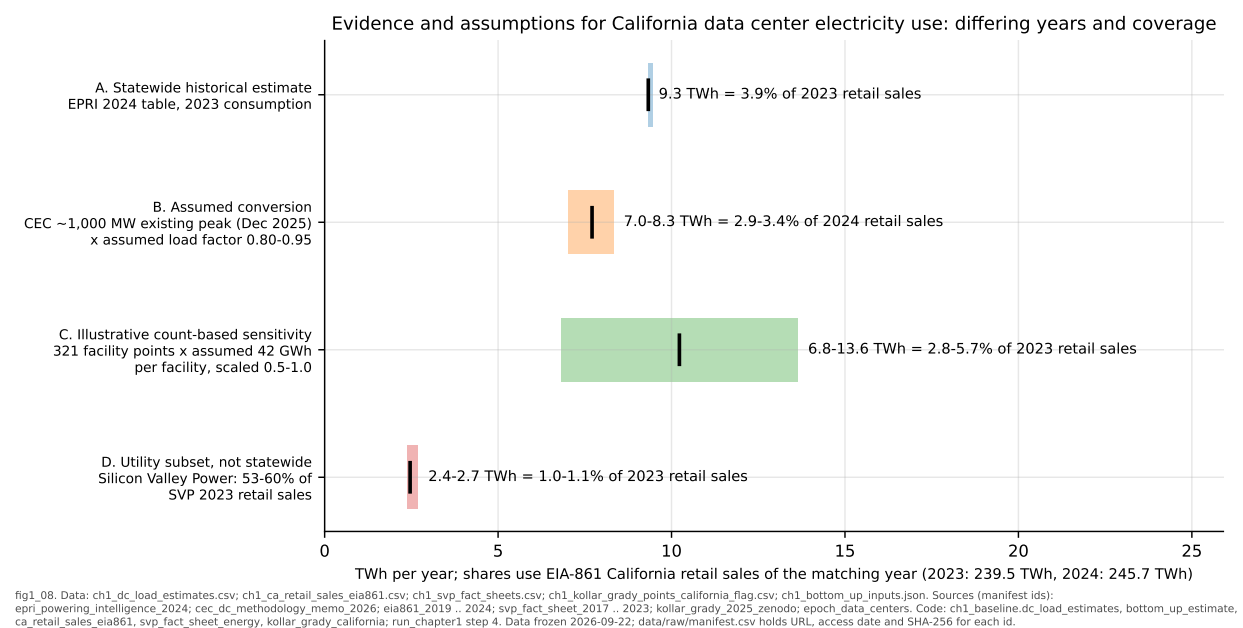


Figure 2.8: Existing data center electricity use in California: one statewide estimate (A), one assumed conversion (B), one illustrative sensitivity (C) and one utility subset (D), with matching-year retail-sales shares. No metered statewide total exists.

Source line: fig1_08. Data: ch1_dc_load_estimates.csv; ch1_ca_retail_sales_eia861.csv; ch1_svp_fact_sheets.csv; ch1_kollar_grady_points_california_flag.csv; ch1_bottom_up_inputs.json. Sources (manifest ids): epri_powering_intelligence_2024; cec_dc_methodology_memo_2026; eia861_2019 .. 2024; svp_fact_sheet_2017 .. 2023; kollar_grady_2025_zenodo; epoch_data_centers. Code: ch1_baseline.dc_load_estimates, bottom_up_estimate, ca_retail_sales_eia861, svp_fact_sheet_energy, kollar_grady_california; run_chapter1 step 4. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

- A. Statewide estimate.** EPRI’s 2024 study estimates California data centers consumed 9,331,619 MWh in 2023, 3.70 percent of state consumption on EPRI’s denominator, the third-largest state total after Virginia and Texas [Electric Power Research Institute, 2024, pp. 13, 28]; against EIA-861 2023 retail sales it is 3.9 percent. This is the only statewide historical estimate in the set.
- B. Assumed conversion.** The CEC’s methodology memo states that “as of December 2025, approximately 1,000 megawatts (MW) of existing data center peak demand operates in

California” [California Energy Commission, 2026d, pp. 2, 18]. Converting a peak into energy needs an annual load factor that no source reports. The CEC’s daily load-factor profiles for existing facilities range from 75 to 100 percent of the daily maximum [California Energy Commission, 2025], and an assumed annual factor of 0.80 to 0.95 turns the 1,000 MW into 7.0 to 8.3 TWh, 2.9 to 3.4 percent of 2024 retail sales. It is a conversion scenario, not a measured total.

3. **C. Illustrative sensitivity.** The Kollar and Grady inventory places 321 of its 2,484 facility points inside California, 102 of them within 12 km of Santa Clara [Kollar and Grady, 2025], but carries no per-facility power, and the Epoch AI hub snapshot contains no California observation [Epoch AI, 2026a], which is a limit of its coverage rather than a count of zero. The sensitivity multiplies the count by an assumed annual energy per facility taken from the Silicon Valley Power cluster: 42 GWh, or 55 percent of SVP’s 2023 retail sales spread over its 58 data centers, an average load of 4.8 MW, scaled by 0.5, 0.75 and 1.0 because the statewide inventory includes many small facilities. The result, 6.8 to 13.6 TWh or 2.8 to 5.7 percent, uses no peak-demand or utilization factor. It is an assumption set whose representativeness is untested, shown to bound the statewide figure rather than to estimate it.
4. **D. Utility subset.** The municipal utility of Santa Clara reports 58 data centers, the largest cluster on the West Coast, taking 55 percent of its power use [Silicon Valley Power, 2026a]; its web page states 53 percent [Silicon Valley Power, 2026b], and city officials cited about 60 percent in 2025 [San José Spotlight, 2025]. Applied to SVP’s 2023 retail sales of 4.48 TWh (Table 2.7) [Silicon Valley Power, 2024] (its purchased and generated supply was 4.59 TWh, which is what a peak of 669.2 MW times a 78.3 percent load factor reconstructs), the cluster used 2.4 to 2.7 TWh, about one percent of statewide 2023 sales; the percentages come from different dates than the sales figure. SVP also reports that its best-performing data center customers reach about 67 percent utilization of requested capacity and that observed utilization was about 64 percent in 2023 [Silicon Valley Power, 2025], the empirical origin of the CEC’s 67 percent factor; that ratio relates operating peak to requested capacity and is not an annual load factor.

The statewide estimate and the assumed conversion agree at 7 to 9 TWh, roughly 3 to 4 percent of retail sales, and the sensitivity admits 3 to 6 percent. One municipal utility hosts about a quarter to a third of that. The defensible statement for the rest of this study is that existing data centers used about 3 to 4 percent of California’s electricity in 2023 to 2025, that the load is geographically concentrated, and that no metered statewide total exists. Against roughly 1 GW of existing peak, the CEC’s December 2025 energization requests located in California total 20,677 MW, or 23,277 MW including the 2,600 MW that VEA reports for Nevada [California Energy Commission, 2026d], a ratio of about 21 that is the California analogue of Finland’s 27x [Manner, 2026] and the subject of Chapter 3.

2.5 Institutional context

The January 2026 forecast decision. The CEC adopted the California Energy Demand 2025–2045 forecast set on January 21, 2026 [California Energy Commission, 2026c]. The set includes three data center scenarios built from the energization tiers. What the joint agencies then chose to do with those scenarios is the institutional fact that matters. The Single Forecast Set Agreement between the CEC, the CPUC and the CAISO, signed in January 2026 and updated on April 15,

2026, directs that CPUC integrated resource planning and CAISO bulk-system transmission studies for the 2026–2027 cycle “will continue to use the 2024 IEPR planning forecast,” that the 2027–2028 cycle will use an IRP portfolio based on the same 2024 forecast, and that system resource adequacy requirements for 2027 “will use the 2025 IEPR planning forecast without known loads” [California Energy Commission, California Public Utilities Commission and California Independent System Operator, 2026, pp. 7–8]. Only local transmission reliability studies use the 2025 IEPR local reliability forecast with known loads. The agencies give two reasons: the approach “manages volatility and promotes stability in electric system resource and infrastructure planning currently underway,” and it “allows CEC staff additional time to collect and review historical data, observe the impacts of known loads, data centers, and recent federal policy changes, and refine forecast approaches as needed” [California Energy Commission, California Public Utilities Commission and California Independent System Operator, 2026, p. 8]. In plain terms, the state now tracks data center energization requests in tiers, converts them into forecast scenarios, and then holds the resulting increment out of the forecast that drives generation procurement and bulk transmission for at least two planning cycles. The 2024 IEPR data center mid case, built from the smaller December 2024 tiers, remains the number in use.

Utility pipeline statements. PG&E told the CEC in February 2025 that it had “a large pipeline of applications to interconnect new data centers” with about 1.5 GW in final engineering or construction and more than half expected within four years [Nelli, 2025]. In July 2025 it announced a 10 GW pipeline over ten years, of which 17 projects totaling about 1.5 GW were in final engineering for operation between 2026 and 2030, and estimated that each gigawatt of new data center demand could lower other customers’ bills by one to two percent [Pacific Gas and Electric Company, 2025]. Its second-quarter 2026 earnings release describes “an overall pipeline of over 12 gigawatts” [PG&E Corporation, 2026]. PG&E has, however, filed a request for confidentiality over its project-level data center data in the IEPR docket every month since August 2025 [Pacific Gas and Electric Company, 2026a], so its pipeline is public only in aggregate. Southern California Edison took the opposite path. It reported 80 MW of existing data center demand in February 2025 with 200 MW of near-term and 400 MW of mid-term growth [Dean, 2025], and it files a public project-level database. The January 2026 vintage lists 115 entries with 8,299 MW of requested peak: 76 MW under signed agreements, 3,313 MW in applications, 1,772 MW in inquiries, and 3,137 MW already canceled [Southern California Edison, 2026]. The CEC’s own summer 2025 tally put SCE’s requests at 5,828 MW and PG&E’s at 11,668 MW [California Energy Commission, 2025]. The canceled share in SCE’s file, 38 percent of everything ever requested, is the first direct California evidence on attrition and is used in Chapter 5.

CPUC large-load proceedings. The proposal could not confirm a CPUC large-load tariff docket; the record now can. On April 9, 2026 the CPUC opened Rulemaking 26-04-009 on Advanced Electric Rate Design, which states that “in coordination with the Senate Bill 57 assessment, this proceeding will consider rate design issues for data centers and other large load customers” [California Public Utilities Commission, 2026, p. 1]. Senate Bill 57 (Padilla), chaptered on October 11, 2025, requires the CPUC to assess whether costs of serving new data center loads shift to other customers and to report to the Legislature by January 1, 2027 [California Legislature, 2025b]. Senate Bill 886 (Padilla), the California Technology Innovation and Ratepayer Protection Act, was chaptered on September 21, 2026 and requires the CPUC, on or before January 1, 2028, to establish new tariffs or update electric rules for the interconnection and service of participating large customers, and to ensure those tariffs “prevent the creation of stranded costs for, or cost

shifts to, nonparticipating customers” [California Legislature, 2026, KQED, 2026]. Assembly Bill 222, which would have required data center energy reporting and efficiency metrics, was held in the Senate Appropriations Committee on August 29, 2025 [California Legislature, 2025a]. On the interconnection side, the CPUC approved an interim PG&E Electric Rule 30 on July 24, 2025 to streamline transmission-level service for large customers that fund the required infrastructure up front [California Public Utilities Commission, 2025]. The sequence is consistent: the state has moved from studying the cost question (SB 57) to mandating a tariff answer (SB 886) within a year, while the forecast that drives procurement still excludes the load the tariff is meant to price.

Chapter 3

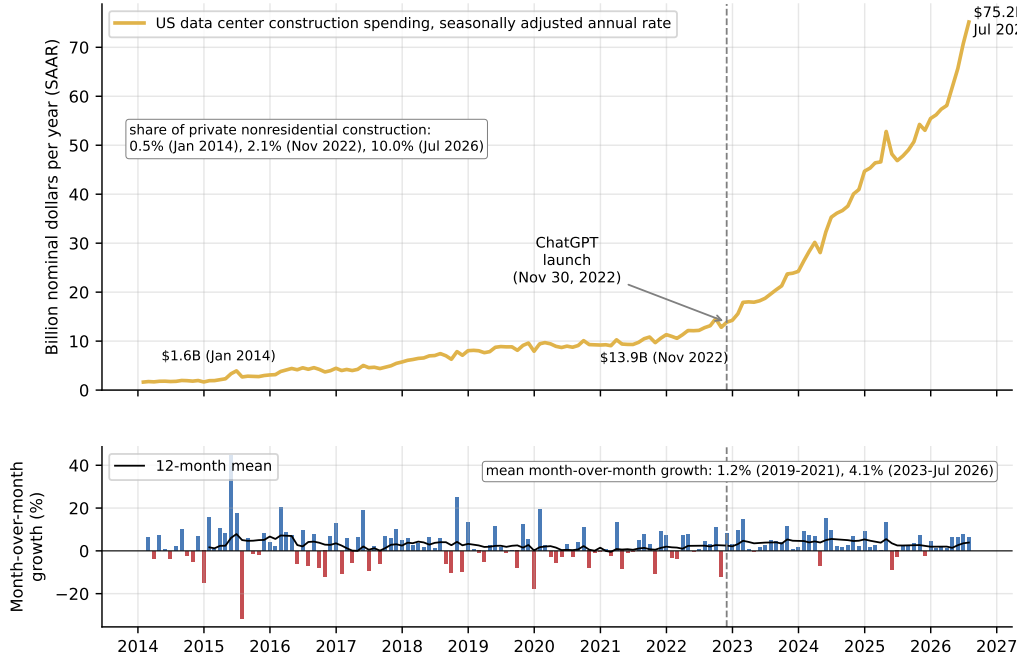
Data Center Growth in California

This chapter answers RQ1: how California’s planned data center capacity varies by stage of development, in absolute terms and relative to the scale of the existing grid. It first reproduces the national construction-spending series that frames public discussion, tests it for structural breaks, and then asks whether California’s own indicators show the same acceleration. It ends with every vintage of the CEC’s energization tiers and the RQ1 table. All code is in `src/ch2_growth.py` and `notebooks/02_data_center_growth.ipynb`.

3.1 The national construction boom, reproduced

The Census Bureau’s Value of Construction Put in Place survey has carried a “data center” line under private nonresidential construction since January 2014 [U.S. Census Bureau, 2026]. Figure 3.1 reproduces the widely circulated chart from the raw seasonally adjusted workbook and adds month-over-month growth. Spending at a seasonally adjusted annual rate rose from \$1.6 billion in January 2014 to \$13.9 billion in November 2022 and \$75.2 billion in July 2026 (preliminary). Data centers were half a percent of private nonresidential construction in 2014, two percent when ChatGPT launched, and ten percent in July 2026. Month-over-month growth averaged 4.1 percent from 2023 through mid-2026 against 1.2 percent in 2019 through 2021. No comparable state-level series exists: the Census publishes this category for the nation only, and the California picture must be built from proxies (Section 3.3).

US private construction spending, data center category (nominal dollars, seasonally adjusted annual rate), Jan 2014 to Jul 2026 (Census C30)



Source: U.S. Census Bureau, Value of Construction Put in Place (privsatime.xlsx), accessed 2026-09-21; p = preliminary

fig2_01. Data: ch2_census_c30_data_center.csv. Sources (manifest ids): census_c30_privsatime; census_c30_privtime. Code: ch2_growth.census_c30; run_chapter2 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 3.1: US private data center construction spending, seasonally adjusted annual rate, with month-over-month growth. Source: Census C30, July 2026 preliminary.

Source line: fig2_01. Data: ch2_census_c30_data_center.csv. Sources (manifest ids): census_c30_privsatime; census_c30_privtime. Code: ch2_growth.census_c30; run_chapter2 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

3.2 Structural breaks and forecasts

The analysis works on the log of the seasonally adjusted series, so that a linear trend is a constant compound growth rate. Three questions are asked in turn: does a break at the known date change the trend (Chow test, Chow, 1960); where does the data itself place breaks when none is imposed (Bai-Perron, Bai and Perron, 1998, 2003); and what does a time-series model project. Growth rates carry Newey-West standard errors. The Bai-Perron search uses 15 percent trimming (a minimum segment of 23 months) and allows up to five breaks, the most the trimming permits, so that the choice of m is never censored by the search. The number of breaks is chosen three ways: by BIC, by the LWZ criterion, and by the sequential $\text{supF}(l+1|l)$ procedure, in which each additional break is tested against the l -break model with a moving-block residual bootstrap p -value. $\text{supF}(0|1)$ gets the same bootstrap under the no-break null, and the single-break date a residual-bootstrap 90 percent interval. The `ruptures` dynamic-programming search [Truong et al., 2020] is run as an independent check.

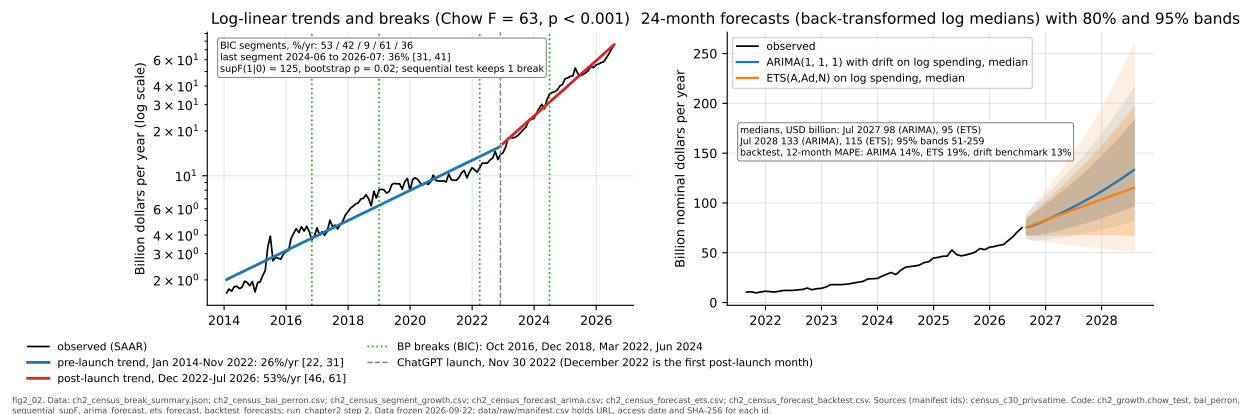


Figure 3.2: Left: log-linear trends before and after December 2022 with Bai-Perron break dates. Right: 24-month ARIMA and ETS forecasts with 80 and 95 percent prediction bands.

Source line: fig2_02. Data: ch2_census_break_summary.json; ch2_census_bai_perron.csv; ch2_census_segment_growth.csv; ch2_census_forecast_arima.csv; ch2_census_forecast_ets.csv; ch2_census_forecast_backtest.csv. Sources (manifest ids): census_c30_privsatime. Code: ch2_growth.chow_test, bai_perron, sequential_supF, arima_forecast, ets_forecast, backtest_forecasts; run_chapter2 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 3.1: Bai-Perron global search on the log Census series: break dates, sum of squared residuals, information criteria, $\text{supF}(0|m)$, and the sequential $\text{supF}(m|m-1)$ with its bootstrap p -value, for $m = 0$ to 5 breaks.

Breaks m	Break dates (first month of new regime)	SSR	BIC	LWZ	$\text{supF}(0 m)$	$\text{supF}(m m-1)$	boot. p
0	—	5.967	-3.165	-3.101	—	—	—
1	2020-11	2.205	-4.061	-3.900	125.4	125.4	0.020
2	2018-12, 2022-03	1.375	-4.433	-4.176	121.1	36.7	0.085
3	2016-10, 2018-12, 2022-03	1.091	-4.565	-4.210	106.5	40.6	0.020
4	2016-10, 2018-12, 2022-03, 2024-06	0.880	-4.680	-4.228	101.9	40.6	0.005
5	2016-02, 2018-01, 2020-04, 2022-05, 2024-06	0.832	-4.636	-4.086	85.7	—	—

Table 3.2: Compound growth by segment of the BIC-selected partition (Newey-West 95 percent intervals).

Segment (BIC partition)	Months	Compound growth (%/yr)	95% interval
2014-01 to 2016-09	33	53	[45, 61]
2016-10 to 2018-11	26	42	[37, 47]
2018-12 to 2022-02	39	9	[7, 10]
2022-03 to 2024-05	27	61	[56, 67]
2024-06 to 2026-07	26	36	[31, 41]

The known-date test is unambiguous. A break in intercept and slope at December 2022, the first full month after the November 30 launch, gives a Chow F of 63 ($p < 0.001$) and the HAC-robust Wald test agrees ($p < 0.001$). Compound growth was 26 percent a year (95 percent interval 22 to 31) through November 2022 and 53 percent a year (46 to 61) from December 2022. Neither

the date annotation nor the test establishes that the launch caused the acceleration; they establish that the trend changed then. Letting the data locate the breaks changes the story in two respects (Tables 3.1 and 3.2). BIC and LWZ both choose four breaks, in October 2016, December 2018, March 2022 and June 2024. The current expansion therefore began eight months before ChatGPT, in the hyperscaler capital-spending ramp of early 2022, and it has already slowed once: the segment from March 2022 to May 2024 grew at 61 percent a year (56 to 67) and the segment from June 2024 to July 2026 at 36 percent a year (31 to 41), still four times the 9 percent of 2019 to early 2022. The sequential procedure is more conservative. $\text{supF}(1|0)$ of 125 rejects the no-break model ($p = 0.02$), but $\text{supF}(2|1)$ of 37 has a bootstrap p -value of 0.085, so at the 5 percent level the procedure stops at one break, in November 2020, even though the third and fourth breaks are each significant once the earlier ones are in place ($p = 0.02$ and $p = 0.005$). `ruptures` reproduces all four BIC dates within one month. The forecasts in Table 3.3 continue the recent trend: an ARIMA(1,1,1) with drift and a damped-trend ETS, both fitted to log spending, put July 2027 at \$95 to \$98 billion and July 2028 at \$115 to \$133 billion, with 95 percent bands from \$51 to \$259 billion by then. The point forecasts are back-transformed log medians, not arithmetic means, and the series is national nominal dollars, so the forecasts say nothing about California megawatts. A rolling-origin backtest over the last 24 months (Table 3.4) shows why they are exploratory: at a twelve-month horizon the mean absolute percentage error is 14 percent for the ARIMA and 19 percent for the ETS against 13 percent for a log random walk with drift, every model ran below the outcomes on average, and the 95 percent bands covered 92 percent of outcomes for the ARIMA and 79 percent for the ETS. The bands, not the point forecasts, are the finding: a series whose growth rate has shifted four times in twelve years, and which a drift benchmark forecasts as well as either model, is not something a planner should extrapolate.

Table 3.3: Forecasts of US data center construction spending, 6, 12 and 24 months ahead: back-transformed log medians with 80 and 95 percent bands, nominal dollars.

Month	ARIMA(1, 1, 1) with drift			ETS(A,Ad,N)		
	Mean	80% band	95% band	Mean	80% band	95% band
Jan 2027	84.5	[70.7, 101.0]	[64.3, 111.0]	84.4	[68.0, 104.7]	[60.6, 117.4]
Jul 2027	98.3	[78.0, 124.0]	[68.9, 140.2]	94.8	[68.4, 131.3]	[57.6, 156.0]
Jul 2028	133.2	[97.4, 182.1]	[82.5, 214.9]	115.1	[67.8, 195.6]	[51.2, 258.9]

Billion dollars per year, seasonally adjusted annual rate.

Table 3.4: Rolling-origin backtest over the last 24 months: mean absolute percentage error on the level scale, root-mean-square log error, mean log error (bias) and band coverage, by horizon, for the two models and two benchmarks.

Model	MAPE (%)	1 month ahead				6 months ahead				12 months ahead			
		RMSE (log)	bias (log)	cover. 80/95		MAPE (%)	RMSE (log)	bias (log)	cover. 80/95	MAPE (%)	RMSE (log)	bias (log)	cover. 80/95
ARIMA (log, drift)	4.6	0.06	-0.02	96 / 100		11.6	0.14	-0.08	75 / 100	14.0	0.20	-0.11	67 / 92
ETS(A,Ad,N) (log)	4.9	0.06	-0.03	96 / 100		13.9	0.18	-0.11	75 / 92	18.8	0.30	-0.20	67 / 79
drift benchmark (log random walk)	4.0	0.06	-0.01	- / -		10.2	0.12	-0.06	- / -	13.0	0.18	-0.10	- / -
naive benchmark (last value)	5.1	0.07	-0.04	- / -		18.2	0.23	-0.21	- / -	31.2	0.41	-0.39	- / -

3.3 California proxies

Three California indicators are subjected to the same battery, with the known break set to the first period after November 2022 (Figure 3.3, Table 3.5).

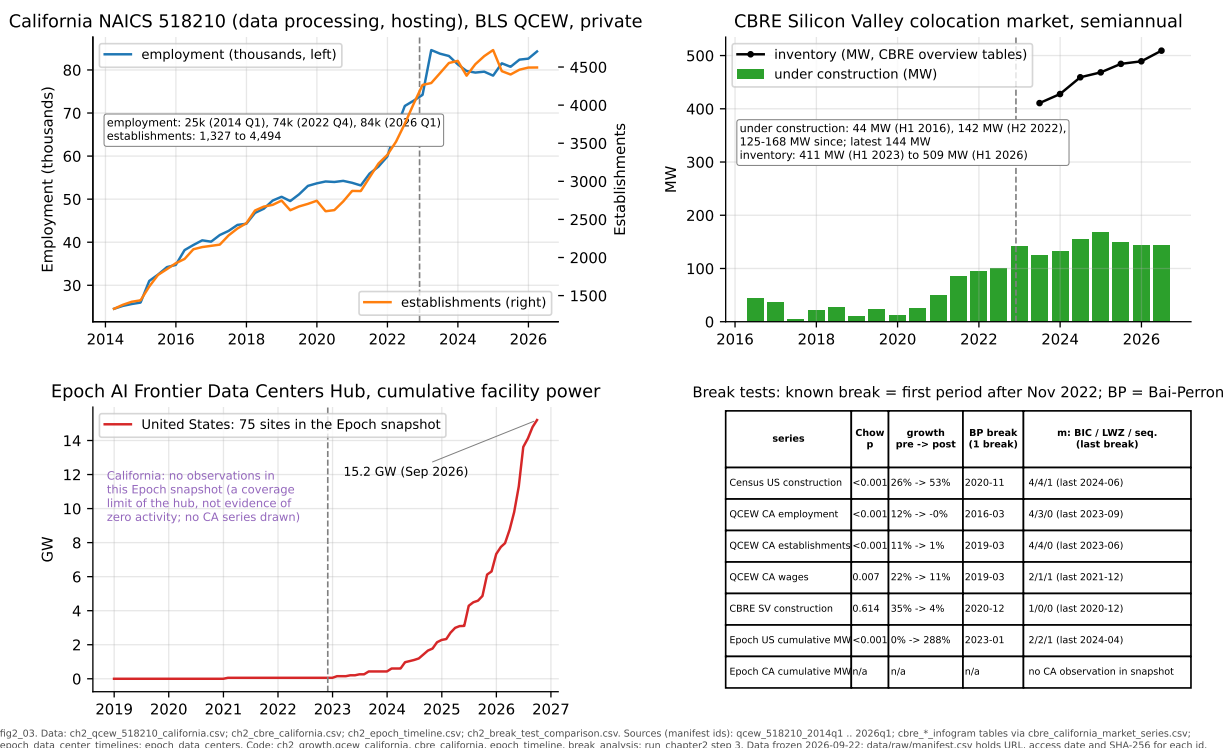


Figure 3.3: California proxies for data center growth: BLS QCEW NAICS 518210 labour indicators (top left), CBRE Silicon Valley construction and inventory (top right), Epoch AI frontier sites, whose snapshot has no California observation (bottom left), and the break-test summary (bottom right); the single-break column is the one-break model's estimate and p-values below 0.001 are shown as such.

Source line: fig2_03. Data: ch2_qcew_518210_california.csv; ch2_cbre_california.csv; ch2_epoch_timeline.csv; ch2_break_test_comparison.csv. Sources (manifest ids): qcew_518210_2014q1 .. 2026q1; cbre_*_infogram tables via cbre_california_market_series.csv; epoch_data_center_timelines; epoch_data_centers. Code: ch2_growth.qcew_california, cbre_california, epoch_timeline, break_analysis; run_chapter2 step 3. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 3.5: Break tests on the Census series and the California proxies. Chow: known break at the first period after November 2022. BP: Bai-Perron single break with residual-bootstrap 90 percent interval, the number of breaks chosen by BIC, LWZ and the sequential procedure, the BIC-selected dates, and the bootstrap p -value of $\sup F(0|1)$.

Series	n	Chow F	p	HAC p	Growth before (%/yr)	Growth after (%/yr)	BP break (1)	90% interval	m: BIC/LWZ/seq.	BP breaks (BIC)	boot. p
Census data center construction (monthly, SAAR)	151	63.0	0.000	0.000	26 [22, 31]	53 [46, 61]	2020-11	2020-11 to 2020-11	4/4/1	2016-10, 2018-12, 2022-03, 2024-06	0.010
QCEW 518210 California employment (quarterly)	49	15.0	0.000	0.000	12 [9, 14]	-0 [-2, 2]	2016-03	2016-03 to 2021-12	4/3/0	2016-03, 2018-09, 2021-03, 2023-09	0.035
QCEW 518210 California establishments (quarterly)	49	11.0	0.000	0.000	11 [9, 13]	1 [-1, 2]	2019-03	2016-12 to 2022-09	4/4/0	2016-03, 2018-03, 2020-12, 2023-06	0.285
QCEW 518210 California wages (quarterly)	49	5.5	0.007	0.000	22 [19, 24]	11 [9, 13]	2019-03	2016-12 to 2019-06	2/1/1	2019-03, 2021-12	0.005
CBRE Silicon Valley MW under construction (semiannual)	21	0.5	0.614	0.000	35 [10, 66]	4 [-2, 10]	2020-12	2020-12 to 2020-12	1/0/0	2020-12	0.140
Epoch frontier sites, United States cumulative MW (monthly)	69	176.1	0.000	0.000	0 [-0, 0]	288 [250, 331]	2023-01	2023-01 to 2023-01	2/2/1	2023-01, 2024-04	0.005
Epoch frontier sites, California cumulative MW	0	no California observation in the Epoch snapshot (0 of 87 sites): a coverage limit of the h									

Employment and establishments. The Quarterly Census of Employment and Wages counts private establishments and jobs in NAICS 518210, computing infrastructure, data processing, hosting and related services, by quarter [U.S. Bureau of Labor Statistics, 2026]. These are labour

indicators of an industry, not counts of buildings or megawatts, and the industry includes hosting and processing firms that do not operate data centers. California employment in the industry rose from 25,000 in 2014 Q1 to 74,000 by the end of 2022, then flattened: 84,000 in 2026 Q1 after a plateau through 2024 and 2025. The Chow test at 2023 Q1 rejects a common trend ($F = 15$, $p < 0.001$), but the break is a deceleration, from 12 percent a year to roughly zero. Establishments show the same shape, 11 percent a year to 0.5 percent ($F = 11$). Data-located breaks in these smooth quarterly series are diffuse: BIC places four in each, the last in 2023 Q3 for employment and 2023 Q2 for establishments, but the single-break bootstrap is significant only for employment ($p = 0.035$ against 0.285 for establishments) and the sequential procedure keeps no break in either. Wages decelerated from 22 to 11 percent a year ($p = 0.007$) with breaks in 2019 Q1 and 2021 Q4 that reflect the industry’s earlier expansion. The county detail is in the notebook; Santa Clara, Los Angeles and San Francisco account for most of the state total.

MW under construction. CBRE’s semiannual market chapters give Silicon Valley colocation capacity under construction from H1 2016 and inventory from H1 2023 [CBRE Research, 2026]. Construction rose from 44 MW in H1 2016 to 142 MW by H2 2022 and has since ranged between 125 and 168 MW, with inventory creeping from 411 to 509 MW. Megawatts under construction are a stock whose change reflects completions as well as starts, and the series covers one colocation market, not California construction spending. The Chow test finds no break at H1 2023 ($p = 0.61$). BIC keeps a single break in H2 2020, when the pipeline stepped up from tens of megawatts to over a hundred, and growth has been 4 percent a year since; LWZ and the sequential procedure keep no break at all (bootstrap $p = 0.14$), which is what a plateau looks like in 21 semiannual points. Cushman & Wakefield’s tables are form-gated and JLL’s report gives a single-period inventory, so the digitized series rests on CBRE; Southern California is available only from H1 2024 and is reported in the notebook.

Frontier facilities. Epoch AI’s hub tracks 89 large AI data centers from satellite imagery and permits (snapshot of September 22, 2026), 75 of them in the United States, with per-site power timelines [Epoch AI, 2026b]. Cumulative US power in these sites went from under 0.5 GW at the end of 2023 to 15.2 GW by September 2026, with a break in January 2023; the curve is a reconstruction from sampled sites, and its breaks can reflect the hub’s coverage and retrospective estimates as much as construction. None of the 89 sites in the snapshot is in California. That is a limit of the hub’s coverage, which Epoch itself describes as incomplete and uneven, not evidence that no frontier facility exists or is being built in California; no California series is drawn and no test is possible.

The three proxies agree on a finding that the national series hides: after 2022 California’s data center industry did not accelerate in jobs, establishments or built Silicon Valley colocation capacity, and the largest AI campuses in Epoch’s sample were sited elsewhere. The growth that California records is in requests to connect, which is the subject of the next section, and the divergence between requests and built capacity is exactly what the tiers are meant to measure.

3.4 Every vintage of the energization tiers

Table 3.6 and Figure 3.4 assemble every published vintage of the CEC’s data center energization requests, followed by the two utility-level series that carry the record past December 2025. The December 2024 data behind the 2024 IEPR Update are known only as PG&E’s and SCE’s agreements

plus applications, 5,808 and 963 MW, because inquiries were not published [California Energy Commission, 2025]. The summer 2025 data behind the preliminary 2025 forecast are published as totals by utility, 21,756 MW across seven utilities; for PG&E and SCE the deck also gives agreements plus applications, 10,080 and 143 MW, so their inquiries were 1,588 and 5,685 MW [California Energy Commission, 2025]. The copy of that deck filed for the November 12, 2025 workshop labelled SCE at 2,492 MW and SVP at 1,382 MW; the published deck corrects these to 143 and 1,375 MW, and the corrected figures are used here. The December 2025 data behind the adopted forecast are the first complete vintage: 5,086 MW signed, 9,587 MW in application and 8,604 MW in inquiry, 23,277 MW in all, with PG&E accounting for 14,747 MW and SCE for 4,624 MW by the memo’s tier rows [California Energy Commission, 2026d,b]. The memo’s footnote states that VEA’s 2,600 MW of applications are for locations in Nevada, inside the CAISO footprint but outside California, and the CEC excludes them from statewide consumption; the pipeline located in California is therefore 20,677 MW (5,086 signed, 6,987 in application, 8,604 in inquiry), and the summer 2025 total contains the same 2,600 MW, leaving 19,156 MW in California. The published marginal totals and the utility rows differ by 1 MW (23,277 against 23,278); both are kept and the marginal totals are used. The August 2026 workshop restated the December 2025 figures and showed June 2026 known-load data only as charts [Smith, 2026]. SCE’s public project database provides the only vintages with cancellations: between its August 2025 and January 2026 versions, inquiries fell from 5,934 to 1,772 MW as about 3,100 MW moved into the application stage, canceled requests rose from 709 to 3,137 MW, and the active total fell from 6,201 to 5,162 MW [Southern California Edison, 2026]. The growth in SCE’s total records between the two snapshots is cancellations, not pipeline; thirty-eight percent of everything SCE had ever logged was canceled within five months, the first direct California evidence on attrition. PG&E’s pipeline table from its second-quarter 2026 earnings, reproduced in its August 2026 presentation to the CEC, uses PG&E’s own stages and excludes inquiries: 5,090 MW in March 2026 and 12,710 MW in June 2026, of which 3,880 MW had a signed work performance agreement, 490 MW an interconnection construction agreement and 140 MW was in construction [Pacific Gas and Electric Company, 2026b]. The jump of almost 8,000 MW in one quarter came from the 2026 cluster study. The Public Advocates Office observes that PG&E had filed agreements covering only about 650 MW with the CPUC for approval [Public Advocates Office, California Public Utilities Commission, 2026], so even the most committed tier has layers of certainty inside it. The three source families of Figure 3.4, the CEC’s statewide vintages, SCE’s database and PG&E’s earnings pipeline, differ in utility coverage, stage definitions and dates, so their bar heights are not one growth series, and signed agreements are commitments to connect, not operating capacity.

Table 3.6: Vintages of the CEC energization tiers found in the record (MW of requested capacity). SCE rows are that utility’s public project database. PG&E rows use PG&E’s own pipeline stages from its Q2 2026 earnings, where WPA+ means a signed work performance agreement or a later stage, and exclude inquiries.

Vintage	Signed	Applications	Inquiries	Agr.+appl. only	Unsplit total	Total active (MW)
Dec 2024 (2024 IEPH Update)	0	0	0	6,771	0	6,771
Summer 2025 (2025 IEPH preliminary)	0	0	7,273	10,223	4,260	21,756
Dec 2025 (2025 IEPH final)	5,086	9,587	8,605	0	0	23,278
Aug 2026 workshop (restates Dec 2025)	0	0	0	0	23,278	23,278
SCE database Aug 29 2025 (SCE only)	51	216	5,934	–	canceled 709	6,201
SCE database Jan 29 2026 (SCE only)	76	3,313	1,772	–	canceled 3,137	5,162
PG&E earnings pipeline Jun 2026 (PG&E only, PG&E stages)	WPA+ 4,510	8,200	n/a	12,710	–	12,710
PG&E earnings pipeline Mar 2026 (PG&E only, PG&E stages)	WPA+ 3,390	1,700	n/a	5,090	–	5,090

Totals are as reported by the utilities and include VEA’s 2,600 MW (summer 2025) and 2,600 MW (December 2025) of requests located in Nevada; California-only totals are 19,156 and 20,678 MW by utility rows (the CEC’s marginal totals are 1 MW lower: 23,277 and 20,677 MW). SCE active totals exclude canceled requests.

Table 3.7: Every utility-by-tier figure behind the December 2024 and summer 2025 vintages in Table 3.6, with its source: the transcription record for the megawatt figures quoted in this section.

Vintage	Source vintage	Utility	Tier or stage	MW	Source (manifest id, page)	Note
2024-12	Dec 2024 (2024 IEPH Update)	PG&E	Agreements + applications (no inquiries)	5,808	cec_prelim_dc_forecast_2025	p.7
2024-12	Dec 2024 (2024 IEPH Update)	SCE	Agreements + applications (no inquiries)	963	cec_prelim_dc_forecast_2025	p.7
2024-12	Dec 2024 (2024 IEPH Update)	all	total (all rows)	6,771	sum of the rows above	
2025-08	Summer 2025 (2025 IEPH preliminary)	Burbank	All tiers	100	cec_prelim_dc_forecast_2025	p.6
2025-08	Summer 2025 (2025 IEPH preliminary)	PG&E	Agreements + applications (no inquiries)	10,080	cec_prelim_dc_forecast_2025	p.7
2025-08	Summer 2025 (2025 IEPH preliminary)	PG&E	Inquiry	1,588	cec_prelim_dc_forecast_2025	p.6 minus p.7; residual: utility total minus agreements + applications
2025-08	Summer 2025 (2025 IEPH preliminary)	Palo Alto	All tiers	85	cec_prelim_dc_forecast_2025	p.6
2025-08	Summer 2025 (2025 IEPH preliminary)	SCE	Agreements + applications (no inquiries)	143	cec_prelim_dc_forecast_2025	p.7; workshop copy TN267165 p.8 labelled SCE 2025 as 2,492 MW; published deck sa
2025-08	Summer 2025 (2025 IEPH preliminary)	SCE	Inquiry	5,685	cec_prelim_dc_forecast_2025	p.6 minus p.7; residual: utility total minus agreements + applications
2025-08	Summer 2025 (2025 IEPH preliminary)	SDG&E	All tiers	100	cec_prelim_dc_forecast_2025	p.6
2025-08	Summer 2025 (2025 IEPH preliminary)	SVP	All tiers	1,375	cec_prelim_dc_forecast_2025	p.6; workshop copy TN267165 p.7 labelled SVP as 1,382 MW; published deck says 1,
2025-08	Summer 2025 (2025 IEPH preliminary)	VEA	All tiers	2,600	cec_prelim_dc_forecast_2025	p.6
2025-08	Summer 2025 (2025 IEPH preliminary)	PG&E	utility total	11,668	sum of the rows above	
2025-08	Summer 2025 (2025 IEPH preliminary)	SCE	utility total	5,828	sum of the rows above	
2025-08	Summer 2025 (2025 IEPH preliminary)	all	total (all rows)	21,756	sum of the rows above	

Table 3.8: The December 2025 vintage by utility and tier (memo Table 1), with utility totals.

Vintage	Source vintage	Utility	Tier or stage	MW	Source (manifest id, page)	Note
2025-12	Dec 2025 (2025 IEPR final)	Burbank	Active application	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	Burbank	Inquiry	100	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	Burbank	Signed agreement	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	PG&E	Active application	3,617	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	PG&E	Inquiry	6,774	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	PG&E	Signed agreement	4,356	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	Palo Alto	Active application	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	Palo Alto	Inquiry	55	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	Palo Alto	Signed agreement	14	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SCE	Active application	3,174	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SCE	Inquiry	1,378	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SCE	Signed agreement	72	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SDG&E	Active application	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SDG&E	Inquiry	100	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SDG&E	Signed agreement	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SVP	Active application	196	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SVP	Inquiry	198	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	SVP	Signed agreement	644	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	VEA	Active application	2,600	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	VEA	Inquiry	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	VEA	Signed agreement	0	cec_dc_methodology_memo_2026	Table 1 p.5
2025-12	Dec 2025 (2025 IEPR final)	Burbank	utility total	100	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	PG&E	utility total	14,747	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	Palo Alto	utility total	69	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	SCE	utility total	4,624	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	SDG&E	utility total	100	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	SVP	utility total	1,038	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	VEA	utility total	2,600	sum of the rows above	
2025-12	Dec 2025 (2025 IEPR final)	all	total (all rows)	23,278	sum of the rows above	
2026-08	Aug 2026 workshop (re-states Dec 2025)	all	Total restated	23,278	cec_tn272026	p.5
2026-08	Aug 2026 workshop (re-states Dec 2025)	all	total (all rows)	23,278	sum of the rows above	

Table 3.9: The utility-level series behind Table 3.6: SCE's public project database, PG&E's earnings pipeline, and the Cal Advocates memo figure.

Vintage	Source vintage	Utility	Tier or stage	MW	Source (manifest id, page)	Note
2025-08	SCE database Aug 29 2025	SCE	Active application	216	cec_tn266008	
2025-08	SCE database Aug 29 2025	SCE	Canceled	709	cec_tn266008	
2025-08	SCE database Aug 29 2025	SCE	Inquiry	5,934	cec_tn266008	
2025-08	SCE database Aug 29 2025	SCE	Signed agreement	51	cec_tn266008	
2025-08	SCE database Aug 29 2025	all	total (active)	6,201	sum of the rows above	excludes canceled
2025-08	SCE database Aug 29 2025	all	total, all records including canceled	6,910	sum of the rows above	
2026-01	SCE database Jan 29 2026	SCE	Active application	3,313	cec_tn268459	
2026-01	SCE database Jan 29 2026	SCE	Canceled	3,137	cec_tn268459	
2026-01	SCE database Jan 29 2026	SCE	Inquiry	1,772	cec_tn268459	
2026-01	SCE database Jan 29 2026	SCE	Signed agreement	76	cec_tn268459	
2026-01	SCE database Jan 29 2026	all	total (active)	5,162	sum of the rows above	excludes canceled
2026-01	SCE database Jan 29 2026	all	total, all records including canceled	8,299	sum of the rows above	
2026-03	PG&E earnings pipeline Mar 2026	PG&E	PG&E: application + preliminary engineering	1,700	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-03	PG&E earnings pipeline Mar 2026	PG&E	PG&E: construction	140	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-03	PG&E earnings pipeline Mar 2026	PG&E	PG&E: final engineering (WPA signed)	3,110	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-03	PG&E earnings pipeline Mar 2026	PG&E	PG&E: interconnection construction agreement	140	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-03	PG&E earnings pipeline Mar 2026	all	total (all rows)	5,090	sum of the rows above	
2026-06	PG&E earnings pipeline Jun 2026	PG&E	PG&E: application + preliminary engineering	8,200	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-06	PG&E earnings pipeline Jun 2026	PG&E	PG&E: construction	140	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-06	PG&E earnings pipeline Jun 2026	PG&E	PG&E: final engineering (WPA signed)	3,880	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-06	PG&E earnings pipeline Jun 2026	PG&E	PG&E: interconnection construction agreement	490	cec_tn272065	p.9 (PG&E Q2 2026 earnin; PG&E stage definitions (PES fee paid and later); no in
2026-06	PG&E earnings pipeline Jun 2026	all	total (all rows)	12,710	sum of the rows above	
2026-09	Cal Advocates comments on the Aug 2026 workshop	PG&E	memo: WPAs filed at the CPUC	650	cec_tn272807 p.3-4	about 650 MW filed; 3,880 MW stated as signed

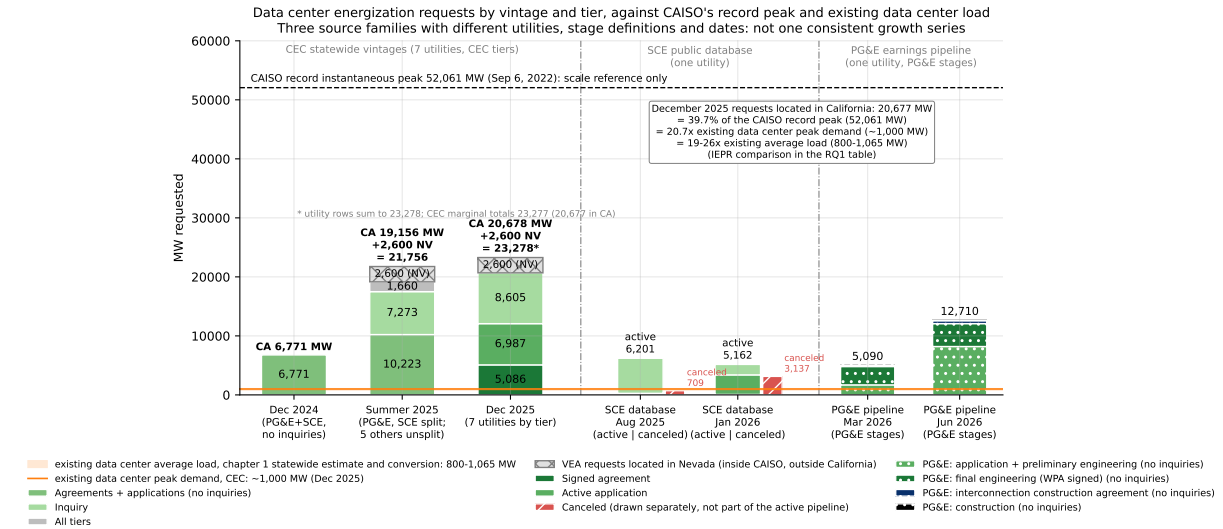


Figure 3.4: Energization requests by vintage and tier against CAISO’s record peak and the existing data center average load from Chapter 2. VEA’s requests in Nevada are hatched and California-only totals are labelled; SCE’s canceled requests are drawn beside, not on top of, its active pipeline; the three source families are divided because they differ in utilities, stages and dates.

Source line: fig2_04. Data: ch2_tier_vintages.csv; ch2_rq1_denominators.json; ch1_dc_load_estimates.csv. Sources (manifest ids): cec_dc_forecast_2024iepr; cec_prelim_dc_forecast_2025; cec_dc_methodology_memo_2026; cec_assembly_hearing_2026_01_28; cec_tn266008; cec_tn268459; cec_tn272026; cec_tn272065; cec_tn272807; caiso_key_statistics_2026_08. Code: ch2_growth.tier_vintages; run_chapter2 steps 4 and 5. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

The sums and ratios are stated explicitly. The three December 2025 tiers total 23,277 MW as reported and 20,677 MW located in California. Against CAISO’s record instantaneous peak of 52,061 MW on September 6, 2022 [California Independent System Operator, 2026b], the California pipeline is 40 percent (40 percent against the hourly EIA-930 peak of 51,104 MW, and 45 percent for the reported total including Nevada). This is a scale comparison, not a coincident-peak ratio and not a measure of usable headroom. Against the roughly 1,000 MW of existing data center peak demand that the CEC reports for December 2025, it is 21 times; against the 800 to 1,065 MW of existing average load implied by the chapter 1 statewide estimate and assumed conversion, 19 to 26 times. Finland’s ratio of registered to installed capacity was 27 [Manner, 2026]; California’s is of the same order.

3.5 RQ1: planned capacity by stage

Table 3.10: RQ1: California’s planned data center capacity by stage (December 2025, requests located in California) relative to the grid, to existing data center load, and to the adopted forecast’s peak and energy growth to 2030; VEA’s Nevada requests and the CEC’s reported total are memo rows.

Stage (Dec 2025)	MW	% of CAISO record peak	% of 2025 peak	× existing DC peak	× existing DC avg. load	% of IEPR peak growth	% of IEPR growth at 67% util.	Energy equiv. (TWh/yr)	% of IEPR energy growth
Signed agreements	5,086	9.8	11.6	5.1x	4.8-6.4x	127	85	26	75
Active applications, California	6,987	13.4	15.9	7.0x	6.6-8.7x	174	116	36	103
Inquiries	8,604	16.5	19.6	8.6x	8.1-10.8x	214	143	44	127
Total, three tiers, California only	20,677	39.7	47.1	20.7x	19.4-25.8x	514	345	107	305
Memo: VEA applications located in Nevada	2,600	5.0	5.9	2.6x	2.4-3.2x	65	43	13	38
Memo: CEC statewide total including VEA	23,277	44.7	53.1	23.3x	21.9-29.1x	579	388	120	344

Denominators: CAISO record peak 52,061 MW (Sept. 6, 2022) and hourly EIA-930 peak 51,104 MW; 2025 hourly peak 43,860 MW; existing data center peak $\approx$ 1,000 MW (CEC); existing average load 800–1,065 MW (chapter 1 range); IEPR planning-scenario CAISO managed net peak 46,479 MW (2025) to 50,498 MW (2030), growth 4,019 MW; IEPR statewide energy 263 to 298 TWh (growth 35 TWh); IEPR data center deliveries 12.1 TWh and data center contribution at the CAISO peak 1,743 MW in 2030 (planning scenario; 4,377 MW in the local-reliability scenario). Energy equivalent = $\text{MW} \times 0.67 \text{ utilization} \times 0.88 \text{ load factor} \times 8760 \text{ h}$.

Table 3.10 answers RQ1. The denominators come from the sources of the preceding chapters: the CAISO record and 2025 peaks, the CEC’s existing-load figure and the chapter 1 range, and the adopted 2025 IEPR planning forecast, whose hourly CAISO forecast gives a managed net peak of 46,479 MW in 2025 and 50,498 MW in 2030, a growth of 4,019 MW, and whose statewide energy to serve load grows from 263 to 298 TWh [California Energy Commission, 2026e]. Signed agreements alone, 5,086 MW, exceed the forecast’s entire five-year peak growth (127 percent of it); the three California tiers together are 5.1 times that growth (5.8 times including the Nevada requests), 3.4 times it even at the CEC’s 67 percent utilization factor, and their energy equivalent at that utilization and a 0.88 load factor, about 107 TWh a year, is 3.1 times the forecast’s statewide energy growth to 2030. Requests are not simultaneous demand: signed agreements are not operating capacity, and the forecast’s confidence and utilization factors exist to discount exactly this. The forecast is not blind to data centers: its planning scenario carries a data center contribution at the CAISO peak that rises from 96 MW in 2025 to 1,743 MW in 2030 (4,377 MW in the local-reliability scenario), and data-center-only deliveries of 12.1 TWh in 2030, roughly a third of projected statewide energy growth. What the forecast does is discount the requests, by the confidence levels and utilization of Chapter 2, to about a tenth of their face value at the peak by 2030. Whether that discount is right is the question the remaining chapters take up with the supply side, the attrition evidence and the adequacy framing.

Chapter 4

Energy Production Growth in California

This chapter builds the supply side of RQ2, which the proposal left thin. It records what California’s generating fleet produced and imported from 2010 to 2025, what capacity it holds and is scheduled to lose, and, from the state’s own planning record, how much of what is planned actually gets built and how late. The last section applies that empirical attrition to the current planned list to give probability-weighted supply for 2030 in three cases, symmetric with the confidence weighting the CEC applies to data center requests. Code is in `src/ch3_supply.py` and `notebooks/03_supply_growth.ipynb`; the pipeline parses sixteen EIA-923 and sixteen EIA-860 annual files and eighteen EIA-860M monthly files directly from the raw archives.

4.1 Generation by resource and imports, 2010–2025

In-state generation comes from the EIA-923 plant-level schedule, with each California plant’s reported fuel and prime mover classified into twelve resource classes [U.S. Energy Information Administration, 2026c]. The EIA-923 totals match the CEC’s in-state totals to within one percent in every year for which both exist. Net imports come from the CEC’s total-system series, which is statewide and includes the utilities outside CAISO [California Energy Commission, 2026a]; the CEC has not yet published 2025 imports, and its 2021 page repeats the 2020 table, so the multi-year table is used for the import series and the CAISO-only net interchange from Chapter 2 is shown as a footprint-limited proxy for 2019 to 2025.

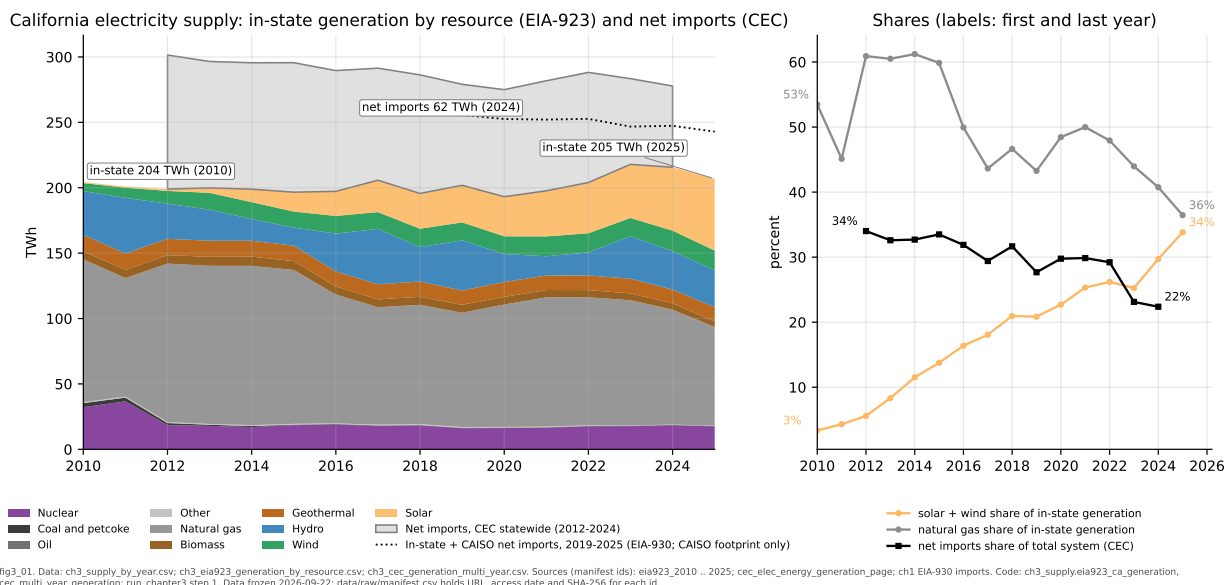


Figure 4.1: California electricity supply, 2010–2025: in-state generation by resource (EIA-923) stacked with CEC statewide net imports (2012–2024), the CAISO-only net imports of Chapter 2 as a proxy, and the shares of solar plus wind, natural gas and imports.

Source line: fig3_01. Data: ch3_supply_by_year.csv; ch3_eia923_generation_by_resource.csv; ch3_cec_generation_multi_year.csv. Sources (manifest ids): eia923_2010 .. 2025; cec_elec_energy_generation_page; ch1 EIA-930 imports. Code: ch3_supply.eia923_ca_generation, cec_multi_year_generation; run_chapter3 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 4.1: In-state generation by resource and net imports, selected years (TWh). Batteries and pumped storage are net of charging. Source: EIA-923, CEC Energy Almanac, EIA-930.

Resource (TWh)	2010	2015	2020	2023	2024	2025
Natural gas	109.2	117.7	93.6	95.8	87.9	75.4
Nuclear	32.2	18.5	16.3	17.7	18.4	17.6
Hydro	33.4	13.8	21.4	32.4	29.5	28.1
Geothermal	12.6	11.9	11.4	11.3	10.5	10.9
Biomass	6.2	6.7	5.8	5.2	4.8	4.3
Wind	6.1	12.2	13.6	14.0	15.6	15.1
Solar	0.8	14.8	30.3	41.0	48.4	54.9
Coal and petcoke	3.1	0.3	0.3	0.2	0.2	0.2
Oil	0.1	0.1	0.0	0.0	0.0	0.0
Other	0.7	0.6	0.6	0.3	0.2	0.3
Batteries	0.0	-0.0	-0.0	-0.9	-1.4	-1.9
Pumped storage	-0.2	0.1	-0.0	-0.4	-0.0	0.2
Total in-state	204.1	196.7	193.1	216.6	214.2	205.1
Net imports, CEC statewide	–	99.0	81.9	65.5	62.2	–
Net imports, CAISO only (EIA-930)	–	–	59.4	28.8	31.7	36.1

In-state generation was 204 TWh in 2010 and 205 TWh in 2025, but its composition inverted (Figure 4.1, Table 4.1). Natural gas fell from 109 to 75 TWh, from 53 to 36 percent of in-state generation, and solar rose from under 1 to 55 TWh, so solar and wind together went from 3 to 34 percent. Nuclear halved with the closure of San Onofre, and hydro swung between 14 TWh in the

2015 drought and 42 TWh in 2017. Statewide net imports fell from 103 TWh in 2012, a third of the total system, to 62 TWh in 2024, 22 percent. The total system therefore shrank from 302 to 278 TWh between 2012 and 2024 while in-state generation grew, which is the same flat-demand picture that Chapter 2 found at the CAISO level, now with the import share falling.

4.2 Capacity, storage and curtailment

Operable nameplate capacity by technology comes from the EIA-860 annual generator files, classified the same way [U.S. Energy Information Administration, 2026a]; battery energy capacity comes from the EIA-860 energy storage file, which begins in 2016; curtailment is CAISO’s monthly wind and solar curtailment series [California Independent System Operator, 2026a].

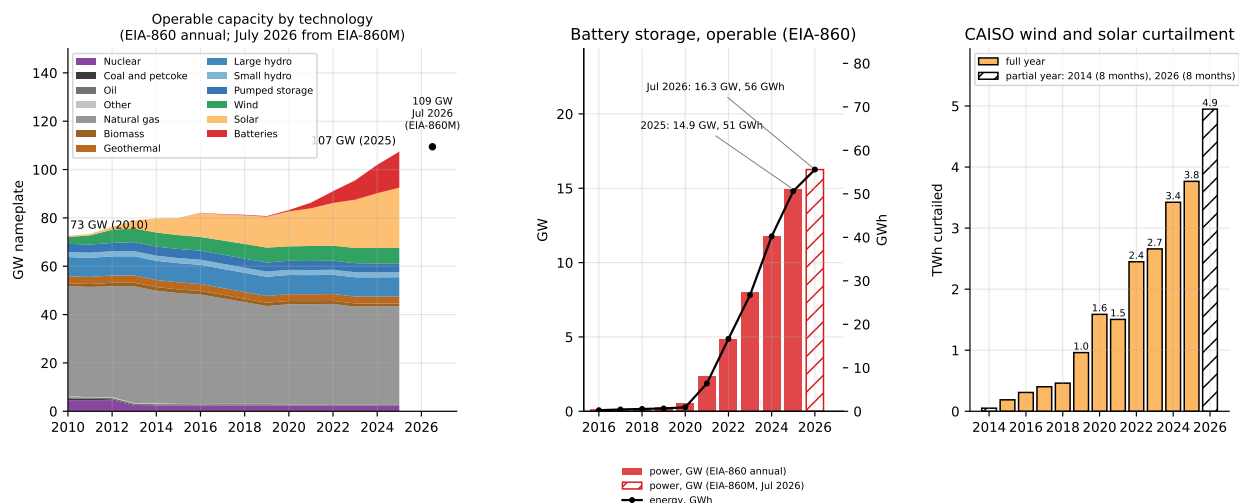


fig3_02. Data: ch3_eia860_capacity_by_resource.csv; ch3_battery_capacity.csv; ch3_caiso_curtailment_annual.csv; ch3_capacity_snapshot_2026_07.json. Sources (manifest ids): eia860_2010 .. 2025; eia860m_2026_07; caiso_curtailments_monthly_csv. Code: ch3_supply.eia860_ca_capacity, eia860_ca_battery_energy, caiso_curtailment_annual, eia860m_capacity_snapshot; run_chapter3 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 4.2: Operable capacity by technology in California (EIA-860, left), battery power and energy capacity (EIA-860 energy storage file, middle) and CAISO wind and solar curtailment by year (right; 2026 covers January to August).

Source line: fig3_02. Data: ch3_eia860_capacity_by_resource.csv; ch3_battery_capacity.csv; ch3_caiso_curtailment_annual.csv; ch3_capacity_snapshot_2026_07.json. Sources (manifest ids): eia860_2010 .. 2025; eia860m_2026_07; caiso_curtailments_monthly_csv. Code: ch3_supply.eia860_ca_capacity, eia860_ca_battery_energy, caiso_curtailment_annual, eia860m_capacity_snapshot; run_chapter3 step 1. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 4.2: Operable nameplate capacity by technology, battery energy capacity and CAISO curtailment, selected years and the July 2026 inventory. Source: EIA-860, EIA-860M, CAISO.

Technology (GW nameplate)	2010	2015	2020	2023	2024	2025	Jul 2026
Natural gas	45.5	45.8	41.2	40.4	40.5	40.4	40.4
Nuclear	4.6	2.3	2.3	2.3	2.3	2.3	2.3
Large hydro	7.9	7.9	7.9	7.9	8.0	8.0	8.0
Small hydro	2.1	2.1	2.1	2.1	2.0	2.0	2.0
Pumped storage	3.4	3.8	3.7	3.7	3.7	3.7	3.7
Geothermal	2.9	2.9	2.8	2.9	2.9	2.9	2.9
Biomass	1.2	1.5	1.4	1.3	1.1	1.1	1.1
Wind	2.8	5.7	6.0	6.3	6.5	6.5	6.5
Solar	0.5	7.1	14.5	19.9	22.6	24.8	25.5
Batteries	–	0.0	0.5	8.0	11.8	14.9	16.3
Coal and petcoke	0.6	0.1	0.1	0.1	0.1	0.1	0.1
Oil	0.7	0.5	0.5	0.5	0.5	0.6	0.6
Other	0.4	0.1	0.1	0.1	0.1	0.1	0.1
Total	72.6	79.9	83.2	95.5	102.0	107.4	109.4
Battery energy capacity (GWh)	–	–	0.9	26.7	40.2	50.6	55.6
CAISO wind and solar curtailment (TWh)	–	0.19	1.59	2.66	3.42	3.77	4.95 (Jan–Aug)

The 2010–2025 columns are the EIA-860 annual files; the July 2026 column is the EIA-860M monthly inventory, which is preliminary.

Capacity rose from 73 GW in 2010 to 107 GW in 2025 (Figure 4.2, Table 4.2). Solar added 24 GW and batteries 15 GW, the latter almost entirely since 2020 and holding 51 GWh of energy across 281 units by 2025; gas capacity fell only from 45 to 40 GW and remains the largest block. The battery fleet is what moved the net peak later in the evening in Chapter 2. Curtailment of wind and solar rose from 0.2 TWh in 2015 to 3.8 TWh in 2025, and reached 4.9 TWh in the first eight months of 2026 alone, so the midday surplus that Chapter 2 measured in prices is now also visible as discarded energy of the order of the output of a gigawatt of solar.

The most recent data. The CEC has not yet published 2025 statewide imports and the EIA-860 annual file ends in 2025, so the latest observations come from three other files, each preliminary (Table 4.3). The EIA-923 monthly file through June 2026, which includes EIA’s state-level increment rows for the plants that report only annually, puts January to June in-state generation at 89.6 TWh against 97.4 TWh in the same months of 2025, with natural gas down 28 percent, hydro down 9 percent and solar up 6 percent; CAISO net imports over the same months rose from 15.7 to 20.6 TWh. The EIA-860M inventory of July 2026 puts operable capacity at 109 GW, with 16.3 GW and 56 GWh of batteries in 305 units, and CAISO curtailment through August 2026 had already exceeded the whole of 2025. The EIA-860M file for August 2026 had not been published when this chapter was built.

Table 4.3: January to June generation by resource, 2025 final against 2026 preliminary, and CAISO net imports over the same months. Source: EIA-923 annual 2025 and monthly June 2026 files, EIA-930.

January–June (TWh)	2025 (final)	2026 (preliminary)	Change (%)
Natural gas	30.6	22.1	-28
Nuclear	8.7	9.5	+9
Hydro	15.1	13.7	-9
Geothermal	5.4	5.2	-3
Biomass	2.1	2.1	+2
Wind	8.6	8.8	+2
Solar	27.5	29.1	+6
Coal and petcoke	0.1	0.0	-88
Oil	0.0	0.0	–
Other	0.1	0.1	+34
Batteries	-0.9	-1.1	+15
Pumped storage	-0.0	-0.2	–
Total in-state	97.4	89.6	-8
of which EIA state-level increment rows	0.2	24.8	–
CAISO net imports, EIA-930 (CAISO footprint)	15.7	20.6	+31

4.3 Retirement schedule

Planned retirements come from the EIA-860M operating list of July 2026, which carries a planned retirement date for each unit that has reported one [U.S. Energy Information Administration, 2026b], overlaid with the compliance schedule of the State Water Board’s once-through-cooling policy as amended in August 2023 [State Water Resources Control Board, 2023]. The two agree where both exist: Alamitos units 3 to 5 and Huntington Beach unit 2 report December 2026 and the policy requires compliance by December 31, 2026; Ormond Beach reports January 2027 against the same policy date; Scattergood units 1 and 2 report 2029 against December 31, 2029. Three cases carry no EIA date: LADWP’s Haynes units 1, 2 and 8 and Harbor unit 5, which the policy requires to comply by December 31, 2029, and Diablo Canyon, whose compliance date SB 846 extended to October 31, 2030 and whose continued operation beyond that depends on federal licence renewal and further state action. The policy dates are used for those units in case C below, and Diablo Canyon is carried as a scenario.

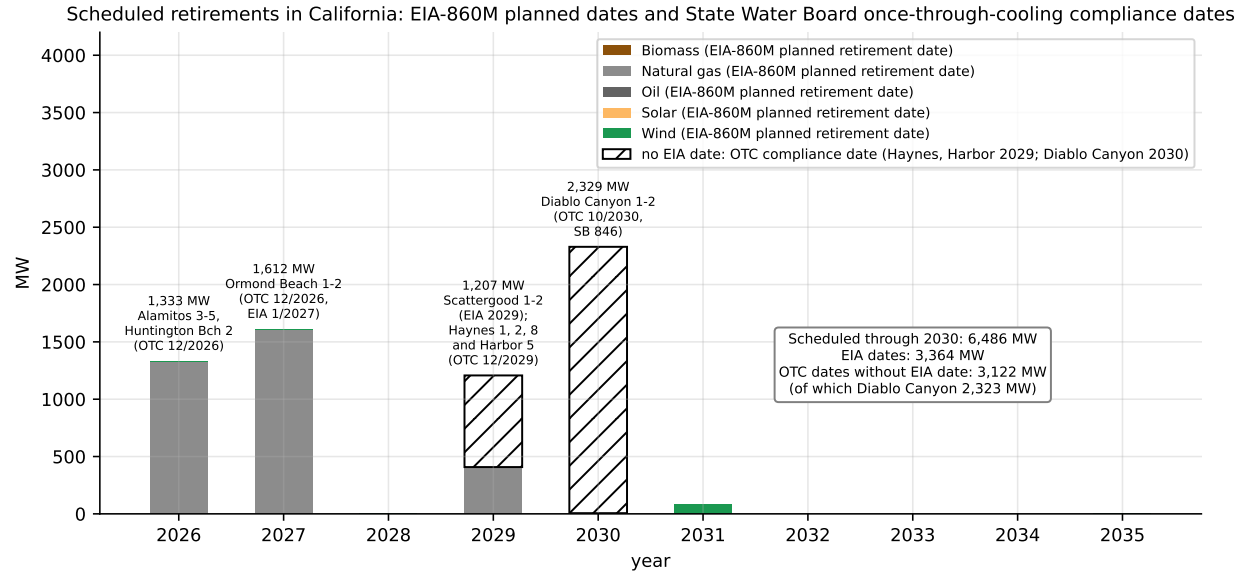


Figure 4.3: Scheduled retirements in California by year: EIA-860M planned retirement dates by technology, with the once-through-cooling compliance dates hatched for the units that report no EIA date.

Source line: fig3_03. Data: ch3_eia860m_planned_retirements.csv; ch3_otc_units.csv. Sources (manifest ids): eia860m_2026_07; swrcb_otc_policy_2023. Code: ch3_supply.retirement_schedule; run_chapter3 step 2. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 4.4: Once-through-cooling units and other scheduled retirements through 2030. Source: EIA-860M July 2026, State Water Board OTC Policy Table 1.

Plant	Units	Technology	MW	In service	OTC compliance date	EIA-860M planned retirement
AES Alamitos LLC	3, 4, 5	Natural Gas Steam Turbine	1,115	1961–1964	2026-12-31	2026
AES Huntington Beach LLC	2	Natural Gas Steam Turbine	218	1958–1958	2026-12-31	2026
Ormond Beach	1, 2	Natural Gas Steam Turbine	1,612	1971–1973	2026-12-31	2027
Haynes	1, 2, 8	Natural Gas Steam Turbine	724	1962–2005	2029-12-31	none
Harbor	5	Natural Gas Fired Combined Cycle	75	1995–1995	2029-12-31	none
Scattergood	1, 2	Natural Gas Steam Turbine	326	1958–1959	2029-12-31	2029
Diablo Canyon	1, 2	Nuclear	2,323	1985–1986	2030-10-31	none
Other units with an EIA date through 2030	5 units	mixed	92	–	–	2028–2030

Scheduled retirements through 2030 total 6.5 GW: 3.4 GW with EIA dates, almost all of it the coastal gas steam units of 2026, 2027 and 2029, plus 0.8 GW of LADWP units at their policy dates and the 2.3 GW of Diablo Canyon. Beyond 2030 the EIA record holds almost nothing, which reflects how retirement dates are reported rather than the age of the fleet.

4.4 A supply-side realization model

The CEC discounts data center requests by a confidence level for each tier. The supply side deserves the same treatment, and the EIA-860M monthly files make it possible to estimate one from California’s own record. For every California unit in the Planned sheet of the January file of each year from 2016 to 2022, the outcome by the December 2025 file is recorded: operating (or operated and since retired), cancelled (the file’s cumulative Canceled or Postponed sheet), still planned, or dropped from the survey without a cancellation record, which is treated as cancelled. Units that appear in more than one January list contribute one observation per vintage, and standard errors are clustered by unit. The sample is 925 unit-vintages of 513 distinct units and 64 GW of planned capacity.

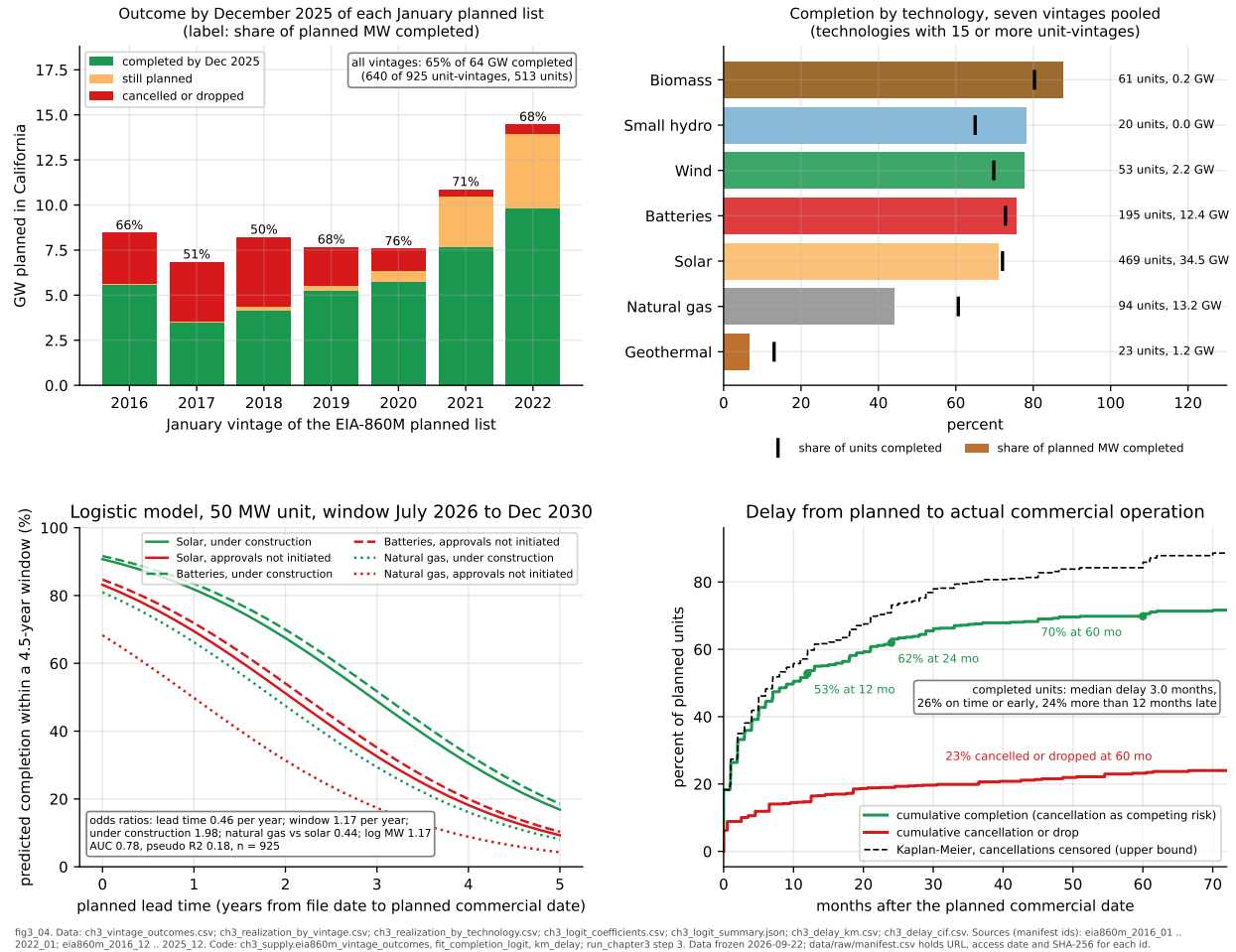


Figure 4.4: The realization model. Top left: outcome by December 2025 of each January planned list, capacity-weighted. Top right: completion by technology, seven vintages pooled. Bottom left: predicted completion within a 4.5-year window from the logistic model for a 50 MW unit. Bottom right: time from the planned to the actual commercial date, as Kaplan-Meier with cancellations censored and as cumulative incidence with cancellation as a competing risk.

Source line: fig3_04. Data: ch3_vintage_outcomes.csv; ch3_realization_by_vintage.csv; ch3_realization_by_technology.csv; ch3_logit_coefficients.csv; ch3_logit_summary.json; ch3_delay_km.csv; ch3_delay_cif.csv. Sources (manifest ids): eia860m_2016_01 .. 2022_01; eia860m_2016_12 .. 2025_12. Code: ch3_supply.eia860m_vintage_outcomes, fit_completion_logit, km_delay; run_chapter3 step 3. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 4.5: Outcome by December 2025 of the California planned lists of January 2016 to 2022, by vintage and by construction status at the vintage.

Vintage or status at the vintage	Unit-vintages	Planned MW	Completed (% units)	Completed (% MW)	Cancelled MW	Still planned MW	Median delay (months)
January 2016	145	8,454	77	66	2,842	20	2
January 2017	110	6,837	66	51	3,292	70	1
January 2018	110	8,209	61	50	3,801	270	2
January 2019	115	7,660	61	68	2,145	270	3
January 2020	127	7,569	72	76	1,245	556	6
January 2021	150	10,860	74	71	372	2,828	4
January 2022	168	14,466	69	68	519	4,146	4
Status: under construction	329	20,536	85	77	4,784	5	2
Status: approvals not initiated	202	16,685	55	54	2,312	5,407	15
Status: approvals pending	172	14,869	63	59	5,214	927	7
Status: approved, not started	150	10,582	53	66	1,798	1,817	5
Status: construction complete	72	1,383	81	92	108	3	0

Table 4.6: Outcome by technology, seven vintages pooled (technologies with at least ten unit-vintages).

Technology	Unit-vintages	Planned MW	Completed (% units)	Completed (% MW)	Cancelled MW	Still planned MW	Median delay (months)
Solar	469	34,524	72	71	5,130	4,878	3
Natural gas	94	13,164	61	44	7,379	0	1
Batteries	195	12,434	73	76	343	2,678	4
Wind	53	2,204	70	78	195	298	5
Geothermal	23	1,158	13	6	1,083	0	7
Biomass	61	164	80	88	15	5	2
Small hydro	20	45	65	78	10	0	28

Across the seven vintages, 69 percent of units and 65 percent of planned megawatts were operating by December 2025 (Table 4.5); by vintage the capacity-weighted completion rate ranges from 50 percent for January 2018 to 76 percent for January 2020, with the 2021 and 2022 lists still carrying 3 to 4 GW of unresolved projects. Technology matters less than one might expect among the resources that dominate the list: solar completed 71 percent of its planned megawatts, batteries 76 and wind 78 (Table 4.6). Natural gas completed 44 percent and geothermal 6 percent, the latter on 23 unit-vintages. Construction status at the vintage is the strongest single signal: units already under construction completed 77 percent of their megawatts, units whose approvals had not been initiated 54 percent.

Table 4.7: Logistic model of completion by December 2025 (unit-vintage observations; standard errors clustered by unit; reference categories solar and approvals not initiated).

Term	Coefficient	Cluster SE	Odds ratio	<i>p</i>
Intercept (solar, approvals not initiated)	0.290	0.554	1.34	0.601
Technology: Batteries	0.116	0.348	1.12	0.739
Technology: Biomass	-0.172	0.654	0.84	0.793
Technology: Geothermal	-2.180	1.122	0.11	0.052
Technology: Natural gas	-0.830	0.466	0.44	0.075
Technology: Other	-2.489	1.169	0.08	0.033
Technology: Small hydro	-0.034	0.961	0.97	0.972
Technology: Wind	-0.094	0.899	0.91	0.917
Status: approvals pending	0.087	0.416	1.09	0.834
Status: approved, not started	-0.273	0.344	0.76	0.429
Status: construction complete	0.257	0.491	1.29	0.601
Status: under construction	0.682	0.355	1.98	0.055
log nameplate MW	0.154	0.092	1.17	0.094
Planned lead time (years)	-0.775	0.189	0.46	<0.001
Observation window (years)	0.158	0.056	1.17	0.005
Observations	925 unit-vintages (513 units)			
Pseudo R^2 / in-sample AUC	0.178 / 0.778			

Table 4.8: Three specifications of the completion model: the main model, the model without construction status, and the literal specification with planned year and vintage fixed effects.

Specification	<i>n</i>	Pseudo R^2	AUC	OR: lead time (A, B) or planned year (C), per year	OR: under construction	OR: gas vs solar	OR: log MW
A. lead time + window + status (main)	925	0.178	0.778	0.46	1.98	0.44	1.17
B. lead time + window, no status	925	0.166	0.774	0.40	–	0.56	1.18
C. planned year + vintage fixed effects + status	925	0.162	0.764	0.55	2.60	0.45	1.11

Table 4.9: Delay from the planned to the actual commercial date, seven vintages pooled.

Statistic	Value
Completed unit-vintages	640
Median delay, planned to actual commercial date	3.0 months
Mean delay	7.4 months
On time or early	26%
More than 12 months late	24%
90th percentile delay	24 months
Cumulative completion by 12 months after the planned date (cancellation as competing risk)	53%
Cumulative completion by 24 months after the planned date (cancellation as competing risk)	62%
Cumulative completion by 36 months after the planned date (cancellation as competing risk)	68%
Cumulative completion by 60 months after the planned date (cancellation as competing risk)	70%
Cumulative cancellation or drop by 60 months	23%

The logistic model (Table 4.7) puts numbers on these patterns while holding the others fixed. Each additional year of planned lead time cuts the odds of completion within the window by more than half (odds ratio 0.46, $p < 0.001$); each additional year of observation raises them by 17

percent ($p = 0.005$), which is the mechanical effect of giving late projects time to finish; being under construction at the vintage doubles the odds ($p = 0.055$); gas and geothermal have lower odds than solar at conventional or near-conventional significance, and size adds a little (odds ratio 1.17 per log-unit of megawatts, $p = 0.09$). The in-sample AUC is 0.78. Lead time and observation window are a reparametrisation of the planned year and the vintage; the literal specification with planned year and vintage fixed effects (Table 4.8) gives the same ranking of technologies and construction status. The delay curves (Table 4.9) describe the projects that do finish and the ones that do not: among completed units the median delay from the planned to the actual commercial date is 3 months, a quarter are on time or early and a quarter are more than a year late. Treating cancellation as a competing risk, 53 percent of planned units are operating within 12 months of their planned date, 62 percent within 24 and 70 percent within 60, while 23 percent have been cancelled or dropped by then; the Kaplan-Meier curve with cancellations censored, an upper bound, reaches about 88 percent at six years.

4.5 Probability-weighted supply for 2030

The July 2026 EIA-860M planned list for California holds 229 units and 20.1 GW with planned commercial dates through 2030: 9.9 GW of batteries, 7.9 GW of solar, 1.6 GW in two pumped-storage projects whose approvals have not been initiated, 0.3 GW of wind and 0.2 GW of gas engines. Each unit is weighted by the model’s completion probability for a window ending December 2030, given its technology, size, lead time and construction status. The weighted list is 13.2 GW, two thirds of the nominal one: 71 percent of the battery and solar megawatts survive, 78 percent of wind, 31 percent of gas and 13 percent of the pumped storage. By status, 88 percent of the megawatts already under construction survive against 36 percent of those whose approvals have not been initiated.

Table 4.10: ELCC values and derates used for peak contribution, with realised 2023–2025 capacity factors used for energy. Source: CPUC/E3-Astrape 2023 MTR ELCC study, EIA-923 and EIA-860; assumptions as stated.

Technology	ELCC or derate	Capacity factor 2023–2025	Basis
Natural gas	0.950	0.244	assumption: firm resource net of forced-outage and ambient derate
Nuclear	0.950	0.879	assumption: firm resource net of forced-outage and ambient derate
Large hydro	0.550	0.342	assumption: summer-peak availability of hydro (CAISO NQC uses historical exceedance)
Small hydro	0.450	0.342	assumption: summer-peak availability of hydro (CAISO NQC uses historical exceedance)
Geothermal	0.950	0.430	assumption: firm resource net of forced-outage and ambient derate
Biomass	0.900	0.470	assumption: firm resource net of forced-outage and ambient derate
Wind	0.147	0.265	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Solar	0.088	0.244	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Batteries	0.765	-0.014	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Pumped storage	0.887	-0.002	CPUC/E3-Astrape 2023 MTR ELCC study, Table 1, Tranche 6 (2028): 4-hour battery, 8-hour PSH, utility solar, in-state wind
Coal and petcoke	0.950	0.419	assumption: firm resource net of forced-outage and ambient derate
Oil	0.900	0.009	assumption: firm resource net of forced-outage and ambient derate
Other	0.800	0.298	assumption: firm resource net of forced-outage and ambient derate

Energy is capacity times the realised 2023 to 2025 capacity factor of each class (batteries and pumped storage contribute no net energy, and for gas the fleet-average factor overstates the energy of the peaking units that retire). Peak contribution uses effective load carrying capability: the CPUC’s 2028 tranche values from the E3 and Astrape mid-term reliability study for solar (8.8 percent), in-state wind (14.7), four-hour batteries (76.5) and pumped storage (88.7) [Energy and Environmental Economics and Astrapé Consulting, 2023, California Public Utilities Commission, Energy Division, 2023], and stated derates for firm resources and hydro (Table 4.10). Three cases are reported (Figure 4.5, Table 4.11): everything on the list builds; the model-weighted list; and the model-weighted list minus the retirements of Section 4.3, with Diablo Canyon at its October 2030 policy date and a sensitivity in which it continues.

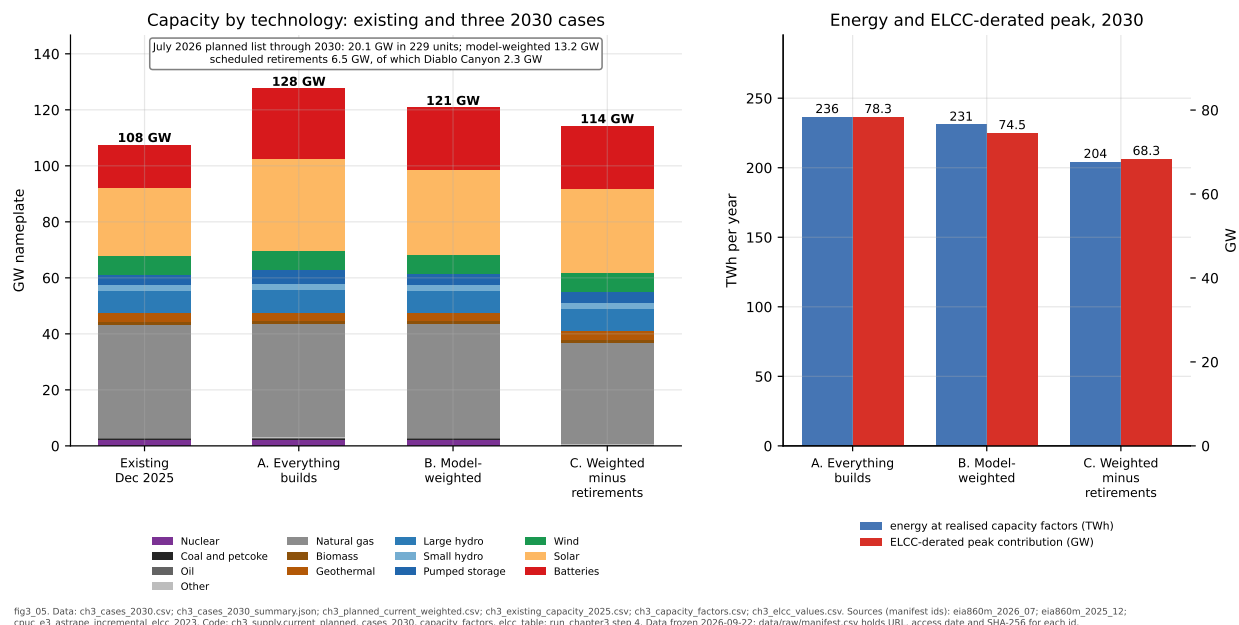


fig3_05. Data: ch3_cases_2030.csv; ch3_cases_2030_summary.json; ch3_planned_current_weighted.csv; ch3_existing_capacity_2025.csv; ch3_capacity_factors.csv; ch3_elcc_values.csv. Sources (manifest ids): eia860m_2026_07; eia860m_2025_12; cpuc_e3_astrape_incremental_elcc_2023. Code: ch3_supply.current_planned, cases_2030, capacity_factors, elcc_table; run_chapter3 step 4. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 4.5: Capacity by technology for the existing fleet and the three 2030 cases (left), and energy at realised capacity factors and ELCC-derated peak contribution by case (right).

Source line: fig3_05. Data: ch3_cases_2030.csv; ch3_cases_2030_summary.json; ch3_planned_current_weighted.csv; ch3_existing_capacity_2025.csv; ch3_capacity_factors.csv; ch3_elcc_values.csv. Sources (manifest ids): eia860m_2026_07; eia860m_2025_12; cpuc_e3_astrape_incremental_elcc_2023. Code: ch3_supply.current_planned, cases_2030, capacity_factors, elcc_table; run_chapter3 step 4. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 4.11: Planned supply for 2030 in three cases: capacity by technology (MW), energy at realised capacity factors and ELCC-derated peak contribution.

Technology	Existing Dec 2025	A. Everything builds	B. Model-weighted	C. Weighted minus retirements
Natural gas	40,414	40,585	40,467	36,308
Nuclear	2,323	2,323	2,323	0
Large hydro	7,979	7,979	7,979	7,979
Small hydro	2,041	2,050	2,047	2,047
Pumped storage	3,704	5,304	3,917	3,917
Geothermal	2,868	2,868	2,868	2,868
Biomass	1,107	1,136	1,127	1,124
Wind	6,530	6,874	6,799	6,799
Solar	24,625	32,556	30,274	30,274
Batteries	15,216	25,137	22,225	22,225
Coal and petcoke	62	62	62	62
Oil	540	540	540	538
Other	103	153	104	104
Total capacity (MW)	107,512	127,566	120,732	114,246
Energy at realised capacity factors (TWh)	—	236	231	204
ELCC-derated peak contribution (MW)	—	78,297	74,467	68,306
Case C with Diablo Canyon continuing: capacity / energy / peak	—	—	—	116,569 MW / 222 TWh / 70,512 MW

If everything builds, California's nameplate capacity reaches 128 GW in 2030, 236 TWh a year at realised capacity factors and 78 GW of ELCC-derated peak contribution. The model-weighted case gives 121 GW, 231 TWh and 74 GW. Subtracting 6.5 GW of retirements leaves 114 GW, 204 TWh and 68 GW, or 117 GW, 222 TWh and 71 GW if Diablo Canyon continues. Net of retirements, the supply side therefore adds about 7 to 9 GW of nameplate and 0 to 3 GW of ELCC-derated peak contribution by 2030 relative to the December 2025 fleet, and its energy falls if Diablo Canyon closes, because almost everything being added is solar and batteries whose peak

value is a fraction of nameplate. Against that stand the 4 GW of managed net peak growth in the adopted forecast and the 20.7 GW of data center requests located in California from Chapter 3. Setting the two probability-weighted sides against each other is the subject of the next chapter.

Chapter 5

Putting It Together

This chapter answers RQ2, RQ4 and RQ3 with the pieces built so far. Section 5.1 turns the December 2025 tiers of Chapter 3 into 2030 demand under four counting rules, on top of the IEPR’s own low, mid and high forecast of everything that is not a data center. Section 5.2 draws 10,000 joint scenarios of that demand and of the probability-weighted supply of Chapter 4 and reports the distribution of the 2030 gap in energy and at the net peak, with the proposal’s upper bound carried in every table. Section 5.3 ranks the inputs by one-at-a-time swings and Sobol indices. Section 5.4 replicates the Duke Nicholas Institute’s curtailment-enabled headroom on CAISO hourly load and sets it against the gap (RQ4). Section 5.5 compares the six data and measurement regimes and restates California’s headline number under each of their counting rules (RQ3), and Section 5.6 adds the emissions of a flat against a curtailable load. Code is in `src/ch4_gap.py`, `src/ch4_regimes.py`, `scripts/run_chapter4.py` and `notebooks/04_gap_model.ipynb` and `05_regimes_flexibility.ipynb`; `make ch4` rebuilds everything in about a minute.

5.1 Demand scenarios for 2030

The tiers and the non-data-center baseline. The starting point is the CEC’s December 2025 request data: 5,086 MW with signed agreements, 9,587 MW in active applications and 8,604 MW of inquiries statewide, of which the 2,600 MW application reported by Valley Electric Association sits in Nevada, so the California-only total is 20,677 MW (Table 5.1). Everything that is not a data center comes from the adopted CED 2025 forms: statewide energy to serve load and the 1-in-2 coincident peak for the Planning forecast, the Baseline (the Planning forecast before additional achievable efficiency and fuel substitution), the Local Reliability scenario and the Local Reliability scenario with known loads [California Energy Commission, 2026c], minus the CEC’s own data center component, which is published as a line of the peak workbook and as the data center allocations of the Planning forecast [California Energy Commission, 2026e,d]. The three cases are labelled low (Planning), mid (Baseline) and high (Local Reliability); the fourth, which adds the utilities’ known loads through a separate adjustment line, is shown for reference. Between 2025 and 2030 non-data-center energy grows by 23.6, 40.5 and 45.3 TWh and the non-data-center statewide peak by 3,625, 4,988 and 5,792 MW in the three cases (Table 5.2). The CEC’s own data center component in 2030 is 1,743 MW at the CAISO peak in the Planning forecast and 4,377 MW in the Local Reliability scenario, of which 160 and 242 MW are VEA’s Nevada load, with 12.1 TWh of deliveries in the Planning forecast.

Table 5.1: Requested capacity by tier, December 2025 vintage. Source: CEC Assembly hearing deck (statewide totals) and the methodology memo’s Table 1 (VEA row).

Tier (December 2025)	All reporting utilities (MW)	VEA, Nevada (MW)	California (MW)
Signed agreement	5,086	0	5,086
Active application	9,587	2,600	6,987
Inquiry	8,604	0	8,604
Total	23,277	2,600	20,677

Table 5.2: Non-data-center demand in 2025 and 2030 by IEPR case: statewide energy to serve load and coincident peak minus the CEC’s data center component. Source: CED 2025 forms (TN 268722, 268727, 268725, 268726), peak components (TN 268124), data center allocations (TN 268824).

Case	CED 2025 scenario	Energy 2025 (TWh)	Energy 2030	Growth	Peak 2025 (MW)	Peak 2030	Growth	CEC DC peak 2030 (MW)	CEC DC energy 2030 (TWh)
low	Planning	262.5	286.1	23.6	57,464	61,089	3,625	1,743	12.1
mid	Baseline	263.3	303.7	40.5	57,464	62,452	4,988	1,743	12.1
high	Local Reliability	262.5	307.8	45.3	57,440	63,233	5,792	4,377	30.3
high plus known loads	Local Reliability plus known	262.5	327.0	64.5	57,440	67,014	9,573	4,382	30.4

Four counting rules. Table 5.3 and Figure 5.1 apply four rules to the same 20,677 MW. The proposal’s upper bound lets every requested megawatt operate flat by 2030: 20,677 MW and 181.1 TWh, or 13,854 MW and 106.8 TWh at the CEC’s utilization factor. The CEC central case is the Commission’s published Planning forecast, 1,583 MW at the CAISO peak once VEA is removed, with 12.1 TWh of deliveries; the Local Reliability scenario gives 4,135 MW and 30.3 TWh. Both are rebuilt from the CEC’s published parameters in Table 5.4: confidence levels of 70/33/0 or 100/50/10 percent times 67 percent utilization, with SVP’s requests exempt from the confidence levels as the methodology memo states [California Energy Commission, 2026d], reproduce the memo’s 2040 endpoints of 4,855 and 7,381 MW to within 1 MW, so the effective tier probabilities are 0.74/0.35/0.02 in the Planning forecast and 1.00/0.51/0.12 in the Local Reliability scenario. The CEC’s own ramp, read as the share of the 2040 CAISO component reached in each year (Table 5.5), puts 36 percent of that demand on line by 2030 in the Planning forecast and 60 percent in the Local Reliability scenario, which gives 1,549 and 3,896 MW for California against the published 1,583 and 4,135 MW, within 2 and 6 percent; the residual is the utility-supplied schedules the CEC uses where they exist. The published values are used as the central and high cases and the replication rows are carried as the check. The ERCOT-calibrated case is described below. The PJM-style case follows the rules PJM applied in its 2026 forecast: only projects with an electric service obligation or construction commitment count before 2030, at 70 percent utilization and a ramp of at least 36 months, and from 2030 half of non-firm requests are added before a national scaling [Mooney, 2025, PJM Interconnection, 2026]; on California’s tiers that is 3,560 MW for the firm-only rule and 9,017 MW with half of the applications and inquiries.

Table 5.3: Data center demand added by 2030 under each counting rule (California-only tiers). Energy uses a 0.88 load factor except where the CEC’s published deliveries are used. Ramps are the share of realized capacity on line by the end of 2030.

Case	P(agreement)	P(application)	P(inquiry)	Utilization	Ramp agr.	Ramp app./inq.	Load factor	Peak 2030 (MW)	Energy 2030 (TWh)
Upper bound: every MW builds, flat load	1.00	1.00	1.00	1.00	1.00	1.00	1.00	20,677	181.1
Upper bound at CEC utilization	1.00	1.00	1.00	0.67	1.00	1.00	0.88	13,854	106.8
CEC central (Planning forecast, published)	0.70	0.33	0.00	0.67	–	–	–	1,583	12.1
CEC central, replicated from parameters	0.74	0.35	0.02	0.67	0.36	0.36	0.88	1,549	11.9
CEC high (Local Reliability, published)	1.00	0.50	0.10	0.67	–	–	–	4,135	30.3
CEC high, replicated from parameters	1.00	0.51	0.12	0.67	0.60	0.60	0.88	3,896	30.0
ERCOT-calibrated stock-flow	0.37	0.06	0.02	0.67	1.00	1.00	0.88	1,622	12.5
ERCOT-calibrated, ERCOT observed utilization	0.37	0.06	0.02	0.64	1.00	1.00	0.88	1,545	11.9
PJM-style: firm only (signed agreements)	1.00	0.00	0.00	0.70	1.00	1.00	0.88	3,560	27.4
PJM-style: firm plus half of non-firm from 2030	1.00	0.50	0.50	0.70	1.00	1.00	0.88	9,017	69.5

Table 5.4: The CEC forecast rebuilt from its published parameters: full-ramp demand with SVP exempt from the confidence levels against the memo’s 2040 endpoints, the effective tier probabilities, the CEC’s 2030 ramp share and the replicated against the published 2030 components.

Scenario	Full-ramp statewide (MW)	Memo 2040 endpoint	Full-ramp California	Effective P (agr. / app. / inq.)	CEC ramp share 2030 (%)	Replicated 2030 California (MW)	Published CAISO 2030	of which VEA	Published California 2030
Planning	4,855	4,855	4,280	0.74 / 0.35 / 0.02	36	1,549	1,743	160	1,583
Local Reliability	7,381	7,381	6,510	1.00 / 0.51 / 0.12	60	3,896	4,377	242	4,135

Table 5.5: The CEC’s ramp profile: CAISO data center component by year and its share of the 2040 value (TN 268124 annual peaks).

CEC ramp profile	2025	2026	2027	2028	2029	2030	2032	2035	2040
Planning: CAISO data center component (MW)	96	288	417	806	1,245	1,743	2,979	4,564	4,817
Planning: share of 2040 (%)	2	6	9	17	26	36	62	95	100
Local Reliability: CAISO data center component (MW)	122	374	684	1,882	3,238	4,377	5,879	7,191	7,314
Local Reliability: share of 2040 (%)	2	5	9	26	44	60	80	98	100

Table 5.6 adds each of the data center cases to each of the three IEPR cases: the 2030 statewide coincident peak runs from 62.7 GW (Planning growth with the CEC central case) to 83.9 GW (Local Reliability growth with the upper bound), and the growth from 2025 from 5.2 to 26.5 GW and from 36 to 226 TWh.

Table 5.6: Each demand case on top of each IEPR non-data-center case: 2030 statewide coincident peak and energy to serve load, and their growth from 2025.

Demand case	IEPR case	Non-DC peak 2030 (MW)	DC peak	Total peak	Growth from 2025	Non-DC energy 2030 (TWh)	DC energy	Total energy	Growth from 2025
Upper bound: every MW builds, flat load	low	61,089	20,677	81,766	24,302	286.1	181.1	467.2	204.8
Upper bound: every MW builds, flat load	mid	62,452	20,677	83,129	25,665	303.7	181.1	484.9	221.6
Upper bound: every MW builds, flat load	high	63,233	20,677	83,910	26,469	307.8	181.1	488.9	226.4
Upper bound at CEC utilization	low	61,089	13,854	74,943	17,479	286.1	106.8	392.9	130.4
Upper bound at CEC utilization	mid	62,452	13,854	76,305	18,841	303.7	106.8	410.5	147.3
Upper bound at CEC utilization	high	63,233	13,854	77,086	19,646	307.8	106.8	414.6	152.1
CEC central (Planning forecast, published)	low	61,089	1,583	62,672	5,208	286.1	12.1	298.2	35.7
CEC central (Planning forecast, published)	mid	62,452	1,583	64,035	6,571	303.7	12.1	315.8	52.6
CEC central (Planning forecast, published)	high	63,233	1,583	64,816	7,375	307.8	12.1	319.9	57.4
CEC high (Local Reliability, published)	low	61,089	4,135	65,224	7,760	286.1	30.3	316.4	54.0
CEC high (Local Reliability, published)	mid	62,452	4,135	66,587	9,123	303.7	30.3	334.1	70.8
CEC high (Local Reliability, published)	high	63,233	4,135	67,368	9,927	307.8	30.3	338.1	75.6
ERCOT-calibrated stock-flow	low	61,089	1,622	62,711	5,247	286.1	12.5	298.6	36.1
ERCOT-calibrated stock-flow	mid	62,452	1,622	64,074	6,610	303.7	12.5	316.2	53.0
ERCOT-calibrated stock-flow	high	63,233	1,622	64,855	7,414	307.8	12.5	320.3	57.8
PJM-style: firm only (signed agreements)	low	61,089	3,560	64,649	7,185	286.1	27.4	313.5	51.1
PJM-style: firm only (signed agreements)	mid	62,452	3,560	66,012	8,548	303.7	27.4	331.2	67.9
PJM-style: firm only (signed agreements)	high	63,233	3,560	66,793	9,352	307.8	27.4	335.2	72.7

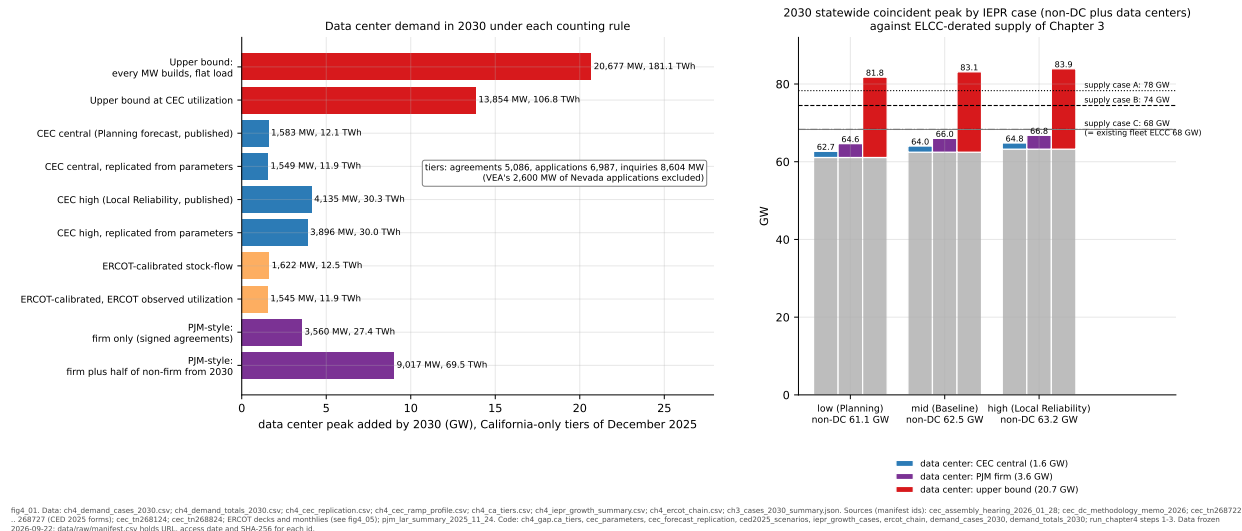


Figure 5.1: Left: data center peak and energy added by 2030 under each counting rule. Right: the 2030 statewide coincident peak by IEPR case, non-data-center demand plus three of the data center cases, against the ELCC-derated supply of Chapter 4 (cases A, B and C) and the existing fleet.

Source line: fig4_01. Data: ch4_demand_cases_2030.csv; ch4_demand_totals_2030.csv; ch4_cec_replication.csv; ch4_cec_ramp_profile.csv; ch4_ca_tiers.csv; ch4_iepr_growth_summary.csv; ch4_ercot_chain.csv; ch3_cases_2030_summary.json. Sources (manifest ids): cec_assembly_hearing_2026_01_28; cec_dc_methodology_memo_2026; cec_tn268722 .. 268727 (CED 2025 forms); cec_tn268124; cec_tn268824; ERCOT decks and monthlies (see fig4_05); pjmlar_summary_2025_11_24. Code: ch4_gap.ca_tiers, cec_parameters, cec_forecast_replication, ced2025_scenarios, iepr_growth_cases, ercot_chain, demand_cases_2030, demand_totals_2030; run_chapter4 steps 1-3. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

The ERCOT-calibrated stock-flow case. ERCOT publishes the stock of large-load requests in each phase of its interconnection process, which makes it possible to estimate how fast projects actually move from one phase to the next. Monthly totals from January 2025 to March 2026, the monthly stocks of projects with approved planning studies and with approval to energize from May 2025 to March 2026, and six snapshots of the queue by status between March and July 2026 were transcribed from ERCOT’s newsletters, its March 2026 TAC report, its December 2025 and June 2026 board decks, its April 2026 legislative decks and its April, June, July and August 2026 operational overviews (Figure 5.6, Section 5.5) [ERCOT Large Load Integration Team, 2026, ERCOT, 2026a, Billo and Hobbs, 2026, ERCOT, 2026b]. Approvals to energize reached 9,456 MW by July 2026, of which 4,370 MW was observed operating that month and 4,316 MW in August. From these, four monthly hazards are estimated (Table 5.7): 2.1 percent a month from no studies submitted to under review, an upper bound because cancellations leave the same stock; 0.75 percent from under review to studies approved; 1.6 percent from studies approved to approval to energize; and 3.1 percent from approval to energize to observed energization. A Markov chain with those hazards gives the probability that a project starting in each phase is energized within 60 months, the distance from the December 2025 vintage to December 2030: 37 percent for a project whose studies are approved, 6 percent for one under review and 2 percent for one without studies (Table 5.8). Mapping inquiries to no studies submitted, applications to under review and signed agreements to approved studies, and applying the CEC’s utilization factor, gives 1,622 MW and 12.5 TWh in 2030, almost the same as the CEC central case. Doubling every hazard, which is what ERCOT’s

move to a batch process is meant to achieve, raises the three probabilities to 72, 25 and 12 percent; a 0.5 percent monthly cancellation hazard lowers them slightly. The rates were measured while ERCOT’s queue quadrupled and its studies fell behind, so they describe study throughput as much as project viability, which is why the Monte Carlo treats the tier probabilities as uncertain over the CEC and ERCOT values rather than fixing them.

Table 5.7: Monthly phase-transition hazards estimated from the ERCOT large-load queue. Source: ERCOT TAC report March 2026 (slide 4), ERCOT Monthly November 2025, TAC snapshot of March 13 2026.

Transition	Flow (MW)	Months	Pool (MW)	Monthly hazard (%)	Annual (%)	Window
Planning studies approved → approved to energize	2,168	10.0	13,860	1.56	17.2	2025-05 to 2026-03
Under ERCOT review → planning studies approved	5,878	10.0	77,982	0.75	8.7	2025-05 to 2026-03
No studies submitted → under ERCOT review (or cancellation)	10,327	3.8	129,152	2.12	22.6	2025-11-18 to 2026-03-13
Approved to energize → observed energized	923	12.0	2,524	3.05	31.0	calendar 2025

Table 5.8: Probability (percent) of energization within 60 months by starting phase, from the Markov chain with the estimated hazards, with every hazard scaled 1, 2 and 3 times and with a monthly cancellation hazard of 0 or 0.5 percent in the pre-approval phases.

Start phase (California tier)	1x, cancel 0%	1x, cancel 0.5%	2x, cancel 0%	2x, cancel 0.5%	3x, cancel 0%	3x, cancel 0.5%
No studies submitted (inquiry)	1.6	1.4	12.2	10.3	30.2	25.6
Under ERCOT review (application)	5.9	5.1	24.5	21.2	45.6	39.6
Planning studies approved (signed agreement)	36.7	33.4	71.9	65.9	88.8	82.2
Approved to energize	84.4	84.4	97.7	97.7	99.7	99.7

5.2 The Monte Carlo gap model

Definitions. The energy gap in 2030 is the growth of demand from 2025 to 2030, non-data-center growth in the drawn IEPR case plus the data center energy of the drawn case, minus the growth of supply, the energy of the planned units that complete at recent capacity factors, minus the energy of the units that retire, plus the change in net imports. The net-peak gap is the growth of the statewide coincident peak times one plus the planning reserve margin, minus the change in ELCC-weighted capacity. Both are increments, so the accounting boundaries of the CEC forecast and of the EIA generation data cancel, and a positive value means that demand grows faster than the resources the weighted pipeline delivers at recent operating patterns; it has to be met by running the existing fleet harder, by importing more, or by resources not yet in the planned list. The absolute balances, which compare 2030 levels directly, are reported alongside; they use in-state ELCC only and so leave out the peak contribution of imports, which in 2025 sat on top of an in-state surplus of about 1.5 GW over the statewide peak with a 16 percent margin.

Inputs. Table 5.9 lists the thirteen inputs and the evidence behind their ranges: the tier probabilities span the CEC’s Planning and Local Reliability confidence levels with lower bounds near the ERCOT chain; utilization runs from ERCOT’s observed 46 percent (the August 2026 monthly peak of approved loads over the approved total) to PJM’s 70 percent with the CEC’s 67 percent as the mode; the share of realized capacity on line by 2030 spans the CEC’s own ramp profile (36 and 60 percent by 2030) and PJM’s 36-month rule; the load factor is the range of Chapter 2; each of the 229 planned generating units completes with its Chapter 4 probability; the hydro capacity factor spans 2010–2025; solar and wind output moves within 10 percent; 2030 net imports span the

CEC’s 2012–2024 statewide series; the IEPR case is drawn with equal weights; Diablo Canyon operates or retires with equal probability, because the Nuclear Regulatory Commission renewed both licences on April 2, 2026 while SB 846 authorises operation only through 2030 and any extension needs an act of the Legislature [Office of the Governor of California, 2026, U.S. Nuclear Regulatory Commission, 2026]; and the reserve margin is uniform between 15 and 17 percent.

Table 5.9: Monte Carlo inputs, distributions and their basis (10,000 draws, seed 20260922).

Input	Distribution	Low	Mode	High	Basis
p_agreement	triangular	0.50	0.70	1.00	CEC 70% (Planning) to 100% (Local Reliability); lower bound above the ERCOT chain value for studies-approved projects (0.37)
p_application	triangular	0.10	0.33	0.50	CEC 33% to 50%; lower bound above the ERCOT chain value for projects under review (0.06)
p_inquiry	triangular	0.00	0.03	0.10	CEC 0% to 10% (0.02 effective in the Planning forecast with SVP exempt); mode above the ERCOT chain value for projects without studies (0.02)
utilization	triangular	0.46	0.67	0.70	ERCOT monthly observed peak over approved (4,316 of 9,456 MW, Aug 2026) to PJM’s 70%; mode CEC 67%
ramp_2030	triangular	0.30	0.55	1.00	share of realized capacity on line by end-2030: the CEC’s own ramp profile reaches 36% (Planning) and 60% (Local Reliability) of full demand by 2030; PJM’s 36-month ramp gives 1.0
load_factor	triangular	0.80	0.88	0.95	chapter 1 assumption range; CEC observed 85-90% (memo p. 11)
gen_realization	per-unit Bernoulli	–	–	–	each planned unit completes with its chapter 3 logit probability p_2030 (229 units, 20.1 GW)
hydro_cf	triangular	0.16	0.34	0.48	hydro capacity factor 2010-2025 range 0.16-0.48, mode 2023-2025 mean
vre_scale	triangular	0.90	1.00	1.10	solar and wind capacity factors within 10% of their 2023-2025 means
imports_twh	triangular	62.16	62.16	99.21	CEC statewide net imports: 2019-2024 minimum 62, 2024 value 62, 2012-2024 maximum 99 TWh
iepr_case	categorical	0.00	1.00	2.00	low = Planning, mid = Baseline, high = Local Reliability, equal weights
diablo_continues	Bernoulli(0.5)	0.00	–	1.00	Diablo Canyon operating in 2030 or retired as in chapter 3 case C: the NRC renewed both licences on April 2 2026, but SB 846 authorises operation only through 2030 and any extension needs legislative action (gov_ca_diablo_license_2026_04_02, nrc_diablo_canyon_rod_2026)
prm	uniform	0.15	0.16	0.17	planning reserve margin applied to the peak: CPUC resource adequacy range 15-17%

Results. The median 2030 energy gap is 50 TWh, with a 5th to 95th percentile range of 19 to 78 TWh, and the median net-peak gap is 7.2 GW (4.2 to 10.1 GW); both are positive in every draw (Table 5.10, Figure 5.2). The data center contribution is 2.3 GW and 18 TWh at the median and 1.4 to 3.6 GW across the 5th to 95th percentiles. The gap is dominated by what is not a data center: non-data-center growth alone is 24 to 45 TWh and 3.6 to 5.8 GW, while the weighted pipeline adds on average only 5 TWh and 1.1 GW of ELCC once retirements are netted, and it removes 18 TWh and 2.2 GW in the half of the draws where Diablo Canyon retires (median gaps of 59 TWh and 8.4 GW with it retired against 42 TWh and 6.2 GW with it operating). Closing the median energy gap with the existing gas fleet would require raising its capacity factor from the 0.24 of 2023–2025 to 0.40, and the 95th percentile to 0.49. The proposal’s upper bound, every requested megawatt operating flat in 2030 with the central supply, gives an energy gap of 202 to 242 TWh and a peak gap of 25.5 to 30.2 GW across the three IEPR cases and both Diablo Canyon settings, four to five times the probability-weighted 95th percentile; those rows appear in every results table.

Table 5.10: Monte Carlo results for 2030: energy gap, net-peak gap and the data center contribution, all draws and by IEPR case, with the proposal's upper bound (every MW builds, flat load; range over the three IEPR cases and Diablo Canyon on or off).

Metric	IEPR case	Mean	P5	P25	P50	P75	P95	Share > 0 (%)
Energy gap (TWh)	all cases	49.4	19.2	37.3	49.7	62.0	78.1	100
Energy gap (TWh)	low	36.3	11.3	26.0	36.7	46.6	60.6	99
Energy gap (TWh)	mid	53.5	28.3	43.0	53.6	64.5	78.8	100
Energy gap (TWh)	high	58.0	33.1	47.7	58.2	68.6	81.7	100
upper bound (every MW builds, flat)	low to high, Diablo on or off	224.2	202.4	—	—	—	242.0	100
Net-peak gap (MW)	all cases	7,182	4,160	5,976	7,169	8,499	10,068	100
Net-peak gap (MW)	low	5,794	3,546	4,630	5,800	6,905	8,143	100
Net-peak gap (MW)	mid	7,426	5,124	6,237	7,444	8,568	9,793	100
Net-peak gap (MW)	high	8,291	6,084	7,144	8,282	9,366	10,620	100
upper bound (every MW builds, flat)	low to high, Diablo on or off	27,979	25,511	—	—	—	30,231	100
Data center peak (MW)	all cases	2,358	1,387	1,862	2,292	2,784	3,556	100
Data center peak (MW)	low	2,356	1,383	1,859	2,293	2,786	3,545	100
Data center peak (MW)	mid	2,365	1,391	1,866	2,301	2,796	3,572	100
Data center peak (MW)	high	2,353	1,388	1,860	2,281	2,768	3,559	100
upper bound (every MW builds, flat)	low to high, Diablo on or off	20,677	20,677	—	—	—	20,677	100
Data center energy (TWh)	all cases	18.1	10.6	14.3	17.6	21.4	27.4	100
Data center energy (TWh)	low	18.1	10.5	14.3	17.6	21.4	27.3	100
Data center energy (TWh)	mid	18.2	10.7	14.4	17.6	21.5	27.5	100
Data center energy (TWh)	high	18.1	10.6	14.3	17.6	21.3	27.4	100
upper bound (every MW builds, flat)	low to high, Diablo on or off	181.1	181.1	—	—	—	181.1	100

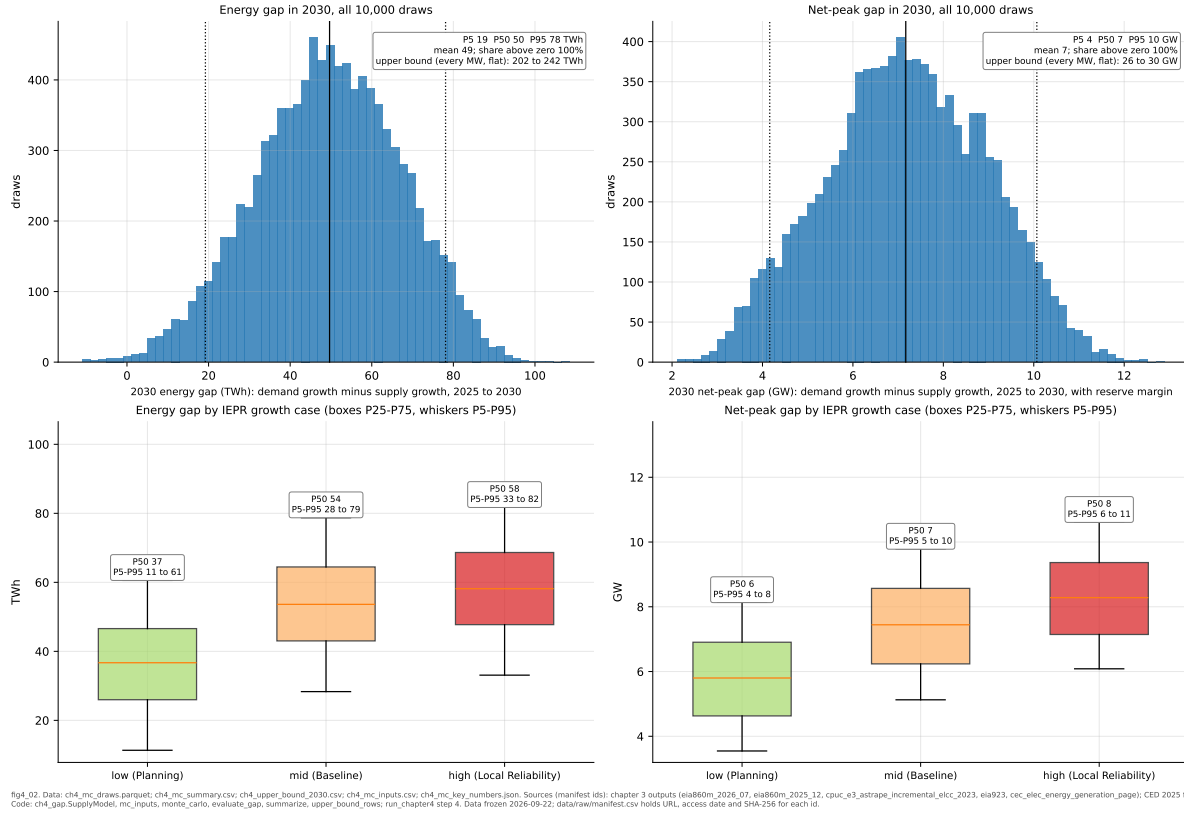


Figure 5.2: Distribution of the 2030 energy gap (left) and net-peak gap (right) over 10,000 draws, with the 5th, 50th and 95th percentiles and the upper-bound range, and by IEPR non-data-center growth case (bottom).

Source line: fig4_02. Data: ch4_mc_draws.parquet; ch4_mc_summary.csv; ch4_upper_bound_2030.csv; ch4_mc_inputs.csv; ch4_mc_key_numbers.json. Sources (manifest ids): chapter 3 outputs (eia860m_2026_07, eia860m_2025_12, cpuc_e3_astrape_incremental_elcc_2023, eia923, cec_elec_energy_generation_page); CED 2025 forms as above. Code: ch4_gap.SupplyModel, mc_inputs, monte_carlo, evaluate_gap, summarize, upper_bound_rows; run_chapter4 step 4. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

5.3 Sensitivity

Table 5.11 moves one input at a time between its 5th and 95th percentiles (low and high case for the IEPR case, retired and operating for Diablo Canyon) with every other input at its median, and Table 5.12 gives the Sobol first-order and total indices from a 15,360-evaluation Sobol design computed with SALib [Herman and Usher, 2017, Sobol', 2001]; both are drawn in Figure 5.3. For the energy gap the supply side and the non-data-center forecast dominate: net imports swing the gap by 28 TWh, the IEPR case by 22, the hydro capacity factor by 20 and Diablo Canyon by 18, while the three tier probabilities, utilization and the load factor together move it by 5 TWh or less each and carry about 3 percent of the variance. For the peak gap the ranking is Diablo Canyon (total index 0.39), the IEPR case (0.35), the data center ramp (0.14) and generation completion (0.07). The first-order indices sum to about one in both cases, so the model is close to additive and the one-at-a-time ranking is reliable.

Table 5.11: One-at-a-time swings of the 2030 gaps (all other inputs at their medians).

Input	Low	High	Energy gap at low (TWh)	at high	Swing	Peak gap at low (MW)	at high	Swing
imports_twh	63.1	91	65	37	28	6,913	6,913	0
iepr_case	0	2	38	60	22	5,332	7,846	2,514
hydro_cf	0.21	0.436	66	46	20	6,913	6,913	0
diablo_continues	0	1	64	46	18	8,016	5,809	2,207
ramp_2030	0.393	0.877	49	63	14	5,962	8,127	2,164
vre_scale	0.932	1.07	61	50	11	6,913	6,913	0
p_application	0.167	0.441	52	58	6	6,467	7,300	833
p_agreement	0.572	0.914	53	58	5	6,571	7,326	755
utilization	0.51	0.681	52	57	5	6,439	7,185	746
gen_realization	0.576	0.728	57	54	3	7,705	6,166	1,539
load_factor	0.824	0.927	54	56	2	6,913	6,913	0
p_inquiry	0.0128	0.0811	55	56	2	6,808	7,063	256
prm	0.151	0.169	55	55	0	6,847	6,979	132
base (all inputs at their medians)			55			6,913		
upper bound (every MW builds, flat; deterministic)			202	242		25,511	30,231	

Table 5.12: Sobol first-order (S1, with bootstrap confidence half-width) and total (ST) indices of the 2030 gaps.

Input	S1 energy	$\pm$	ST energy	S1 peak	$\pm$	ST peak
iepr_case	0.276	0.039	0.278	0.343	0.044	0.349
diablo_continues	0.255	0.040	0.254	0.391	0.047	0.391
imports_twh	0.240	0.042	0.242	0.000	0.000	0.000
hydro_cf	0.114	0.030	0.110	0.000	0.000	0.000
ramp_2030	0.059	0.022	0.059	0.130	0.034	0.136
vre_scale	0.034	0.015	0.033	0.000	0.000	0.000
p_application	0.008	0.009	0.009	0.018	0.012	0.021
utilization	0.008	0.009	0.009	0.018	0.013	0.020
p_agreement	0.008	0.007	0.008	0.018	0.011	0.018
gen_realization	0.002	0.004	0.002	0.072	0.023	0.069
load_factor	0.002	0.003	0.001	0.000	0.000	0.000
p_inquiry	0.001	0.002	0.001	0.002	0.004	0.002
prm	0.000	0.000	0.000	0.000	0.002	0.001

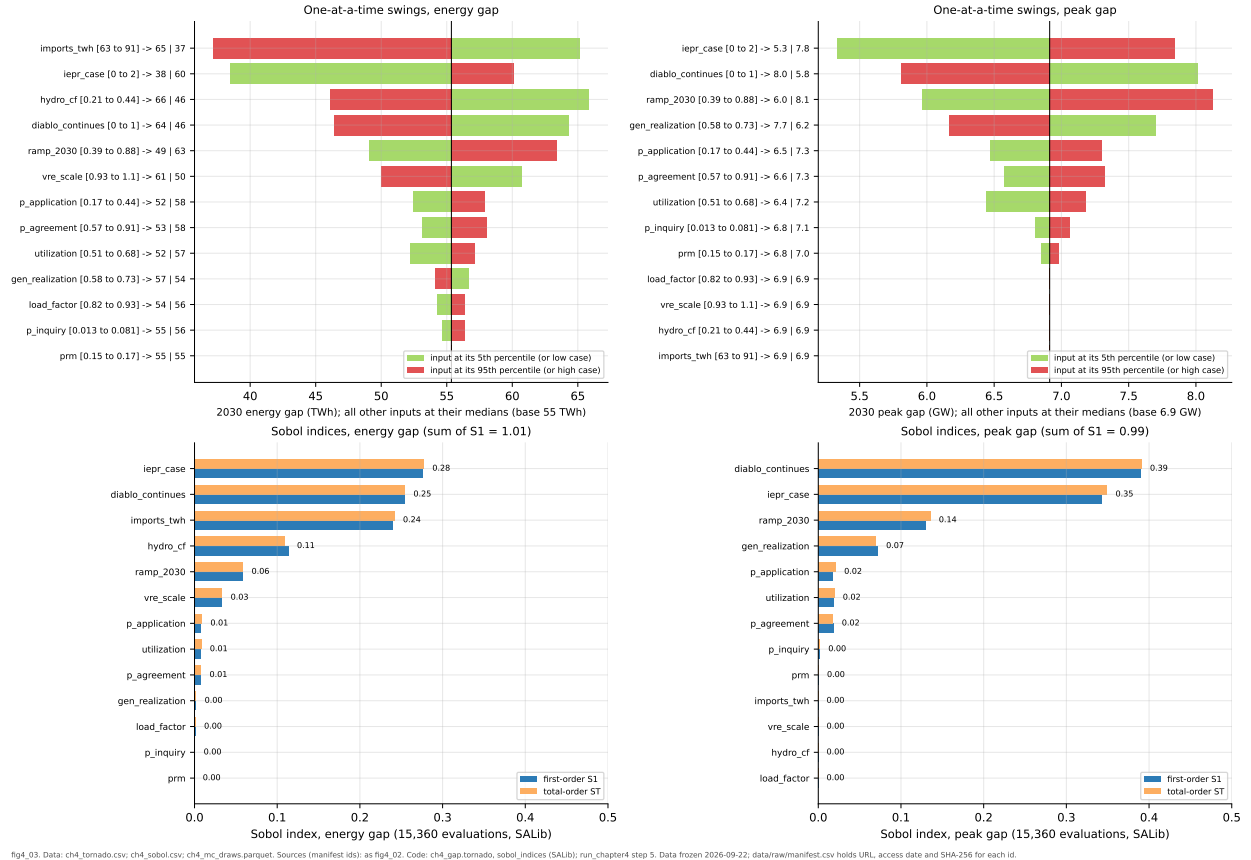


Figure 5.3: Sensitivity of the 2030 energy gap (left) and net-peak gap (right): one-at-a-time swings (top) and Sobol indices (bottom).

Source line: fig4_03. Data: ch4_tornado.csv; ch4_sobol.csv; ch4_mc_draws.parquet. Sources (manifest ids): as fig4_02.

Code: ch4_gap.tornado, sobol_indices (SALib); run_chapter4 step 5. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

5.4 Headroom from load flexibility (RQ4)

Method. The Duke Nicholas Institute asked how much constant new load a balancing authority could absorb without exceeding the demand its planners already serve, provided the new load accepts a small amount of curtailment [Norris et al., 2025]. Their method is replicated on the cleaned CAISO hourly demand of Chapter 2 for 2019–2025: a constant load L is added to every hour, the excess of demand plus L over the seasonal peak threshold is counted as curtailment, the thresholds are the maximum winter and non-winter demand across all years with winter defined as November to February for Pacific-coast balancing authorities, the annual curtailment rate is averaged over the years, and L is solved for curtailment limits of 0.25, 0.5, 1 and 5 percent. Missing hours are interpolated as in Duke’s cleaning. Three variants test the definition: a December-to-February winter, the 2022–2025 subsample, and a single annual threshold. The headroom is also solved year by year for 2019 to 2025 under two criteria: Duke’s, curtailed energy under 0.25, 0.5, 1 or 5 percent of the load’s potential energy, and the stricter one in the phase plan, the hours in which demand plus the addition exceeds the historical peak threshold kept under the same shares of the year’s hours, which has a closed form from the sorted margins to the threshold.

Results. The headroom is 3.1 GW at 0.25 percent curtailment, 3.8 GW at 0.5 percent, 4.6 GW at 1 percent and 8.0 GW at 5 percent (Table 5.13, Figure 5.4), against Duke’s 4.2, 5.0 and 5.9 GW for CAISO on 2016–2024 data. The shortfall against Duke comes from the sample of years and from the winter threshold, 31,226 MW here: 97 to 99 percent of the curtailment falls in November to February, when winter evening demand sits close to the winter peak, and the 2022–2025 subsample gives 2.7 to 4.1 GW. With a single annual threshold of 51,104 MW the headroom would be 11.6 to 16.0 GW, which shows how much of the answer is a definition of season. At the 0.5 percent headroom the new load is curtailed in about 197 hours a year, in 134 of which less than 90 percent of it is available and in 18 less than half, and in the worst hour of the seven years the whole 3.8 GW is cut (Table 5.14). Solved year by year (Table 5.15), the energy criterion gives 3.3 to 4.2 GW at 0.5 percent, 2.4 to 3.7 GW at 0.25 percent and 4.2 to 4.9 GW at 1 percent, lowest in 2025 and highest in 2023 and 2024; the hours criterion is stricter, 2.0 to 2.9 GW at 0.5 percent and 2.6 to 3.3 GW at 1 percent, because it charges a full hour for any excess however small. With a single historical peak instead of seasonal thresholds the same criteria give 7.0 to 11.7 GW and 11.4 to 16.2 GW at 0.5 percent (Table 5.16).

Against the gap. The 0.5 percent headroom is 2.4 times the CEC’s central data center addition and 2.3 times the ERCOT-calibrated one, about equal to the PJM firm-only case, 92 percent of the CEC’s high case, and 18 percent of the upper bound (Table 5.17). Set against the Monte Carlo net-peak gap, which includes everything else that grows, it covers 53 percent of the median and 38 percent of the 95th percentile, and 13 percent of the upper-bound peak gap; the 1 percent headroom covers 65 percent of the median gap. Flexibility is therefore enough to place the probability-weighted data center load, and about half of the whole probability-weighted peak gap, without new capacity, but not the upper bound. On the energy side the argument is stronger (Table 5.18): the SCE load-aggregation point had 877 hours of negative day-ahead prices in 2025 and 1,131 in 2024, 78 percent of them between 10 and 16 h (77 to 94 percent across the three load-aggregation points), and CAISO curtailed 3.8 TWh of wind and solar in 2025 and 4.9 TWh in January to August 2026 alone, the equivalent of a 430 MW load running flat all year, so a load that can follow the midday surplus adds energy the system is already discarding.

Table 5.13: Curtailment-enabled headroom (MW) on CAISO hourly demand by variant and curtailment limit, with Duke’s CAISO values.

Variant	0.25%	0.5%	1%	5%
Duke seasons (Nov-Feb winter), 2019-2025	3,112	3,798	4,641	8,031
Dec-Feb winter, 2019-2025	3,244	4,023	4,977	9,001
Duke seasons, 2022-2025 only	2,684	3,322	4,093	7,187
single annual threshold, 2019-2025	11,593	13,578	16,033	24,334
Duke (2016-2024, EIA-930 CISO)	4,200	5,000	5,900	—

Table 5.14: What the headroom means for the new load, Duke seasons 2019–2025, averages over the seven years.

Limit (%)	Headroom (MW)	Hours curtailed per year	Share in Nov-Feb (%)	Hours below 90% available	below 75%	below 50%	Largest hourly cut (MW)
0.25	3,112	95	97	65	34	11	2,227
0.5	3,798	197	98	134	68	18	2,912
1	4,641	383	99	272	144	34	3,756
5	8,031	1511	99	1225	796	236	7,145

Table 5.15: Headroom solved year by year on CAISO demand with the Duke seasonal thresholds (MW): hours above the peak under the limit (left) and curtailed energy under the limit (right).

Year	Hours above the peak under 0.25 / 0.5 / 1 / 5% of the year (MW)				Curtailed energy under 0.25 / 0.5 / 1 / 5% (Duke, MW)			
2019	1,945	2,315	2,870	5,040	2,954	3,657	4,585	8,299
2020	2,223	2,508	3,102	5,156	3,316	3,916	4,784	8,535
2021	1,561	2,071	2,834	5,067	2,523	3,466	4,545	8,347
2022	2,001	2,534	3,140	4,885	3,193	3,845	4,660	8,029
2023	2,619	2,873	3,315	4,850	3,657	4,196	4,898	7,974
2024	2,635	2,945	3,330	4,815	3,650	4,173	4,867	7,849
2025	1,345	1,985	2,641	4,438	2,433	3,264	4,152	7,331
range	1,345–2,635	1,985–2,945	2,641–3,330	4,438–5,156	2,433–3,657	3,264–4,196	4,152–4,898	7,331–8,535

Table 5.16: The same, with a single historical peak threshold of 51,104 MW in every month (MW).

Year	Hours above the peak under 0.25 / 0.5 / 1 / 5% of the year (MW)				Curtailed energy under 0.25 / 0.5 / 1 / 5% (Duke, MW)			
2019	8,995	9,602	10,806	16,353	12,128	13,784	16,122	24,933
2020	6,559	8,056	9,811	15,770	10,318	12,492	15,355	24,088
2021	9,630	10,438	11,314	15,757	12,745	14,255	16,306	24,410
2022	4,992	6,962	9,428	14,719	8,401	11,354	14,149	23,037
2023	9,107	10,416	12,041	17,377	13,160	15,141	17,666	25,561
2024	7,425	8,506	10,145	15,199	10,752	12,705	15,035	23,450
2025	10,454	11,738	13,084	17,324	14,436	16,219	18,286	24,577
range	4,992–10,454	6,962–11,738	9,428–13,084	14,719–17,377	8,401–14,436	11,354–16,219	14,149–18,286	23,037–25,561

Table 5.17: Headroom as a share of the data center cases and of the Monte Carlo net-peak gap (percent).

Limit (%)	Headroom (MW)	of CEC central DC (%)	of ERCOT-calibrated	of PJM firm	of CEC high	of upper-bound DC	of peak gap P50	of peak gap P95	of upper-bound peak gap
0.25	3,112	197	192	87	75	15	43	31	10
0.5	3,798	240	234	107	92	18	53	38	13
1	4,641	293	286	130	112	22	65	46	15
5	8,031	507	495	226	194	39	112	80	27

Table 5.18: The energy-side argument: negative-price hours and curtailed energy. Source: Chapter 2 day-ahead prices, Chapter 4 curtailment.

Item	Value	Unit	Share in 10–16 h (%)	Energy a 1 GW flexible load could absorb (GWh)	Source
PG&E DLAP hours with negative day-ahead price, 2024	190	hours	88	190	ch1_price_stats
PG&E DLAP hours with negative day-ahead price, 2025	218	hours	94	218	ch1_price_stats
SCE DLAP hours with negative day-ahead price, 2024	1,131	hours	77	1,131	ch1_price_stats
SCE DLAP hours with negative day-ahead price, 2025	877	hours	78	877	ch1_price_stats
SDG&E DLAP hours with negative day-ahead price, 2024	989	hours	77	989	ch1_price_stats
SDG&E DLAP hours with negative day-ahead price, 2025	748	hours	80	748	ch1_price_stats
NP15 hours with negative day-ahead price, 2024	271	hours	86	271	ch1_price_stats
NP15 hours with negative day-ahead price, 2025	288	hours	94	288	ch1_price_stats
SP15 hours with negative day-ahead price, 2024	1,169	hours	76	1,169	ch1_price_stats
SP15 hours with negative day-ahead price, 2025	976	hours	78	976	ch1_price_stats
ZP26 hours with negative day-ahead price, 2024	1,151	hours	78	1,151	ch1_price_stats
ZP26 hours with negative day-ahead price, 2025	952	hours	80	952	ch1_price_stats
CAISO wind and solar curtailment, 2024	3,423	GWh	–	–	ch3_caiso_curtailment_annual
CAISO wind and solar curtailment, 2025	3,766	GWh	–	–	ch3_caiso_curtailment_annual
CAISO wind and solar curtailment, Jan-Aug 2026	4,949	GWh	–	–	ch3_caiso_curtailment_annual
flat load equivalent of 2025 curtailment (GWh / 8,760 h)	430	MW	–	–	derived

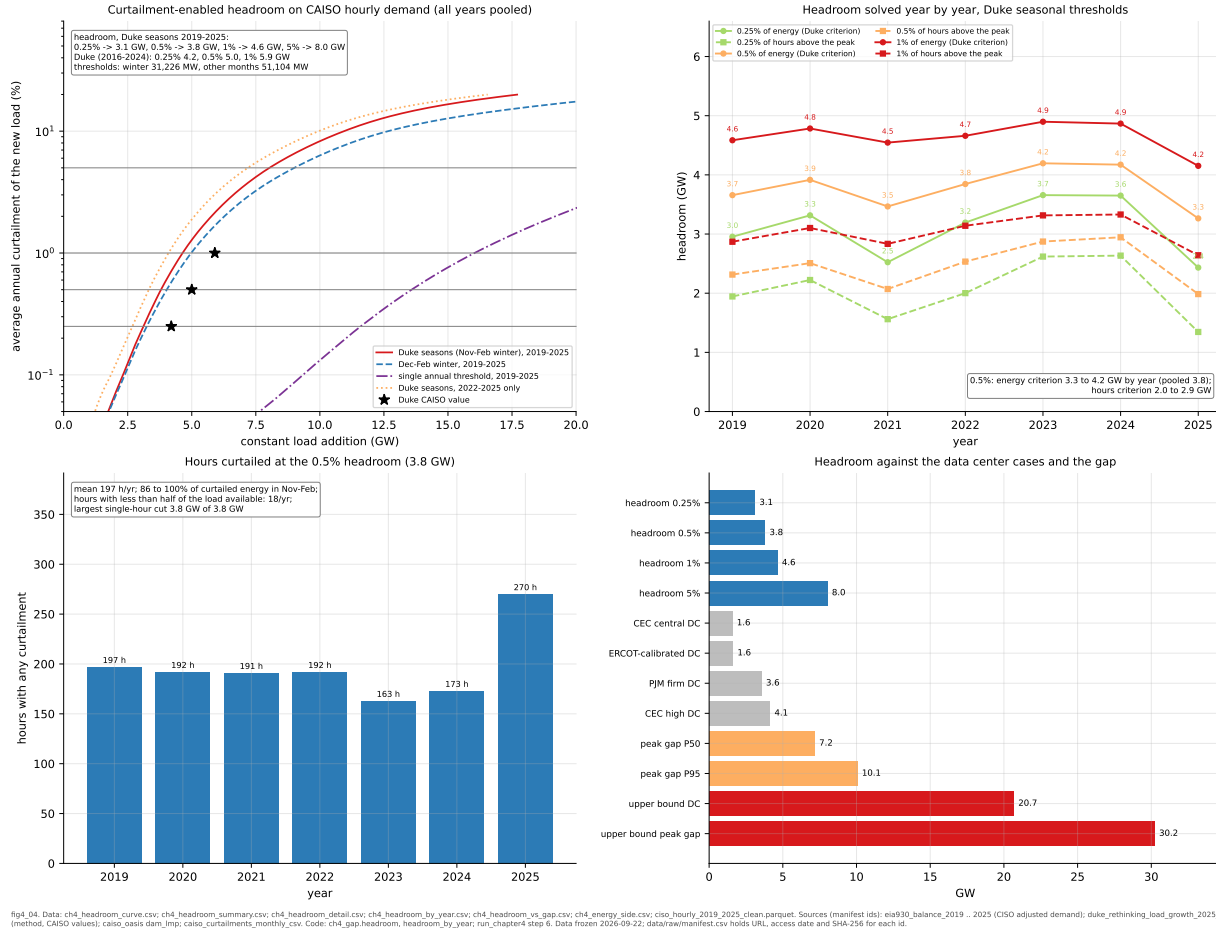


Figure 5.4: Top left: average annual curtailment of a constant load addition on CAISO demand 2019–2025 under four definitions, with Duke’s CAISO values. Top right: the headroom solved year by year under the energy and the hours criteria. Bottom left: hours curtailed each year at the 0.5 percent headroom. Bottom right: the headroom against the data center cases and the net-peak gap.

Source line: fig4_04. Data: ch4_headroom_curve.csv; ch4_headroom_summary.csv; ch4_headroom_detail.csv; ch4_headroom_by_year.csv; ch4_headroom_vs_gap.csv; ch4_energy_side.csv; ciso_hourly_2019_2025_clean.parquet. Sources (manifest ids): eia930_balance_2019 .. 2025 (CISO adjusted demand); duke_rethinking_load_growth_2025_mirror (method, CAISO values); caiso_oasis_dam_lmp; caiso_curtailments_monthly_csv. Code: ch4_gap.headroom, headroom_by_year; run_chapter4 step 6. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

5.5 Data and measurement regimes (RQ3)

Crosswalk. Table 5.19 compares six regimes on granularity, stage taxonomy, verification, time-liness, coverage and public access: the CEC’s energization tiers [California Energy Commission, 2026d, Smith, 2026, Southern California Edison, 2026], ERCOT’s large-load interconnection reporting [ERCOT Large Load Integration Team, 2026, ERCOT, 2026b], PJM’s load forecast adjustments [Mooney, 2025, PJM Interconnection, 2026], EIA’s pilot data center surveys [U.S. Energy Information Administration, 2026f], Texas SB 6 with the PUCT rules that implement it [Texas Legislature, 2025, Vegas, 2026], and FERC Docket RM26-4 [Federal Energy Regulatory Commission, 2026a,

California Independent System Operator, 2025, North American Electric Reliability Corporation, 2026, Federal Energy Regulatory Commission, 2026b]. Only ERCOT reports monthly and by phase; only PJM publishes a firm and non-firm split with the discounts it applies; only EIA measures consumption rather than requests, and it has not yet published; Texas and the FERC docket set thresholds (75 and 20 MW) and verification duties (fees, security, disclosure) rather than data; and California’s tiers have no size threshold, no public deposit or site-control test, and project-level detail only where SCE chose to publish it. Table 5.20 and Figure 5.5 score each cell from 0 to 3 under a stated rubric: ERCOT scores highest (13 of 18) on the strength of its monthly, phase-by-phase reporting, the CEC tiers 10 with full coverage but weak verification and annual timing, PJM 8, Texas SB 6 and the FERC docket 7 each because they set duties rather than publish data, and the EIA pilot 4 because it is facility-level but unpublished and one-off. Table 5.21 maps the five stages that every regime distinguishes in some form, and Figure 5.6 draws the ERCOT time series that the stock-flow case was estimated from: 63 GW of tracked requests in December 2024, 239 GW in March 2026 before that month’s submissions, 411 GW two weeks later and 467 GW by July 2026, against approvals to energize that rose from 6.9 to 9.5 GW and observed load that stayed near 4 GW. PJM’s Table B-9 adds 33.7 GW of large-load adjustments above the embedded load to its 2030 summer peak (38.8 GW of total load associated with adjustments) against requests of roughly 60 GW in its preliminary November 2025 summary, of which about 35 GW were firm (Table 5.22).

Table 5.19: Crosswalk of six large-load data and measurement regimes.

Regime	Granularity	Stage taxonomy	Verification	Timeliness	Coverage	Public access
CEC energiza- tion tiers (Calif- ornia)	MW by utility and tier, published as aggregates; project level only in SCE’s public database (TN 266008, 268459)	three groups: signed agreement, active application, inquiry (2026 IEPH proposes a longer milestone list from inquiry to ramp)	utility-reported; no deposit or site-control test in the public record; confidence levels 70/33/0 or 100/50/10 applied by CEC staff	annual IEPH data request (December 2024, summer 2025, December 2025 vintages); SCE database twice a year	no size threshold; seven utilities reported for the 2025 IEPH (VEA’s 2,600 MW are in Nevada)	docket documents and workshop decks; project-level detail confidential except SCE
ERCOT large load intercon- nection (Texas)	MW by status, load zone, TSP, project type, size class and in-service year; monthly snapshots	no studies submitted; under ERCOT review; planning studies approved (Section 9.4/9.5 requirements met); approved to energize but not operational; observed energized	TSP planning studies reviewed by ERCOT; observed energization measured from telemetry (non-simultaneous peak); Batch Zero (July 2026) adds attestations and a load information form; PUCT rule 25.194 proposes USD 50,000/MW security	monthly (ERCOT Monthly, operational overview, NPRR1267 status report approved July 2025); board and legislative decks quarterly	loads of 75 MW or more at a point of interconnection (SB 6 threshold); 20 MW co-located with a resource	aggregated public reports; customer identities confidential
PJM load fore- cast adjust- ments	MW by zone and year 2026-2046 (Tables B-9, B-9b); firm and non-firm by zone in the LAS summary	firm (electric service obligation or construction commitment) versus non-firm (letters of authorization, feasibility studies, other agreements)	utility submissions reviewed by PJM: 70% utilization unless supported, ramps of at least 36 months, non-firm zero before 2030 and 50% from 2030 with national scaling	annual cycle (requests July, review September to November, forecast January)	large load adjustments submitted by transmission owners; no MW threshold stated in the summary	load report, tables and the Load Analysis Subcommittee decks are public; project lists are not
EIA pilot data center surveys (federal)	facility-level consumption for at least one data center per company (196 companies identified)	operating facilities only; measures energy use, energy sources, site characteristics, servers and cooling	voluntary web survey (Texas, Washington) and interviews (Northern Virginia and DC)	pilot launched March 25 2026; no publication schedule yet	three regions; California not included	results not yet published; EIA states the aim of faster cycles and finer detail
Texas SB 6 (2025)	statute: disclosure duties per large load customer to the utility and ERCOT	interconnection standards to be set by the PUCT; PUCT rule 25.370 (March 2026) counts only loads with an executed interconnection agreement in the forecast after 2026	disclosure of similar requests elsewhere in the state and of on-site backup generation; flat study fee of at least USD 100,000; proposed USD 50,000/MW financial security (rule 25.194)	standards and forecast rule in force from 2026; monthly ERCOT reporting continues	demand of 75 MW or more (the PUCT may lower it); curtailment service for loads of at least 75 MW during emergencies	utilities barred from selling or sharing customer submissions except to the PUCT and ERCOT
FERC Docket RM26-4 (fed- eral)	no data collection yet; the ANOPR (Oct 23 2025) proposes interconnection procedures for loads above 20 MW	study, agreement and transition provisions modelled on Order No. 2023 (CAISO comments); June 18 2026 show cause orders name five categories of reform	to be set in each RTO/ISO tariff; the orders give the six RTOs/ISOs 60 days to justify or file revisions (CAISO Docket EL26-71-000)	ANOPR Oct 2025; comments Nov 2025; FERC committed to act by June 2026 (Apr 16 2026) and issued the orders Jun 18 2026	transmission-connected loads above 20 MW in the six RTO/ISO regions, about two-thirds of jurisdictional load	docket filings public through eLibrary (ferc.gov rejects non-browser clients, so only the CAISO and NERC filings and two news pages are in the raw store)

Table 5.20: Scores 0 to 3 per criterion under the rubric below; the cells of Table 5.19 are the evidence.

Regime	Granularity	Stage taxonomy	Verification	Timeliness	Coverage	Public access	Total (of 18)
CEC energization tiers (California)	1	2	1	1	3	2	10
ERCOT large load interconnection (Texas)	2	3	2	3	2	1	13
PJM load forecast adjustments	2	1	2	1	1	1	8
EIA pilot data center surveys (federal)	3	0	0	0	1	0	4
Texas SB 6 (2025)	0	1	3	1	2	0	7
FERC Docket RM26-4 (federal)	0	1	1	1	2	2	7

Rubric: *granularity*: 0 no data product; 1 aggregates by utility, zone or tier; 2 aggregates by phase, zone, size class and year; 3 project or facility level; *stage taxonomy*: 0 operating facilities only; 1 one split (firm/non-firm or agreement/no agreement); 2 three groups; 3 five or more phases; *verification*: 0 none or voluntary self-report; 1 utility or staff review without money at risk; 2 study or agreement gating; 3 fees, financial security, disclosure duties and curtailment obligations; *timeliness*: 0 none or one-off pilot; 1 annual; 2 twice a year or quarterly; 3 monthly; *coverage*: 0 none; 1 partial (three regions or submitted adjustments only); 2 all loads above a threshold in the jurisdiction; 3 all requests regardless of size; *public access*: 0 not published or confidential; 1 aggregates only; 2 aggregates plus some project-level or docket detail; 3 full project-level detail.

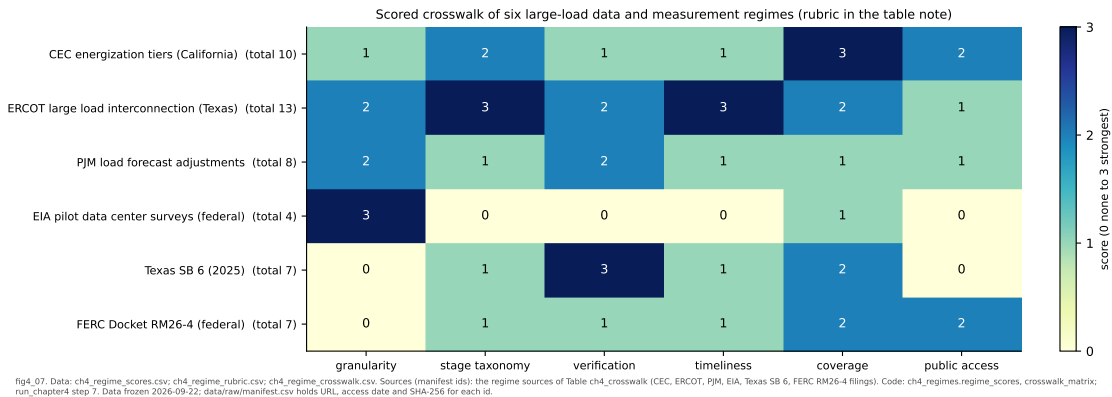


Figure 5.5: The scored crosswalk.

Source line: fig4_07. Data: ch4_regime_scores.csv; ch4_regime_rubric.csv; ch4_regime_crosswalk.csv. Sources (manifest ids): the regime sources of Table ch4_crosswalk (CEC, ERCOT, PJM, EIA, Texas SB 6, FERC RM26-4 filings). Code: ch4_regimes.regime_scores, crosswalk_matrix; run_chapter4 step 7. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 5.21: A common taxonomy of pipeline stages and how each regime names them.

common_stage	CEC tiers	ERCOT phases	PJM firm / non-firm	EIA pilot survey	Texas SB 6 / PUCT	FERC RM26-4
1 Inquiry or screening	Inquiry (Group 3)	No studies submitted	not counted (no adjustment without an agreement)	not applicable (operating facilities)	screening study after the flat study fee	request submitted; study queue
2 Application and study	Active application (Group 2)	Under ERCOT review	non-firm: feasibility study, letter of authorization	not applicable	interconnection standards; intermediate agreement with site control and security (proposed rule 25.194)	study process (60-day study proposed for flexible loads)
3 Agreement or financial commitment	Signed agreement (Group 1)	Planning studies approved; Section 9.4/9.5 requirements met	firm: electric service obligation or construction commitment	not applicable	interconnection agreement; USD 50,000/MW fee converts to non-refundable	interconnection agreement; cost responsibility for upgrades
4 Approved to energize or under construction	not distinguished (construction and initial energization in the 2026 milestone list)	Approved to energize but not operational	firm with ramp of at least 36 months	not applicable	curtailment protocol before interconnection	transition provisions for loads already under study
5 Energized and observed	existing load (about 1,000 MW peak, Dec 2025)	Observed energized (all-time non-simultaneous peak)	embedded in the historical load	surveyed consumption, energy sources, servers, cooling	curtailment during emergencies; telemetry to ERCOT	flexible service options

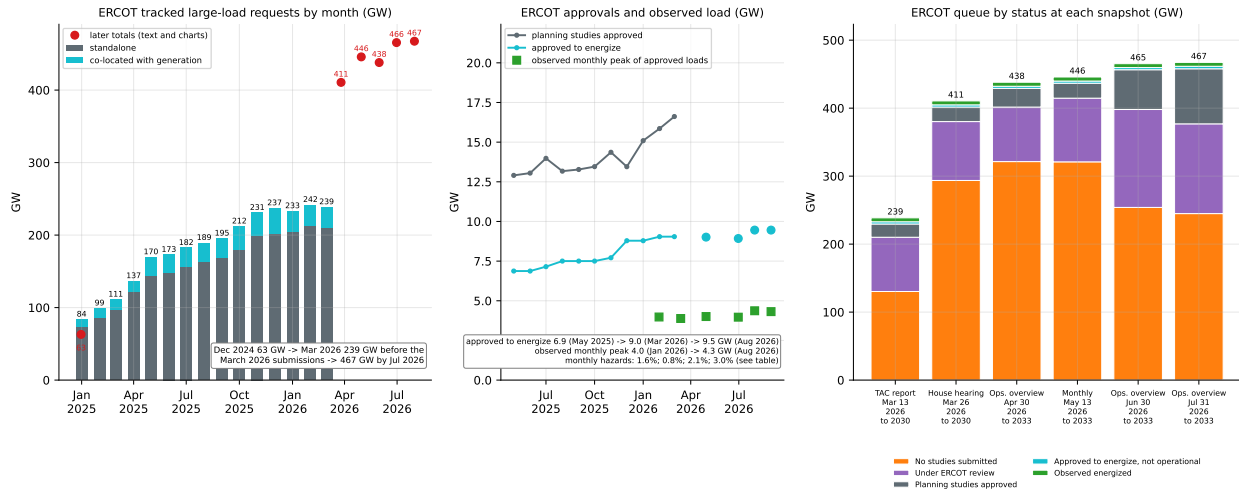


fig1_05. Data: ch4_ercot_queue_monthly.csv; ch4_ercot_approvals_monthly.csv; ch4_ercot_status_snapshots.csv; ch4_ercot_narrative.csv; ch4_ercot_transition_rates.csv. Sources (manifest ids): ercot_board_2025_12_system_planning; ercot_tac_2026_03_large_load_status; ercot_monthly_2025_07..2026_06; ercot_house_hearing_2026_04_09; ercot_board_2026_05_interconnection_update; ercot_ops_overview_2026_04..2026_08. Code: ch4_gap.ercot_series, ercot_transition_rates; run_chapter4 steps 2 and 7. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Figure 5.6: ERCOT's large-load queue: tracked requests by month (left), planning studies approved, approvals to energize and observed monthly peaks of approved loads (middle), and the queue by status at each 2026 snapshot (right).

Source line: fig4_05. Data: ch4_ercot_queue_monthly.csv; ch4_ercot_approvals_monthly.csv; ch4_ercot_status_snapshots.csv; ch4_ercot_narrative.csv; ch4_ercot_transition_rates.csv. Sources (manifest ids): ercot_board_2025_12_system_planning; ercot_tac_2026_03_large_load_status; ercot_monthly_2025_07..2026_06; ercot_house_hearing_2026_04_09; ercot_board_2026_05_interconnection_update; ercot_ops_overview_2026_04..2026_08. Code: ch4_gap.ercot_series, ercot_transition_rates; run_chapter4 steps 2 and 7. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

Table 5.22: PJM RTO large load adjustments to the summer peak (MW): the 2026 load forecast tables and the preliminary November 2025 summary chart.

PJM RTO large load adjustment (MW)	2026	2027	2028	2029	2030	2035	2040	2046
B-9 adjustments above embedded (RTO, Excel table)	6,403	10,785	16,242	22,290	33,707	63,796	77,450	81,810
B-9b total load associated with adjustments (RTO, Excel table)	11,469	15,856	21,280	27,361	38,805	68,968	82,648	87,186
Requested (LAS chart, Nov 2025, approximate)	14,000	21,000	32,000	46,000	60,000	98,000	110,000	115,000
Firm (LAS chart)	13,000	18,000	23,000	27,000	35,000	53,000	58,000	61,000
Non-firm (LAS chart)	0	0	0	0	5,000	21,000	25,000	27,000

California’s headline number under each rule. Table 5.23 restates the same requests under each regime’s counting rule. The CEC and ERCOT both count every request regardless of study status, so the headline stays at 20,677 MW (23,277 with VEA). The Texas forecast rule that admits only executed interconnection agreements after 2026 and the PJM firm-only rule both give 5,086 MW, or 3,560 MW of demand at PJM’s utilization; ERCOT’s realized phase-transition rates would energize about 2,400 MW by 2030; and a consumption survey of the EIA type would report the roughly 1,000 MW that exists. Size thresholds cannot be applied to the CEC aggregates, but in SCE’s project-level database 76 percent of active requested megawatts are in projects of 75 MW or more and 95 percent in projects of 20 MW or more (Table 5.24), so the Texas and FERC thresholds would leave most of the capacity in scope while removing most of the projects.

Table 5.23: California’s December 2025 requests under each regime’s counting rule.

Counting rule	California (MW)	What is counted
CEC: all three tiers	20,677	requested capacity, signed agreements plus applications plus inquiries, California-only (Dec 2025)
CEC: all tiers including VEA (Nevada)	23,277	the CEC’s published statewide total
CEC: agreements plus applications	12,073	the comparison the CEC used between the 2024 and 2025 IEPR vintages
ERCOT-style: every tracked request	20,677	ERCOT reports all requests of 75 MW or more regardless of study status; the CEC tiers carry no size split, so the total is the same
ERCOT-style: energized by 2030 at the ERCOT hazards	2,421	capacity expected to be energized within 60 months at the monthly phase-transition rates (before utilization)
PUCT rule 25.370 (Texas forecast): executed agreements only	5,086	only loads with an executed interconnection agreement enter the forecast after 2026
PJM-style: firm only	5,086	electric service obligation or construction commitment; non-firm zero before 2030
PJM-style: firm at 70% utilization (demand)	3,560	PJM converts requests to demand at 70% utilization
Texas SB 6 threshold (75 MW and over)	–	not computable from the CEC aggregates; in SCE’s project-level database 76% of active requested MW is in projects of 75 MW or more
FERC RM26-4 threshold (above 20 MW)	–	not computable from the CEC aggregates; in SCE’s database 95% of active requested MW is in projects of 20 MW or more
EIA pilot-style: measured consumption of existing facilities	1,000	existing data center peak of about 1,000 MW (CEC, Dec 2025); 7.0 to 9.3 TWh a year in chapter 1

Table 5.24: Active requests in SCE’s public data center database (January 29 2026) by size class.

Size class	Projects	Requested (MW)	Share of active MW (%)
under 20 MW	28	180	3
20 to under 75 MW	23	1,009	20
75 MW and over	21	3,899	76

5.6 Emissions of a flat against a curtailable load

The CAISO hourly accounting intensity of Chapter 2 gives the emissions of a 1 MW load that is flat, curtailed to zero in the highest-intensity 5, 10 or 25 percent of hours, or shifted, with the same energy moved from those hours into the lowest-intensity hours (Table 5.25, Figure 5.7). A flat load carried 183 t of CO₂ per GWh in 2025, down from 239 in 2023. Curtailing the highest-intensity quarter of hours lowers the intensity of what the load still consumes by 20 percent; shifting that energy into the lowest-intensity quarter, whose mean accounting intensity was 44 g/kWh in 2025, lowers it by 34 percent to 121 t per GWh. Flexibility therefore buys both the headroom of Section 5.4 and a third of the load’s emissions.

Table 5.25: Emissions intensity (t CO₂ per GWh consumed) of a 1 MW load that is flat, curtailed or shifted, CAISO accounting intensity 2023–2025.

Strategy	Share of hours (%)	2023	2024	2025	Change vs flat, 2025 (%)
flat	0	239	195	183	0
curtail	5	233	187	175	4
shift	5	223	176	165	10
curtail	10	227	180	168	8
shift	10	212	162	150	18
curtail	25	210	159	146	20
shift	25	190	133	121	34

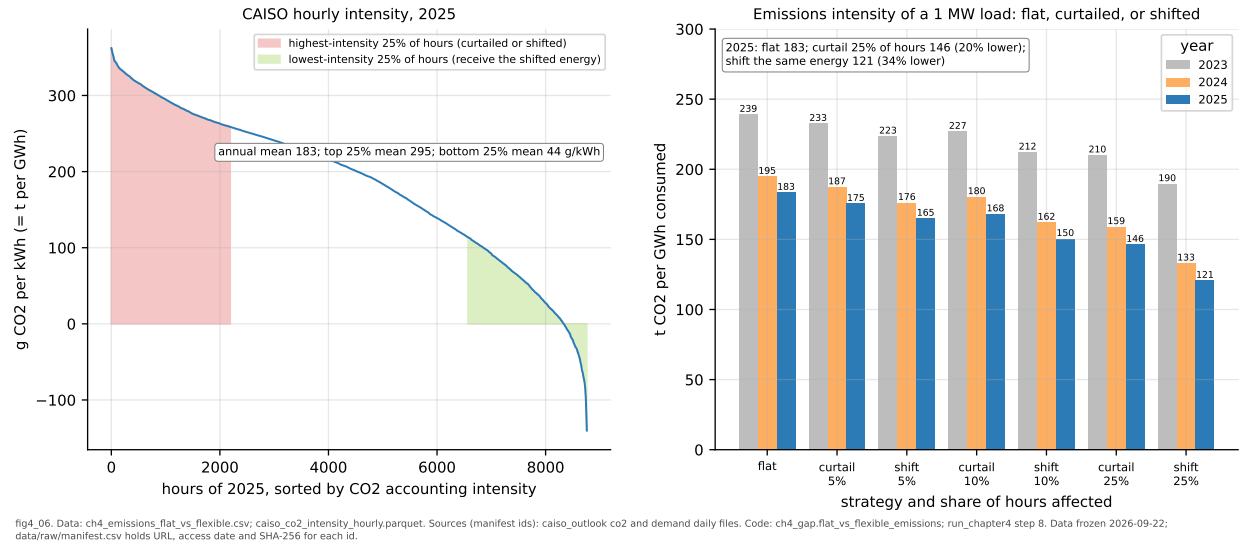


Figure 5.7: Left: CAISO hourly accounting intensity in 2025 sorted from highest to lowest, with the quarter of hours that a flexible load would avoid and the quarter that would receive the shifted energy. Right: emissions intensity of a flat, curtailed or shifted load, 2023–2025.

Source line: fig4_06. Data: ch4_emissions_flat_vs_flexible.csv; caiso_co2_intensity_hourly.parquet. Sources (manifest ids): caiso_outlook co2 and demand daily files. Code: ch4_gap.flat_vs_flexible_emissions; run_chapter4 step 8. Data frozen 2026-09-22; data/raw/manifest.csv holds URL, access date and SHA-256 for each id.

5.7 What the chapter answers

RQ2: with the CEC’s tiers weighted by realization, utilization and ramp and the supply side weighted by California’s own completion record, the 2030 gap is 50 TWh (19 to 78) and 7.2 GW (4.2 to 10.1) at the net peak, of which data centers are 18 TWh and 2.3 GW at the median; the rest is ordinary load growth and the retirements and slow additions of the supply side. The proposal’s upper bound of 202 to 242 TWh and 25 to 30 GW is four to five times the weighted 95th percentile. RQ4: curtailment-enabled headroom of 3.8 GW at 0.5 percent (3.3 to 4.2 GW year by year, or 2.0 to 2.9 GW if every hour above the peak counts) is enough to place the probability-weighted data center load two times over and half of the whole weighted peak gap, but 18 percent of the upper bound, and the surplus midday energy that a flexible load could use is real and growing. RQ3: the same requests are 20,677 MW under the CEC’s and ERCOT’s rules, 5,086 MW under the PJM and Texas forecast rules, about 2,400 MW at ERCOT’s realized transition rates and about 1,000 MW of measured load; the regimes differ most in timeliness (monthly in ERCOT, annual elsewhere), verification (deposits and agreements in Texas and PJM, none in California’s public record) and public access (aggregates everywhere, project-level detail only from SCE).

Chapter 6

Discussion and Policy

6.1 The contribution restated

This report is an independent replication and audit of the California Energy Commission’s data center forecast method, carried out on public data with every assumption written down, varied and tested against outside evidence. It does not propose a competing forecast. It shows what the CEC’s published parameters produce when applied to the CEC’s published request data, where the result matches the Commission’s own numbers and where it does not, what the same discipline does to the supply side, and how much of the resulting gap the assumptions that people argue about actually move.

Three results carry the argument. The CEC’s confidence levels of 70/33/0 and 100/50/10 percent, its 67 percent utilization factor and the exemption of Silicon Valley Power’s requests reproduce the Commission’s 2040 endpoints of 4,855 and 7,381 MW to within 1 MW, and its ramp profile reproduces the 2030 components within 2 and 6 percent (Table 5.4). The forecast is therefore transparent in the strict sense: anyone with the request table and the memo can rebuild it. Second, when the supply side is discounted with a completion model estimated on California’s own planned generators, a third of the 2030 planned list disappears and the weighted fleet adds on average 5 TWh and 1.1 GW of ELCC by 2030 once retirements are netted (Table 5.10), so the gap that opens is mostly not a data center gap. Third, the proposal’s upper bound of every requested megawatt operating flat is four to five times the probability-weighted 95th percentile, which is the quantitative form of the point Manner made for Finland: the registry number and the load that will exist are different quantities, and the difference is the object of study rather than an error to be corrected.

6.2 What the audit found about each assumption

Confidence levels. The CEC’s 70/33/0 percent for agreements, applications and inquiries sit above what ERCOT’s realized phase-transition rates would give over the same 60 months (37, 6 and 2 percent, Table 5.8) and, for agreements, below the 100 percent that PJM’s firm rule and the CEC’s own Local Reliability scenario assign. The ERCOT rates were measured while its queue quadrupled, so they measure study throughput as much as project viability, but they are the only empirical phase-transition evidence in the public record, and the CEC’s central values are not conservative against it. The sensitivity analysis shows that this matters less than it seems: the three tier probabilities together carry about 3 percent of the variance of the energy gap (Table 5.12).

Utilization. The 67 percent factor is the upper end of what SVP observed among its existing customers, by the memo’s own account. ERCOT’s observed monthly peaks of approved loads run at 46 to 64 percent of approved capacity depending on the month and the definition (Table 5.7, Figure 5.6); PJM uses 70 percent unless a utility supports more. The CEC’s value is in the middle of that evidence and is defensible.

Ramp. The CEC’s ramp puts 36 percent of the Planning forecast’s full demand on line by 2030 and 60 percent of the Local Reliability scenario’s (Table 5.5). PJM’s minimum 36-month ramp would put all of a December 2025 agreement on line by 2030. The ramp is the data center assumption with the largest effect on the peak gap (a 2.2 GW swing between its 5th and 95th percentiles, Table 5.11), which argues for publishing the utility schedules it rests on.

The SVP exemption and the Nevada request. Two features of the CEC forecast are not visible in its headline numbers. SVP’s 1,038 MW of requests enter at full confidence, which is 350 MW of the Planning forecast’s 2040 endpoint. VEA’s 2,600 MW application is in Nevada but inside the CAISO balancing area, so it appears in the CAISO data center component (160 and 242 MW in 2030) and in the 23,277 MW statewide total. Both are legitimate choices; both should be stated where the totals are published.

The supply side. The CEC discounts requests but the resource planning process does not discount planned generation in the same public way. California’s own record says it should: 65 percent of the megawatts on the January planned lists of 2016 to 2022 were operating by December 2025, with gas at 44 percent and geothermal at 6 percent (Table 4.6), and the model-weighted 2030 list loses a third of its capacity. The realization model is the symmetric counterpart of the confidence levels and is the part of this audit most worth carrying into the state’s own process.

6.3 What the gap is, and what it is not

The gap reported in Chapter 5 is an increment: the growth of demand from 2025 to 2030 minus the growth of supply at recent operating patterns. Its median of 50 TWh and 7.2 GW is not a forecast of blackouts. It is the amount that has to come from somewhere other than the probability-weighted pipeline: from the existing gas fleet running harder (a capacity factor of 0.40 rather than 0.24 would close the median energy gap), from imports that have fallen from 103 to 62 TWh since 2012 and could rise again, from Diablo Canyon operating past 2030, from load flexibility, or from resources that are not yet on the planned list. The 2025 system already meets its peak with in-state resources plus imports, and the incremental gap says that the same position in 2030 needs 4 to 10 GW more than the weighted pipeline delivers. Data centers are about a third of that at the median and a smaller share of its uncertainty. The energy gap moves with imports, the non-data-center forecast, hydro and Diablo Canyon; the peak gap with Diablo Canyon, the forecast case, the ramp and generation completion (Figure 5.3).

The upper bound is different in kind. Twenty gigawatts of flat load added to a 46 GW system is not a scenario the state’s planning process can absorb, and no counting regime in the crosswalk treats it as one. Its value is as the ceiling that the confidence levels, utilization and ramp are discounting from, and the audit shows the discount to be a factor of thirteen in the CEC’s central case.

6.4 Policy implications

For the CEC: publish the request data at the level the audit needed. Everything in Section 6.2 was possible only because the memo states its parameters and SCE chose to publish its project-level database. The rest of the tiers are aggregates by utility. Publishing every request with its size class, requested energization year, tier and status history, as SCE already does and as ERCOT does in aggregate, would let the confidence levels be estimated from California’s own attrition rather than asserted, would make the size thresholds of Texas and FERC applicable (in SCE’s database 76 percent of active megawatts are in projects of 75 MW or more), and would make duplicate requests across utilities detectable.

For the CEC: report monthly and by phase. ERCOT scores highest on the crosswalk because it publishes its queue every month by phase, and that series is what made the stock-flow case possible (Figure 5.6). An annual vintage cannot distinguish a project that advanced from one that was replaced. A monthly or quarterly publication of the tiers with the common taxonomy of Table 5.21 would cost little and would let the state estimate its own transition rates within two years.

For the CEC and the CPUC: separate firm from non-firm and say so. PJM’s rule, that only projects with a service obligation or construction commitment count in the near term, and the Texas forecast rule, that only executed interconnection agreements enter the forecast after 2026, both give California 5,086 MW (Table 5.23). The CEC’s Planning forecast already effectively does this through the zero weight on inquiries and the third weight on applications, but it presents the result as a single number. Publishing the firm and non-firm components separately, with the discount applied to each, would make the forecast comparable with PJM’s and would make the choice of confidence level visible.

For the CPUC and CAISO: offer the flexible service the headroom pays for. The replicated headroom of 3.1 to 4.6 GW at 0.25 to 1 percent curtailment (Table 5.13) is larger than every probability-weighted data center case and about equal to PJM’s firm-only case, and almost all of the curtailment falls in a few hundred winter evening hours. FERC’s June 2026 orders ask CAISO to propose new transmission services for flexible large loads [Federal Energy Regulatory Commission, 2026b]; the headroom numbers are the evidence for what such a service could offer in California, and the negative-price hours and 3.8 TWh of curtailed wind and solar are the energy the same load could absorb.

For the resource planning process: weight the supply side. A completion probability by technology and construction status, estimated as in Chapter 4 and updated each year from EIA-860M, should sit next to the confidence levels in the same forecast set. The two pumped-storage projects on the 2030 list, whose approvals have not been initiated, are the clearest case: 1,600 MW of nameplate that the model weights at 13 percent.

For the Legislature: the Diablo Canyon decision is the largest single lever. The plant’s status moves the median energy gap by 18 TWh and the peak gap by 2.2 GW, more than any data center assumption. The federal licences are in hand and state law stops at 2030; the decision belongs to the Legislature and should be taken with the 2030 balance in view, not after it.

For the federal data effort. EIA’s pilot survey covers Texas, Washington and Northern Virginia and not California [U.S. Energy Information Administration, 2026f]. A California sample would replace the four-way range of existing use in Chapter 2 (7.0 to 9.3 TWh, about 1,000 MW) with a measurement.

6.5 Limitations

The boundaries do not all coincide. Load, prices and carbon are CAISO’s; generation, capacity and the forecast are statewide; the incremental form of the gap cancels the level mismatch but not the geography, and the absolute peak balance leaves out imports’ contribution at the peak. The ELCC values come from the CPUC’s 2023 study and the hydro and firm derates are stated assumptions. The ERCOT hazards were estimated over ten to twelve months during a surge and the no-studies hazard is an upper bound. Several ERCOT and PJM values were transcribed from chart images and are labelled as such. The reserve margin is an assumed range with no raw file behind it. Data center energy in 2030 is the annual rate at the end-2030 level, which overstates calendar-year 2030 energy for a load still ramping. Data center load is treated as flat at the system peak, which the CEC’s own profiles support at 85 to 90 percent. The headroom replication uses seven years of load rather than Duke’s nine and a cleaning of its own, and the winter threshold decides most of the answer. No project-level California request data exist outside SCE, so the tier probabilities could not be estimated from California’s own attrition. The CEC’s utility-supplied ramp schedules are not public, which is why the replicated 2030 components differ from the published ones by 2 to 6 percent.

6.6 Future work

The monthly snapshot job that has run since the start of the project will, by the end of 2027, hold enough vintages of the CEC docket, ERCOT’s queue, PJM’s forecast and EIA-860M to estimate California’s own tier transition rates and to test the ERCOT-calibrated case against what ERCOT’s Batch Zero actually energizes. Matching SCE’s project-level database to Epoch AI’s facility timelines would give the first California check of paper requests against visible construction. Hourly profiles of the existing data center load, which the CEC has from utility interval meters but has published only as ratios, would replace the flat-load assumption in both the gap and the headroom. And the 2026 IEPR, which proposes a longer milestone list from inquiry to ramp, will be the first vintage against which the taxonomy of Table 5.21 can be tested.

6.7 Conclusion

California’s data center pipeline is 20,677 MW in three tiers, 40 percent of the record CAISO peak and twenty times the load that exists (RQ1). Weighted by the CEC’s own parameters, which this report reproduces, and set against a supply side weighted by California’s own completion record, the 2030 gap between demand growth and supply growth is 50 TWh and 7.2 GW at the median, with data centers 18 TWh and 2.3 GW of it, against an upper bound of 202 to 242 TWh and 25 to 30 GW (RQ2). The regimes that count planned load differ most in timeliness, verification and public access, and the same requests are 20,677 MW under California’s and Texas’s counting rules, 5,086 MW under PJM’s and the Texas forecast rule, about 2,400 MW at ERCOT’s realized rates and about 1,000 MW as measured load (RQ3). The existing system holds 3.8 GW of headroom at 0.5 percent curtailment, enough for the probability-weighted data center load twice over and for

half of the weighted peak gap, but for less than a fifth of the upper bound (RQ4). The method is the result: a forecast that can be rebuilt, a supply side discounted the same way, and a gap with its uncertainty and its drivers named.

Appendix A

Provenance of Figures and Quoted Numbers

Every figure in this report is drawn by a script from files in `data/processed/`, which are themselves produced from raw downloads recorded in `data/raw/manifest.csv` with their URL, access date and SHA-256. The tables below list, for each figure, the processed files it is drawn from, the raw sources behind them and the code that produces them. Headline values quoted in the text are printed on the figures or appear in the chapter tables; the few numbers that appear only in prose (source-document figures such as EPRI’s state total, and ratios derived from tabulated values) are listed with their sources in `docs/DATA_NOTES.md`, which also records every data limit found during the audits. Rebuilding is one command per chapter (`make ch1`, `make ch2`, `make ch3`, `make ch4`); the standalone bundle reproduces every output byte for byte.

Table A.1: Chapter 2 figures: processed files, raw sources and code.

Figure	Content	Processed files (data/processed)	Raw sources (manifest ids)	Code
fig1_01	Load duration curves; annual demand range, record peak, load factor	ch1_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet	eia930_balance_2019_jan_jun .. 2025_jul_dec; caiso_outlook demand and fuelsource (fills, artifact filter)	ch1_baseline.load_eia930_ciso, clean_demand_against_caiso, fill_from_caiso, duration_curves; run_chapter1 step 1
fig1_02	Net-load duration curves; minimum net load; hours below zero with and without battery charging	ch1_net_load_duration_curves.csv; ch1_ciso_annual_summary.csv; ciso_hourly_2019_2025_clean.parquet	eia930 balance files; caiso_outlook fuelsource (Batteries)	ch1_baseline.add_storage_adjusted, duration_curves; run_chapter1 step 1
fig1_03	Seasonal daily profiles; 12-14h and 20h means for 2019 and 2025	ch1_seasonal_profiles.csv	eia930 balance files	ch1_baseline.seasonal_profiles; run_chapter1 step 1
fig1_04	Top-100 net-load hours by month and hour; median start, Jul-Sep and 17-21h shares	ch1_top100_net_load_hours.csv; ch1_top100_net_load_timing.csv	eia930 balance files	ch1_baseline.top_hours, top_hours_timing; run_chapter1 step 1
fig1_05	Day-ahead price duration and diurnal means; extremes; 11-14h and 18-20h mean price ranges	caiso_dam_lmp_hourly.parquet; ch1_price_duration_curves.csv; ch1_price_by_hour_of_day.csv; ch1_price_stats.csv	caiso_oasis dam_lmp monthly files (PRC_LMP DAM, DLAP nodes)	ch1_baseline.load_dam_lmp, price_duration, price_by_hour, price_stats; run_chapter1 step 2
fig1_06	Negative-price hours: full years, matched Jan 1 to Sep 22 windows, by month; share in 10-16h	ch1_negative_price_hours_windows.csv; ch1_negative_price_hours_by_month.csv; ch1_price_stats.csv	caiso_oasis dam_lmp monthly files	ch1_baseline.negative_hours_by_month, price_stats; run_chapter1 step 2
fig1_07	CO2 accounting intensity: annual, duration, diurnal; percentiles; eGRID reference	ch1_carbon_intensity_annual.csv; ch1_carbon_intensity_duration.csv; ch1_carbon_intensity_diurnal.csv; ch1_egrid_camx_2023.json; caiso_co2_intensity_hourly.parquet	caiso_outlook co2 and demand daily files; epa_egrid2023_rev1	ch1_baseline.load_caiso_outlook_hourly, carbon_summary, egrid_camx; run_chapter1 step 3
fig1_08	Existing data center electricity use: statewide estimate, assumed conversion, sensitivity, utility subset	ch1_dc_load_estimates.csv; ch1_ca_retail_sales_eia861.csv; ch1_svp_fact_sheets.csv; ch1_kollar_grady_points_california_flag.csv; ch1_bottom_up_inputs.json	epri_powering_intelligence_2024; cec_dc_methodology_memo_2026; eia861_2019 .. 2024; svp_fact_sheet_2017 .. 2023; kollar_grady_2025_zenodo; epoch_data_centers	ch1_baseline.dc_load_estimates, bottom_up_estimate, ca_retail_sales_eia861, svp_fact_sheet_energy, kollar_grady_california; run_chapter1 step 4

Table A.2: Chapter 3 figures: processed files, raw sources and code.

Figure	Content	Processed files (data/processed)	Raw sources (manifest ids)	Code
fig2_01	Census data center construction spending with ChatGPT annotation; month-over-month growth; shares of nonresidential construction	ch2_census_c30_data_center.csv	census_c30_privsatime; census_c30_privtime	ch2_growth.census_c30; run_chapter2 step 1
fig2_02	Log-linear trends, Chow test, Bai-Perron breaks, segment growth, sequential test; ARIMA and ETS forecasts with bands and backtest	ch2_census_break_summary.json; ch2_census_bai_perron.csv; ch2_census_segment_growth.csv; ch2_census_forecast_arima.csv; ch2_census_forecast_ets.csv; ch2_census_forecast_backtest.csv	census_c30_privsatime	ch2_growth.chow_test, bai_perron, sequential_supF, arima_forecast, ets_forecast, backtest_forecasts; run_chapter2 step 2
fig2_03	California proxies: QCEW employment and establishments, CBRE Silicon Valley construction and inventory, Epoch cumulative power; break-test summary	ch2_qcew_518210_california.csv; ch2_cbre_california.csv; ch2_epoch_timeline.csv; ch2_break_test_comparison.csv	qcew_518210_2014q1 .. 2026q1; cbre_*_infogram tables via cbre_california_market_series.csv; epoch_data_center_timelines; epoch_data_centers	ch2_growth.qcew_california, cbre_california, epoch_timeline, break_analysis; run_chapter2 step 3
fig2_04	Energization requests by vintage and tier; Nevada separated; SCE cancellations; PG E pipeline; ratios to CAISO record peak and existing load	ch2_tier_vintages.csv; ch2_rq1_denominators.json; ch1_dc_load_estimates.csv	cec_dc_forecast_2024iepr; cec_prelim_dc_forecast_2025; cec_dc_methodology_memo_2026; cec_assembly_hearing_2026_01_28; cec_tn266008; cec_tn268459; cec_tn272026; cec_tn272065; cec_tn272807; caiso_key_statistics_2026_08	ch2_growth.tier_vintages; run_chapter2 steps 4 and 5

Table A.3: Chapter 4 figures: processed files, raw sources and code.

Figure	Content	Processed files (data/processed)	Raw sources (manifest ids)	Code
fig3_01	In-state generation by resource and CEC net imports 2010-2025; shares	ch3_supply_by_year.csv; ch3_eia923_generation_by_resource.csv; ch3_cec_generation_multi_year.csv	eia923_2010 .. 2025; cec_elec_energy_generation_page; ch1 EIA-930 imports	ch3_supply.eia923_ca_generation, cec_multi_year_generation; run_chapter3 step 1
fig3_02	Operable capacity by technology; battery power and energy; CAISO curtailment; July 2026 snapshot	ch3_eia860_capacity_by_resource.csv; ch3_battery_capacity.csv; ch3_caiso_curtailment_annual.csv; ch3_capacity_snapshot_2026_07.json	eia860_2010 .. 2025; eia860m_2026_07; caiso_curtailments_monthly_csv	ch3_supply.eia860_ca_capacity, eia860_ca_battery_energy, caiso_curtailment_annual, eia860m_capacity_snapshot; run_chapter3 step 1
fig3_03	Scheduled retirements: EIA-860M planned dates and once-through-cooling compliance dates	ch3_eia860m_planned_retirements.csv; ch3_otc_units.csv	eia860m_2026_07; swrcb_otc_policy_2023	ch3_supply.retirement_schedule; run_chapter3 step 2
fig3_04	Realization model: outcomes by vintage and technology; logistic model; delay and cumulative-incidence curves	ch3_vintage_outcomes.csv; ch3_realization_by_vintage.csv; ch3_realization_by_technology.csv; ch3_logit_coefficients.csv; ch3_logit_summary.json; ch3_delay_km.csv; ch3_delay_cif.csv	eia860m_2016_01 .. 2022_01; eia860m_2016_12 .. 2025_12	ch3_supply.eia860m_vintage_outcomes, fit_completion_logit, km_delay; run_chapter3 step 3
fig3_05	Capacity, energy and ELCC-derived peak in 2030: existing fleet and three cases	ch3_cases_2030.csv; ch3_cases_2030_summary.json; ch3_planned_current_weighted.csv; ch3_existing_capacity_2025.csv; ch3_capacity_factors.csv; ch3_elcc_values.csv	eia860m_2026_07; eia860m_2025_12; cpuc_e3_astrape_incremental_elcc_2023	ch3_supply.current_planned, cases_2030, capacity_factors, elcc_table; run_chapter3 step 4

Table A.4: Chapter 5 figures: processed files, raw sources and code.

Figure	Content	Processed files (data/processed)	Raw sources (manifest ids)	Code
fig4_01	Data center demand in 2030 under each counting rule; statewide peak by IEPR case against ELCC supply	ch4_demand_cases_2030.csv; ch4_demand_totals_2030.csv; ch4_cec_replication.csv; ch4_cec_ramp_profile.csv; ch4_ca_tiers.csv; ch4_iepr_growth_summary.csv; ch4_ercot_chain.csv; ch3_cases_2030_summary.json	cec_assembly_hearing_2026_01_28; cec_dc_methodology_memo_2026; cec_tn268722 .. 268727 (CED 2025 forms); cec_tn268124; cec_tn268824; ERCOT decks and monthlies (see fig4_05); pjmlar_summary_2025_11_24	ch4_gap.ca_tiers, cec_parameters, cec_forecast_replication, ced2025_scenarios, iepr_growth_cases, ercot_chain, demand_cases_2030, demand_totals_2030; run_chapter4 steps 1-3
fig4_02	Monte Carlo distributions of the 2030 energy and net-peak gap, by IEPR case; upper bound	ch4_mc_draws.parquet; ch4_mc_summary.csv; ch4_upper_bound_2030.csv; ch4_mc_inputs.csv; ch4_mc_key_numbers.json	chapter 3 outputs (eia860m_2026_07, eia860m_2025_12, cpuc_e3_astrape_incremental_elcc_2023, eia923, cec_elec_energy_generation_page); CED 2025 forms as above	ch4_gap.SupplyModel, mc_inputs, monte_carlo, evaluate_gap, summarize, upper_bound_rows; run_chapter4 step 4
fig4_03	One-at-a-time tornado and Sobol indices for both gaps	ch4_tornado.csv; ch4_sobol.csv; ch4_mc_draws.parquet	as fig4_02	ch4_gap.tornado, sobol_indices (SALib); run_chapter4 step 5
fig4_04	Curtailed-enabled headroom on CAISO hourly demand; hours curtailed; headroom against the cases and the gap	ch4_headroom_curve.csv; ch4_headroom_summary.csv; ch4_headroom_detail.csv; ch4_headroom_by_year.csv; ch4_headroom_vs_gap.csv; ch4_energy_side.csv; ciso_hourly_2019_2025_clean.parquet	eia930_balance_2019 .. 2025 (CISO adjusted demand); duke_rethinking_load_growth_2025_mirror (method, CAISO values); caiso_oasis_dam_lmp; caiso_curtailments_monthly_csv	ch4_gap.headroom, headroom_by_year; run_chapter4 step 6
fig4_05	ERCOT tracked requests by month, approvals and observed load, queue by status at each snapshot	ch4_ercot_queue_monthly.csv; ch4_ercot_approvals_monthly.csv; ch4_ercot_status_snapshots.csv; ch4_ercot_narrative.csv; ch4_ercot_transition_rates.csv	ercot_board_2025_12_system_planning; ercot_tac_2026_03_large_load_status; ercot_monthly_2025_07 .. 2026_06; ercot_house_hearing_2026_04_09; ercot_board_2026_05_interconnection_update; ercot_ops_overview_2026_04 .. 2026_08	ch4_gap.ercot_series, ercot_transition_rates; run_chapter4 steps 2 and 7
fig4_06	CAISO hourly intensity 2025 and emissions of a flat, curtailed or shifted 1 MW load	ch4_emissions_flat_vs_flexible.csv; caiso_co2_intensity_hourly.parquet	caiso_outlook_co2_and_demand_daily_files	ch4_gap.flat_vs_flexible_emissions; run_chapter4 step 8
fig4_07	Scored crosswalk of the six regimes (0-3 per criterion)	ch4_regime_scores.csv; ch4_regime_rubric.csv; ch4_regime_crosswalk.csv	the regime sources of Table ch4_crosswalk (CEC, ERCOT, PJM, EIA, Texas SB 6, FERC RM26-4 filings)	ch4_regimes.regime_scores, crosswalk_matrix; run_chapter4 step 7

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